

MF-17

Inverse-semigroup growth

Complete proposed proof, attempt history,
failed routes, constants, and open checks

A handoff for the next research agent

22 September 2026

Current proposed rate

$$G_M(t) \asymp_M [\log \log(t + e^e)]^{\frac{2}{\pi} \arccos(1/M)}, \quad M > 1.$$

Unresolved refinement: equality of the normalized lower and upper tail constants. The saved reports supply a proposed analytic proof and finite arithmetic certificates. This compilation preserves those arguments and their qualifications; it is not an independent validation of the universal theorem.

Scope decision. The latest SDP-asymptotics branch is deferred. The earlier exact finite representation is retained only as an optional continuation tool.

Deliverables. This standalone LaTeX document, its compiled PDF, the original available source reports, and the retained exact checkers and data are packaged together. No original mathematical source has been silently replaced by a new claim.

Contents

I	Orientation and attempt ledger	4
H.1	Scope, status, and the continuation target	4
H.2	Chronology of the mathematical progress	6
H.3	Failure ledger: what to retain and what not to repeat	8
H.4	Proof dependencies and the highest-priority audit points	11
II	Inherited reductions and obstructions	13
R0.1	The target, the baseline, and the actual outcome	13
R0.2	The constraints that every successful approach must respect	14
R0.3	Resolvent normalization: removing unbounded generators	14
R0.4	Finite nilpotents recover the full envelope	16
R0.5	The semidefinite characterization and its unresolved optimization	17
R0.6	A direct sum removes one possible quantifier obstruction	19
R0.7	Foguel–Hankel models: a proved obstruction to a natural construction	19
R0.8	Why the logarithmic Besov norm did not turn into a lower bound	21
R0.9	Other exploratory routes and their specific limitations	22
R0.10	What was actually checked computationally	23
R0.11	The precise unfinished task	24
III	Damping, a finite witness, and variation	25
R1.1	The target and the outcome	25
R1.2	A continuous damping estimate	26
R1.3	An exact 48-dimensional witness	27
R1.4	Why a bounded-variation phase-lattice construction cannot be packed arbitrarily densely	29
R1.5	Reproducibility and the remaining gap	31
IV	The first doubly logarithmic upper proof	32
U.1	Statement and the remaining gap	32
U.2	A three-factor shifted-window estimate	32
U.3	The scalar oscillatory estimate	34
U.4	Applying the estimate to inverse semigroups	36
U.5	Two consequences with different quantifiers	37
U.6	What this proves, and what it does not	38

V	The binary-cut intermediate route	39
B.1	The intermediate binary-cut proof	39
VI	Complete proposed sharp-rate proof	42
S.1	Statement	42
S.2	Sharp geometry of all coordinate cuts	43
S.3	Approximate cuts and two-sided summation	44
S.4	Oscillatory blocks and sparse resolvent cuts	46
S.5	Completion of the upper bound	48
S.6	A replication lemma for one active phase	49
S.7	Rounded sector weights at the endpoint	50
S.8	Realization by exponentially stable generators	53
S.9	Endpoints, consequences, and scope	55
S.A	Proof of the oscillatory-window estimate	57
S.B	Higher-order checks on the lower construction	59
S.C	An exact check for real-Gram phase models	61
S.D	Numerical diagnostics and reproducibility	61
VII	Constants and threshold bookkeeping	63
Q.1	All-time constants: dependency recipe from source Q	63
Q.2	Threshold trade-offs and the true tail constants	66
VIII	Complete eventual-constant optimization	68
O.1	Which constants are threshold costs rather than asymptotic prefactors? . . .	69
O.2	Exact finite-degree Poisson comparison	69
O.3	Exact use of the partial-sum constraints	70
O.4	Asymptotically free augmentation and two residue classes	70
O.5	Replacing the astronomical oscillatory constant	71
O.6	Combining the optimized upper constants	73
O.7	Lower normalization: preserve the leading coefficient	74
O.8	Optimize the test polynomial instead of fixing a Dirichlet kernel	74
O.9	A sharper phase-replication certificate	76
O.10	The symbolic lower optimization	78
O.11	A simpler explicit lower recipe	78
O.12	Fully explicit representative at M=2	79
O.13	Threshold conventions and what has actually been optimized	80
O.A	The beta-profile moments	81

IX	Latest tail-coefficient construction	82
T.1	Yosida–Poisson normalization	82
T.2	A compact semidefinite program	83
T.3	A uniform truncation error	84
T.4	Exact expressions for the two requested constants	87
T.5	A better scalar concentration bound	87
T.6	Polar factorization and direct row estimates	88
T.7	The coefficient of one real oscillatory block	89
T.8	Averaging shifted logarithmic grids removes the residue-class loss	90
T.9	Retaining the Hardy-space information	90
T.10	The six exact moments for one Hardy branch	91
T.11	A finite matrix recipe for the exact boundary certificates	92
T.12	Different replication numbers for the two coordinate types	94
T.13	A strictly verified numerical consequence at $M=2$	95
T.14	What has been determined	96
T.15	A structural bound on the possible tail oscillation	96
T.16	Why shrinking the certificate interval is not a convergence proof	97
X	Continuation and reproducibility	98
Z.1	Continuation protocol	98
Z.2	Evidence, retained artifacts, and reproducibility	99
Z.3	Final state to preserve	100
	Sources and recorded external dependencies	102

Part I

Orientation and attempt ledger

Source and scope. Editorial synthesis of sources P, R0, R1, U, S, Q, O, T, and the displayed conversation. This part is new organization of the record, not a new research proof.

H.1 Scope, status, and the continuation target

H.1.1 What this handoff contains

This is a source-grounded consolidation of the recoverable MF-17 research record, prepared for a future agent. It contains the complete saved sharp-rate proof, both versions of the lower construction, the successive upper-bound improvements, the attempted routes that did not produce a solution, the constant and threshold calculations, and the exact finite certificates. It is not a transcript of private reasoning. Where an exploration was reported but no formula, matrix, or log survives, that limitation is stated rather than filled in.

The latest **SDP-asymptotics branch is deliberately set aside**. That is a change of research direction, not a finding that the branch is false. The finite SDP representation preceding that branch is retained in report T, Sections 1–4, as an optional tool. None of the fixed-degree asymptotics, central-path asymptotics, bandwidth bounds, or second-order finite-section refinements from the last branch is needed below.

The source material makes a strong proposed claim:

$$G_M(t) \asymp_M L(t)^{a(M)}, \quad L(t) = \log \log(t + e^e), \quad a(M) = \frac{2}{\pi} \arccos(M^{-1}), \quad M > 1.$$

This is the current proposed proof, not a theorem independently certified by this handoff. Reports S, Q, O, and T explicitly distinguish their analytic arguments from peer review and from finite arithmetic checks. Their mathematical assertions and proof steps are preserved here so that a successor can examine them. The document does not promote repeated assertions, successful compilation, or arithmetic test results into an independent verification of the universal theorem.

There are two different continuation tasks. First, audit the claimed sharp exponent and its fixed- M endpoint construction. Second, conditional on those arguments, determine whether the normalized envelope has a limit:

$$\underline{K}_M = \liminf_{t \rightarrow \infty} \frac{G_M(t)}{L(t)^{a(M)}}, \quad \overline{K}_M = \limsup_{t \rightarrow \infty} \frac{G_M(t)}{L(t)^{a(M)}}, \quad \underline{K}_M \stackrel{?}{=} \overline{K}_M.$$

The record does **not** prove this equality or evaluate either true leading coefficient. Improved lower and upper certificates must not be relabeled as the true constants.

H.1.2 Exact problem and non-negotiable quantifiers

For a fixed real number $M > 1$, $\mathcal{A}_M$ consists of pairs (H, A) with H a complex Hilbert space and A the densely defined generator of a strongly continuous semigroup satisfying

$$\|e^{sA}\| \leq M e^{-s} \quad (s \geq 0).$$

The generator may be unbounded. Exponential stability gives $0 \in \rho(A)$ and $A^{-1} \in \mathcal{L}(H)$, so $e^{tA^{-1}}$ is the norm-convergent exponential of a bounded operator. The target is

$$G_M(t) = \sup_{(H,A) \in \mathcal{A}_M} \|e^{tA^{-1}}\|.$$

The original problem asks for a rate with matching positive multiplicative bounds for all sufficiently large t . Constants may depend on the **fixed** M , but not on t , A , H , dimension, or the particular finite witness. This is the statement in source P; leading-constant equality is a subsequent, stronger research question.

All inner products in the main proof are linear in the first argument. Circle measure is normalized unless a source explicitly says otherwise. The bar in $\bar{H}$ is entrywise boundary conjugation, not an adjoint. In the lower construction the compatible convention is

$$a = Hu, \quad b = \bar{H}v, \quad g = \bar{H}H^{-1}, \quad \langle u, v \rangle_H = \langle ga, b \rangle_{L^2}.$$

The original TeX sources, rather than text extracted from PDFs, are authoritative for overlines, starred operators, and subscripts. No mathematical correction has been silently inferred from a PDF extraction artifact.

H.1.3 Reading routes

For the **complete exponent proof**, read report S in order, including its oscillatory-window appendix and its phase-replication lemma. It is the most self-contained baseline construction. The newer pure-power route in T improves coefficients but introduces an additional spectral-diagonalization dependency; it should not replace the baseline during an initial audit.

For the **best retained leading-coefficient bounds**, read O Sections 2, 4–5, and 8 with its beta-profile appendix, then T Sections 5–13. T replaces O’s scalar concentration coefficient, rearrangement step, separate resolvent windows, phase-by-phase estimate, residue-class split, and rounded-weight replication bound. The source dependencies are explicitly identified in both reports.

For **failed approaches and design constraints**, read the chronological ledger below, then R0 Sections 2 and 6–10 and R1 Sections 2–4. Report U explains the first change from a logarithmic to a doubly logarithmic upper rate. Part B reconstructs the intermediate binary-splitting argument that was recorded in the conversation but not retained as a separate source file.

For **constants and thresholds**, use Part Q and the archived constant ledger. For computational evidence, use Part Z and the bundled scripts. For an optional finite model independent of the sharp exponent, use T Sections 1–4, not the deferred SDP-asymptotics report.

H.1.4 Source labels and preservation policy

Label	Retained source	Role and status
P	<code>problem.pdf</code>	User-supplied problem and historical bibliography. Its historical status check is not an updated literature review.
R0	<code>MF-17-progress-and-obstacles.tex</code> in the initial ZIP	Inherited reduction and obstruction report. Full mathematical body reproduced.
R1	<code>MF-17-follow-up.tex</code> in <code>MF-17-results.zip</code>	Damping proof, exact 48-dimensional witness, variation obstruction. Full mathematical body reproduced.
U	<code>MF-17-doubly-logarithmic-upper-bound.tex</code>	First double-log upper proof. Full body reproduced.
S	<code>MF-17-sharp-fixed-bound-rate.tex</code>	Complete proposed sharp fixed- M proof and appendices. Full body reproduced.

Label	Retained source	Role and status
Q	MF-17-explicit-constants.pdf, constant-ledger.json	All-time constant trace. Part Q transcribes its dependency recipe and the conversation's threshold discussion; the complete original PDF is bundled.
O	optimized-constants-proof.md	Complete first eventual-prefactor optimization, including its profile-moment appendix.
T	MF-17-tail-constants-research.md	Complete last pre-SDP-asymptotics report: finite representation, refined upper coefficient, pure-power lower route, and remaining equality question.
B	Conversation-only intermediate argument	Reconstructed below from the displayed argument; not an independently saved research report.

Source-local section and equation numbers are prefixed in this handoff: for example, S.7.9 corresponds to equation (7.9) of source S. Bracketed references such as “[S, (7.9)]” retain that source-local meaning. Explicit equation tags in the Markdown reports are similarly prefixed. Old conclusions such as “the gap remains” are retained as historical statements of that stage, not as contradictions of a later proposed refinement.

The body of each saved proof is preserved apart from typesetting, cross-reference namespaces, and moving bibliographies to the end. The supplied original sources are included in the companion bundle. Part Q is an identified transcription/synthesis because its full LaTeX source is not mounted; it is not represented as an exact reproduction. No new external literature search or independent mathematical audit is claimed for this compilation.

H.2 Chronology of the mathematical progress

H.2.1 Initial reduction: the spectral difficulty is not eigenvalue complexity

R0 first removes unbounded generators through the resolvent normalization

$$C_\lambda = (\lambda + 1)(\lambda - A)^{-1}, \quad B_\lambda = \lambda C_\lambda - (\lambda + 1)I, \quad B_\lambda^{-1} = \frac{\lambda A^{-1} - I}{\lambda + 1}.$$

The semigroup integral proves $\|C_\lambda^k\| \leq M$, with one factor M , not M^k . An initially exponential-in- t perturbation bound is replaced by an exact commuting identity, yielding a constant-factor comparison with λ linear in t .

A tensor shift and a cyclic invariant subspace then reduce the envelope to finite nilpotents without losing the power bound. Ordinary compression was not used: it need not preserve powers. A single negative eigenvalue can therefore suffice in each finite near-extremizer. This does not imply bounded growth of the envelope; the matrix and its dimension may vary with t .

The Gram-matrix SDP and a countable direct sum establish useful exact reductions. They do not by themselves evaluate an asymptotic rate. Direct sums take a supremum of block norms, not a sum, so they do not accumulate favorable scales.

H.2.2 First quantitative continuation: a finite witness and a restricted obstruction

R1 derives the energy inequalities and the damping estimate

$$0 \leq G_M(t_2) - G_M(t_1) \leq \frac{M^2}{2} \log(t_2/t_1), \quad 0 < t_1 \leq t_2,$$

with

$$G_M(t) \leq 1 + M\sqrt{t/2}, \quad G_M(t) \leq 1 + M^2 + \frac{M^2}{2} \log \frac{t}{2M^2} \quad (t \geq 2M^2).$$

This improves a logarithmic coefficient, not its order.

An exact 48-dimensional certificate proves $G_2(100) > 5/2$. The generator is $400J_{48} - 401I$ in the supplied Q -inner product, not in the unweighted Euclidean norm. It refutes a universal estimate $G_M(t) \leq M$ but supplies no asymptotic lower family.

The variation obstruction proves that phase-lattice points with a fixed scalar-variation budget occupy only $O(\log \log Y)$ effective intervals up to Y . Dense clustering does not avoid that certificate-level restriction. It does **not** show that an arbitrary bounded semigroup has a bounded scalar-variation budget.

H.2.3 Three semigroup factors: the first double logarithm

U uses a translated Fourier-window identity with three factors of the bounded semigroup. Instead of charging separately for dyadic windows, it controls a complete interval of the form $[2N, CN^{4/3}]$ at bounded cost. Iterating $N \mapsto N^{4/3}$ requires only $O(L(t))$ blocks. This yields the proposed uniform bound

$$G_M(t) \leq 1 + CM^3[1 + L(t)].$$

The Bessel remainder is already integrable, so adding further terms to that expansion cannot produce the next exponent improvement.

At this stage the lower exponent recorded from the literature was $(M-1)/(M+49)$. Its approach to one as $M \rightarrow \infty$ proved sharpness of the **universal power** one, not a sharp exponent at prescribed M . The normalized envelope $\sup_{M>1} G_M(t)/M^3$ was bounded between constant multiples of $L(t)/(\log L(t))^3$ and $L(t)$; that was an exact logarithmic-growth index, not a fixed- M resolution.

H.2.4 Binary separation: a genuine but nonsharp power saving

The next conversation argument uses a resolvent to separate early and late oscillatory blocks. Their operator-norm separation errors have a summable bound. A binary angle argument gives a recurrence $F(2n) \leq q_M F(n) + O(M^4)$ with $q_M < 2$, and hence a power smaller than one. A convenient stated form was

$$G_M(t) \leq CM^4 L(t)^{1-[32(\log 2)M^2]^{-1}}.$$

This implies every fixed logarithmic saving relative to $L(t)$ for a fixed finite M . It does not determine the correct exponent. The proof's loss is structural: it retains one cut at a time rather than the entire family of ordered cuts. The reconstructed argument is included in Part B so that this discarded route is not lost.

H.2.5 All-cut geometry and endpoint realization

S replaces binary splitting by a scalar Hardy-space concentration argument obtained by stationarizing the full Gram matrix. The exact projection angle is $\rho_M = \sqrt{1 - M^{-2}}$, and the proposed exponent becomes

$$a(M) = \frac{2}{\pi} \arcsin \rho_M = \frac{2}{\pi} \arccos(M^{-1}).$$

Sparse resolvent cuts, orthogonal augmentation, and two-sided summation provide the upper estimate at this endpoint rather than only at $a(M) + \varepsilon$.

For the lower estimate, S uses the rounded analytic weight

$$h(z) = (1-z)^{-\alpha/2} \exp\{\eta[(1-z)^\nu - 1]\}, \quad \nu = (1-\alpha)/2, \quad \eta = \alpha/4.$$

It has the endpoint sector angle but an integrable quantitative margin. Paired coordinates yield a multiplier of size n^α . Orthogonal replication controls an arbitrary one-coordinate phase without increasing M . A polynomial finite-size margin absorbs the approximation to a scalar semigroup. Quadratically separated integer phases give observation times with $L(t_n) = O_M(n)$. These are the steps that were claimed to close the fixed- M exponent gap.

H.2.6 Constant trace and the switch from all-time to eventual coefficients

Q gives deliberately large explicit constants, conditional on the analytic proof:

$$c_M = 2^{-32}M^{-3}, \quad C_M = 2^{154} \max\{M^7, (M-1)^{-3/2}\}, \quad t_M = 0.$$

Its tighter $M = 2$ recipe is $c_2 = 10^{-6}$ and $C_2 = 2 \cdot 10^{45}$. These formulas certify one triple; they are not intrinsic constants of the envelope. The symbols c_M and C_M were reused for multiple auxiliary quantities in S, which motivated a separate dependency ledger.

The threshold discussion identifies the optimal tail constants as the infimum and supremum of $G_M(t)/L(t)^\alpha$ over $t \geq \tau$. Larger thresholds can improve these values, but their ratio tends to one only if a true leading coefficient exists. A limsup bound permits coefficients with strict slack; it does not necessarily permit the endpoint coefficient itself at all large times.

O removes bounded remainders, the augmentation penalty, crude block-count constants, attenuation, and much of the replication cost. It obtains the eventual pair $1/200$ and 15000 at $M = 2$. Its scalar and small-matrix optimizations are optimal only inside their stated relaxations.

H.2.7 Latest pre-asymptotic refinement

T keeps quadratic synthesis, factors blocks by polar decomposition, estimates the real Bessel oscillation together, and averages shifted logarithmic grids. The recorded upper coefficient is

$$\overline{K}_M \leq \frac{\kappa_* M^5 (1 + \rho_M)^2}{\alpha^\alpha (1 - \alpha)^{1-\alpha} [\log(4/3)]^\alpha}, \quad \kappa_* = \inf_{\substack{g \in C_c^\infty(0,1), \\ \|g\|_2=1}} \frac{\|g\|_1^2}{\|g * g\|_2^2}, \quad 1 \leq \kappa_* < \sqrt{2}.$$

On the lower side, T returns to the pure power weight but retains the actual Hardy-space constraints. An imported generalized-Hilbert-matrix diagonalization is used to calculate inverse-energy norms of finitely many boundary functionals. The finite forms remain meaningful even though the critical energy is not bounded below. At $M = 2$, the resulting arithmetic certificates allow three replicas of one coordinate type and two of the other.

The last saved pre-asymptotic coefficients are

$$0.38836928068929 < \underline{K}_2 \leq \overline{K}_2 < 683.37560548915,$$

with the convenient eventual statement

$$\frac{3}{8}L(t)^{2/3} \leq G_2(t) \leq 700L(t)^{2/3} \quad (t \geq t_2)$$

for some finite, unspecified t_2 . They depend on all analytic steps in T and its source dependencies. Successful exact arithmetic does not independently prove those steps.

H.3 Failure ledger: what to retain and what not to repeat

H.3.1 Logical failures with explicit obstructions

F1 — Treating a large calculus norm as a lower witness. The radial Besov quantity of $F_{c,1+\varepsilon}(z) = \exp[-c/(1+\varepsilon-z)]$ grows like $e^{-1-c/2} \log(1/\varepsilon)$, but the fixed-angle radial variation

is uniformly at most π . An integral of separate angular maxima can accumulate scales without a single operator realizing them. The source supplies both calculations. Lesson: a norm used in an upper functional calculus has no automatic reverse inequality. [R0, Sections 7–8.]

F2 — Classical scalar Foguel–Hankel operators. The power-bounded family $\begin{pmatrix} S^* & \Gamma_h \\ 0 & S \end{pmatrix}$ has a particular Peller-type calculus controlled by the fixed-angle radial quantity. The report deduces a uniform bound for the MF-17 functions throughout this family. Lesson: this scalar-symbol route is genuinely obstructed, subject to its stated imported calculus theorem. It does not exclude operator-valued Hankel, general triangular, or arbitrary nilpotent models. [R0, Section 7.]

F3 — Packing phase-lattice points with fixed scalar variation. R1 gives an explicit occupancy bound and a unit-interval clustering argument. Lesson: density alone does not avoid the variation budget; the actual operator norm can be much smaller than scalar variation, so a successful construction must not confuse the two. [R1, Section 4.]

F4 — Uniform similarity to contractions. If $A = LBL^{-1}$ with a uniformly bounded condition number and contractive inverse semigroup for B , then $\|e^{tA^{-1}}\|$ is uniformly bounded by that condition number. Lesson: models whose nonnormality remains uniformly removable cannot explain unbounded inverse growth. [R0, Section 2.]

F5 — Hoping for $G_M(t) \leq M$. The exact 48-dimensional witness has $G_2(100) > 5/2$. Lesson: bounded forward amplification does not cap inverse amplification at the same number. The certificate is finite-time only. [R1, Section 3.]

H.3.2 Explorations that did not yield an asymptotic construction

F6 — Triangular commutators. Block similarities produce off-diagonal entries $K_{ij}[f(z_j) - f(z_i)]$. Entrywise cancellation for the forward functions did not give dimension-uniform operator estimates while retaining large inverse values. Lesson: increasing K can increase the forward or similarity bound. This is an unsuccessful search, not a no-go theorem. [R0, Section 9.2.]

F7 — Sparse shifts, dyadic Hankel interactions, and guessed square-function gains. Neither orthogonality nor coherent alignment follows from being in a Hilbert space. The record contains no certified asymptotic family yielding the guessed logarithmic or square-root-logarithmic gains. Lesson: prove the cross-scale geometry before fitting an exponent. [R0, Section 9.3.]

F8 — General Foguel, renewal, and operator-valued lacunary trials. The conversation reports these explorations after the initial report. In the operator-valued lacunary attempt, disjoint coordinate intervals controlled the forward cocycle uniformly. No inverse-growth lower theorem emerged. No full matrix family or reproducible asymptotic certificate for these particular trials is mounted. Lesson: disjointness sufficient for the forward estimate may also prevent the inverse contributions from adding. Do not regard the unretained experiments as exclusions of those entire classes. [Conversation record; no complete retained construction.]

F9 — Lower spectral spacing chosen only at the inverse observation time. Alternating phases are easy to prescribe at one time; controlling every forward time is the real restriction. The successful proposed S route instead tolerates quadratic separation and exploits the $\log \log t$ conversion. Lesson: test the forward transitions, not only the chosen inverse phases. [R0, Section 9.1; S, Sections 6–8.]

H.3.3 Valid reductions that were mistaken for possible endpoints

F10 — Naive compression. A compressed operator’s powers need not be compressions of the original powers. The repair is tensoring with a nilpotent shift and then restricting to an invariant cyclic subspace. Dependent cyclic vectors require a positive-semidefinite Gram quotient, not a false linear-independence assumption. [R0, Section 4; T, Section 2.]

F11 — Large resolvent-discretization costs. A direct norm perturbation produces an e^{Mt} factor. Exact commutation identities avoid using that bound when making a constant-factor reduction. Lesson: preserve algebraic identities before expanding a general perturbation estimate. [R0, Section 3.]

F12 — A finite SDP as an evaluated asymptotic rate. The reduction is exact or quantitatively controlled, but its unevaluated optimum is still an extremal problem. Small matrices give lower certificates, not universal upper bounds. The most recent SDP-asymptotics branch is now deliberately deferred. [R0, Section 5; T, Sections 1–4.]

F13 — Direct sums as accumulation. They realize a supremum and can combine near-extremizers into a single generator, but cannot add block norms. Lesson: a direct sum repairs quantifiers, not the size of the lower rate. [R0, Section 6.]

H.3.4 Losses replaced in the successful proposed route

F14 — More Bessel terms as the hoped-for exponent improvement. The first omitted term is already integrable. Higher primitives reduce already summable errors, while stationary phase shows a single window can have a nonvanishing limiting contribution. Lesson: the improvement must exploit relations among windows. [S, Appendices A–B.]

F15 — Binary splitting. It produces a real power saving but loses the exact exponent. Stationarizing all cuts and using the exact angle replaces the M^{-2} -type deficit by the proposed M^{-1} -type deficit. [Part B; S, Sections 2–3.]

F16 — Fixed nonzero cut error. It leaves augmentation or endpoint-logarithm costs. Sparse tails make the errors super-exponentially small in the first retained block index. Auxiliary lengths can then vanish while the exponent perturbation is $o(1/\log n)$. [S, Sections 3–5; O, Section 4; T, Section 6.]

F17 — Pure weights estimated only pointwise. Some reciprocal pointwise-margin integrals diverge. S rounds the weight; T instead computes the true Hardy-functional norms. Lesson: divergence of an overestimate does not prove divergence of the quantity being bounded. The pure-power refinement needs its own spectral and inverse-form audit. [S, Section 7; T, Sections 9–11.]

F18 — Uncontrolled active complex phase. A bound for coordinate cuts does not automatically cover an active multiplier z with $|z| \leq 1$. Replication and a strictly positive finite-size margin address this. Replacing M by $M + \varepsilon$ would lose the fixed- M endpoint. [S, Section 6; O, Section 9; T, Sections 11–12.]

F19 — Coarse constants absorbed too early. The universal package, bounded prefixes, floor losses, attenuation, and separate phase costs produced enormous all-time coefficients. O and T remove these only after preserving the large-time limit and fixing auxiliary parameters in the correct order. Lesson: eventual coefficients and all-time coefficients solve different optimization problems. [Q; O; T.]

F20 — Conflating power sharpness with fixed- M sharpness. Exponents approaching one as $M \rightarrow \infty$ do not meet at a prescribed M . Nor does matching order imply a unique leading coefficient. Increasing a threshold alone cannot prove $\underline{K}_M = \overline{K}_M$. [U, Section 5; Part Q; T, Section 16.]

H.3.5 Numerical and record-keeping failures

The inherited report mentions an unretained dimension-24 candidate near a norm of 2.01, but no matrix or feasibility certificate survives. It is not a certified result. By contrast, the later 48-dimensional witness has exact integer and rational data.

The source S records that a naive global high-frequency quadrature overstated finite-degree amplification. Composite quadrature and independent Dirichlet moments were used to diagnose it. Those checks remain numerical diagnostics. Near-singular Gram matrices also make tiny negative residual eigenvalues and apparent feasibility unreliable.

Some planned larger Gram checks were not completed or their scripts are not mounted. Interrupted continuation turns with no mathematical output do not supply evidence for or against a theorem. The handoff lists the available scripts and reruns precisely; it does not repeat an absent computation as though it had been recovered.

H.4 Proof dependencies and the highest-priority audit points

H.4.1 Baseline exponent chain

The upper chain is

cut angle $\rightarrow$ Hardy concentration $\rightarrow$ all-cut synthesis
 $\rightarrow$ augmentation/products $\rightarrow$ sparse resolvent cuts $\rightarrow$ Bessel blocks.

The lower chain is

rounded weight $\rightarrow$ finite-degree deficit $\rightarrow$ boundary functionals
 $\rightarrow$ phase replication $\rightarrow$ quadratic spacing $\rightarrow$ normalization/scaling.

Both chains must preserve the exact M . The upper must be uniform over every admissible H, A ; the lower only needs an admissible family but its constants and replication count must be independent of dimension.

H.4.2 New leading-coefficient dependencies

The current upper coefficient additionally needs the exact Poisson Toeplitz comparison from O, its uniform stationary-phase estimate for fixed $1 < p < 4/3$, the limiting translated-window scalar constant, and T's polar-factor and averaged-grid arguments. The order of limits is fixed auxiliary parameters first, then $t \rightarrow \infty$, then optimization of the parameters. An unjustified interchange could change an eventual coefficient while leaving a coarser order estimate intact.

The current lower coefficient additionally imports a unitary diagonalization of the generalized Hilbert matrix and uses it only on specific inverse-energy feature vectors. The critical energy has spectrum approaching zero. The calculation must establish integrability of each inverse-form moment and the positivity of every Schur-complement denominator; it must not write a bounded inverse on the entire Hilbert space. The arithmetic certificate checks the resulting finite formulas, not those analytic premises.

H.4.3 Audit checklist for a successor

1. Recheck the weighted Hardy projection-to-angle step, Nehari approximation, winding-number argument, and endpoint passage in S. Verify the real/reflection symmetrization used in T's stronger concentration coefficient.
2. Check every strong-integral and generator-domain identity. Norm continuity of an unbounded generator's semigroup is never available by default. The Hilbert-space adjoint semigroup must be treated explicitly.
3. Verify scalar frequency-tail summation and the uniform stationary-phase remainder, including frequencies outside the stationary set and observation times entering the amplitudes. A numerical window test cannot establish that uniformity.

4. Check that the polar-factor estimates supply cuts for exactly the row and column families used by the quadratic synthesis theorem; polar factors themselves need not commute with the resolvents.
5. Recheck the replication quadratic forms with the stated inner-product convention and normalized basis. Verify strictness at both endpoints and an arbitrary active phase before approximating the forward semigroup.
6. For T's pure-power route, rederive the spectral measure, the six moment formulas, the forbidden-constant restrictions, the positive rank update, and the minimizer restriction. Distinguish restrictions on the genuine Hardy space from relaxations to a larger feature space.
7. Check that polynomial margins dominate all spectral-approximation errors and that time conversion gives a lower bound for **every** sufficiently large time, not only a subsequence.
8. Keep the unresolved leading-coefficient equality separate. Optimizing a sufficient certificate, passing exact arithmetic, or observing monotonicity cannot establish convergence of the normalized envelope.

These are audit tasks, not newly demonstrated counterexamples or newly claimed gaps. The source reports assert their own proofs; this compilation does not silently upgrade or repair them.

Part II

Inherited reductions and obstructions

Source and scope. Full mathematical body of the initial progress-and-obstacles report, originally dated 17 September 2026.

Its statements about the then-unresolved gap are historical. The original source and diagnostics are bundled. Finite representations here are retained as historical tools, not as a renewed SDP-asymptotics project.

R0.1 The target, the baseline, and the actual outcome

Fix $M > 1$. Let $\mathcal{A}_M$ contain all complex Hilbert spaces H and generators A of strongly continuous semigroups $T(s)$ satisfying

$$\|T(s)\| \leq Me^{-s} \quad (s \geq 0). \quad (\text{R0.1.1})$$

The generator may be unbounded. Its inverse exists and is bounded. The canonical MF-17 target is to determine an asymptotic growth rate, with matching positive multiplicative constants depending only on M , for

$$G_M(t) = \sup_{(H,A) \in \mathcal{A}_M} \|e^{tA^{-1}}\| \quad (t \rightarrow \infty). \quad (\text{R0.1.2})$$

This is the fixed-bound formulation in the repository [MF17].

The external benchmark is

$$\frac{1}{250}(\log \log(t + e^e))^{\alpha_M} \leq G_M(t) \leq 2M^2(2 - e^{-1} + e^{-1} \log t), \quad t > 1, \quad (\text{R0.1.3})$$

where

$$\alpha_M = \frac{M - 1}{M + 49}. \quad (\text{R0.1.4})$$

The upper bound is Batty–Gomilko–Tomilov [BGT, Corollary 5.7]. The lower bound follows from Lorist–Meyries–Veraar [LMV, Theorem 1.2(2)], using their $C = 50$ and $1 + 50\alpha_M/(1 - \alpha_M) = M$. Their Problem 1.3 asks for the optimal growth. *Neither attempt improved the two sides of (R0.1.3).*

Approach	Result retained	Unresolved issue
Resolvent normalization	Same- M reduction to power-bounded operators	No growth estimate for their function calculus
Finite cyclic models	Finite nilpotents recover the full envelope	The matrix dimension and geometry can depend on time
Semidefinite program	A finite optimum approximates $G_M(t)$ within $2e$	Its asymptotic value has not been determined
Direct sums	One generator attains the envelope at every time	Direct sums select a supremum; they do not add growth
Scalar Foguel–Hankel models	The inverse exponentials are uniformly bounded in this family	The conclusion does not cover general power-bounded operators
Triangular, spectral, and numerical searches	Useful diagnostics and design constraints	No matching asymptotic construction or upper certificate

R0.2 The constraints that every successful approach must respect

R0.2.1 The constant M cannot grow with the construction

A family satisfying $\|T_n(s)\| \leq M_n e^{-s}$ with $M_n \rightarrow \infty$ does not supply a lower bound in a fixed class $\mathcal{A}_M$. Likewise, choosing a parameter to make an inverse-growth exponent large is invalid when that choice also makes the forward bound exceed M . Formula (R0.1.4) illustrates this issue exactly: one cannot send the exponent of the cited lower construction to one while keeping M fixed.

A related danger is replacing a semigroup or power bound by a one-step norm bound. From $\|C\| \leq M$ one gets only $\|C^k\| \leq M^k$, not the uniform bound needed below. The dependence on the entire orbit is essential.

R0.2.2 A matrix can be stable without being harmless

Eigenvalue location alone cannot control $\|e^{tA^{-1}}\|$ in a nonnormal family. Section R0.4 shows that all near-extremizing behavior can already be represented by finite matrices with a single negative real eigenvalue. Thus complicated spectra are not necessary. Conversely, this does not make the problem a scalar spectral calculation: nonorthogonal invariant directions carry the difficulty.

There is also a quantifier distinction. A fixed finite admissible matrix has $\operatorname{Re} z < 0$ for every eigenvalue z ; hence $\operatorname{Re}(1/z) < 0$, and its inverse exponential eventually decays. This does not prevent $G_M(t)$ from growing: the maximizing matrix and its dimension may change with t .

R0.2.3 Contractive and uniformly similar models cannot explain the growth

The endpoint is exactly

$$G_1(t) = 1 \quad (t \geq 0). \quad (\text{R0.2.1})$$

Indeed, $\|T(s)\| \leq e^{-s}$ implies $\operatorname{Re}\langle Ay, y \rangle \leq -\|y\|^2$ for $y \in D(A)$. Writing $x = Ay$ shows that A^{-1} is dissipative. Since it is bounded, differentiating $\|e^{tA^{-1}}x\|^2$ proves contractivity of its exponential. The reverse inequality for the supremum follows from the scalar generators $A = -aI$, $a \geq 1$, because $e^{-t/a} \rightarrow 1$ as $a \rightarrow \infty$.

More generally, if $A = LBL^{-1}$ and B generates a contractive, exponentially stable semigroup, then

$$\|e^{tA^{-1}}\| \leq \|L\| \|L^{-1}\|. \quad (\text{R0.2.2})$$

Therefore a family with a *uniformly controlled* similarity to contractions cannot provide unbounded inverse growth. This was a recurring limitation of simple shift and weighted-translation models. It is not legitimate to assume that every member of $\mathcal{A}_M$ has such a similarity with a condition number controlled by M .

R0.3 Resolvent normalization: removing unbounded generators

Define the class of power-bounded operators

$$\mathcal{P}_M = \{C : C \text{ is a bounded Hilbert-space operator and } \sup_{k \geq 0} \|C^k\| \leq M\}.$$

For $C \in \mathcal{P}_M$ and $\lambda > 0$, set

$$B_\lambda = \lambda C - (\lambda + 1)I. \quad (\text{R0.3.1})$$

The exponential series gives

$$\|e^{sB_\lambda}\| \leq e^{-(\lambda+1)s} \sum_{k=0}^{\infty} \frac{(\lambda s)^k}{k!} \|C^k\| \leq M e^{-s}. \quad (\text{R0.3.2})$$

Thus every such B_λ is admissible with the original value of M .

The reverse construction is equally important. For an admissible A , define

$$C_\lambda = (\lambda + 1)(\lambda I - A)^{-1}. \quad (\text{R0.3.3})$$

For $k \geq 1$, the strong resolvent integral yields

$$(\lambda I - A)^{-k}x = \frac{1}{(k-1)!} \int_0^\infty s^{k-1} e^{-\lambda s} T(s)x \, ds, \quad \|(\lambda I - A)^{-k}\| \leq \frac{M}{(\lambda + 1)^k}.$$

Consequently $C_\lambda \in \mathcal{P}_M$. Notice that integrating the semigroup before estimating gives M , rather than the false substitute M^k .

The associated bounded generator satisfies

$$B_\lambda = (\lambda + 1)A(\lambda I - A)^{-1},$$

and a bounded-operator multiplication proves the exact identity

$$\boxed{B_\lambda^{-1} = \frac{\lambda A^{-1} - I}{\lambda + 1}}. \quad (\text{R0.3.4})$$

All expressions here are well-defined even when A is unbounded: the resolvent maps into $D(A)$, and $A(\lambda I - A)^{-1}$ is bounded.

The strong integral

$$A^{-1}x = - \int_0^\infty T(s)x \, ds$$

gives $\|A^{-1}\| \leq M$. Hence

$$\|B_\lambda^{-1} - A^{-1}\| \leq \frac{M+1}{\lambda+1}, \quad \|e^{tB_\lambda^{-1}} - e^{tA^{-1}}\| \leq \frac{t(M+1)e^{Mt}}{\lambda+1}. \quad (\text{R0.3.5})$$

This proves convergence at each fixed t as integer $\lambda \rightarrow \infty$.

R0.3.1 The first quantitative obstacle and its repair

Estimate (R0.3.5) is too expensive for a useful large-time reduction: used literally, it can demand an exponentially large λ . The repair was not a sharper perturbation estimate. It was to use the commutativity in (R0.3.4) directly.

For $t \geq 1$, put

$$\lambda = (M+1)t, \quad r = 1 + \lambda^{-1}, \quad c = \frac{1}{M+1}, \quad f_t(z) = \exp\left(-\frac{c}{r-z}\right), \quad (\text{R0.3.6})$$

and define

$$\mathcal{F}_M(t) = \sup_{C \in \mathcal{P}_M} \|f_t(C)\|.$$

Since $\sigma(C) \subseteq \overline{\mathbb{D}}$ and $r > 1$, all these function-calculus expressions are given by absolutely convergent Taylor series.

Proposition R0.3.1. *For $M > 1$ and $t \geq 1$,*

$$\mathcal{F}_M(t) \leq G_M(t) \leq e \mathcal{F}_M(t). \quad (\text{R0.3.7})$$

Proof. For any $C \in \mathcal{P}_M$, equations (R0.3.1) and (R0.3.6) give $f_t(C) = e^{tB_\lambda^{-1}}$, proving the first inequality. Conversely, for $C = C_\lambda$ in (R0.3.3), rearrange (R0.3.4) to get

$$A^{-1} = \left(1 + \frac{1}{\lambda}\right) B_\lambda^{-1} + \frac{1}{\lambda} I.$$

The terms commute, and $\|B_\lambda^{-1}\| \leq M$ by (R0.3.2). Thus

$$\|e^{tA^{-1}}\| \leq e^{t/\lambda} \|e^{(t/\lambda)B_\lambda^{-1}}\| \|e^{tB_\lambda^{-1}}\| \leq e^{(M+1)t/\lambda} \mathcal{F}_M(t) = e \mathcal{F}_M(t).$$

□

What this establishes. Unbounded domains and the approximation parameter can be removed without enlarging M , and a linear-in- t choice of λ suffices for constant-factor comparison.

What remains difficult. The remaining quantity is still a supremum of $\|f_t(C)\|$ over all power-bounded Hilbert-space operators. No bound sharper than the original growth gap follows merely from this change of variables.

R0.4 Finite nilpotents recover the full envelope

The next issue was to make the operator optimization finite-dimensional without losing the uniform power bound. Ordinary compression is not adequate in general: $(PCP)^k$ need not equal PC^kP , so a power bound for C need not survive. The successful replacement was a shift on copies of the original space, followed by restriction to a finite cyclic invariant subspace.

Lemma R0.4.1 (Finite cyclic model). *Let $C \in \mathcal{P}_M$, let x be a unit vector, and suppose $f(z) = \sum_{k \geq 0} a_k z^k$ with $\sum_k |a_k| < \infty$. For each $n \geq 1$ there are a nilpotent operator N_n on a space of dimension at most n and a unit vector v_n such that*

$$N_n^n = 0, \quad \sup_{k \geq 0} \|N_n^k\| \leq M, \quad (\text{R0.4.1})$$

and

$$\|f(N_n)v_n\|^2 = \frac{1}{n} \sum_{j=0}^{n-1} \left\| \sum_{k=0}^j a_k C^k x \right\|^2. \quad (\text{R0.4.2})$$

In particular, $\|f(N_n)v_n\| \rightarrow \|f(C)x\|$.

Proof. Let S_n be the nilpotent right shift on $\mathbb{C}^n$, so that $S_n e_j = e_{j+1}$ for $j < n-1$ and $S_n e_{n-1} = 0$. On $H \otimes \mathbb{C}^n$ set

$$U_n = C \otimes S_n, \quad v_n = x \otimes \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} e_j.$$

Restrict U_n to the invariant cyclic subspace

$$E_n = \text{span}\{v_n, U_n v_n, \dots, U_n^{n-1} v_n\}.$$

Its dimension is at most n . The restriction N_n satisfies (R0.4.1) because $U_n^n = 0$ and $\|U_n^k\| \leq M$. The component of $f(U_n)v_n$ at shift coordinate j equals $n^{-1/2} \sum_{k=0}^j a_k C^k x$. Orthogonality of the shift coordinates proves (R0.4.2). Absolute convergence gives convergence of the partial sums to $f(C)x$, and their squared norms have the same Cesàro limit. □

Write $\mathcal{N}_M$ for the finite-dimensional nilpotents power bounded by M . Taking suprema in Lemma R0.4.1 gives

$$\sup_{C \in \mathcal{P}_M} \|f(C)\| = \sup_{N \in \mathcal{N}_M} \|f(N)\|. \quad (\text{R0.4.3})$$

Combining this with Section R0.3 proves the following exact result.

Theorem R0.4.2 (Exact finite-matrix reduction). *For every $t \geq 0$,*

$$G_M(t) = \sup_{\lambda \in \mathbb{N}, \lambda \geq 1} \sup_{N \in \mathcal{N}_M} \left\| \exp\left(-t((\lambda + 1)I - \lambda N)^{-1}\right) \right\|. \quad (\text{R0.4.4})$$

Allowing all real $\lambda > 0$ gives the same value. Every generator on the right has the single eigenvalue $-(\lambda + 1)$.

Proof. Every matrix on the right is admissible by (R0.3.2). For the reverse inequality, apply (R0.4.3) to $f(z) = \exp(-t/(\lambda + 1 - \lambda z))$ and C_λ , then use (R0.3.5) as integer $\lambda \rightarrow \infty$. $\square$

What this establishes. Finite matrices suffice, and neither an unbounded domain nor a broad set of eigenvalues is intrinsically required for near-extremizers. The proof preserves the bound M exactly.

What remains difficult. The reduction does not provide a bounded matrix size independent of t , or a useful description of the extremizing inner products. All of the nontrivial behavior has moved into the geometry of a nilpotent orbit. This is a global reduction, not a growth estimate.

R0.5 The semidefinite characterization and its unresolved optimization

Keep the parameters in (R0.3.6). Define

$$R = 1 + \frac{1}{2\lambda}, \quad d = \left\lceil \frac{\log(40eM\lambda)}{\log R} \right\rceil, \quad n = 2(d + 1). \quad (\text{R0.5.1})$$

Write $f_t(z) = \sum_{k \geq 0} a_k z^k$ and $a = (a_0, \dots, a_{n-1})^\top$. These real coefficients are specified by

$$a_{-1} = 0, \quad a_0 = e^{-c/r}, \quad a_{k+1} = \frac{(2rk - c)a_k - (k - 1)a_{k-1}}{r^2(k + 1)} \quad (k \geq 0). \quad (\text{R0.5.2})$$

This follows by equating coefficients in $(r - z)^2 f'_t(z) + c f_t(z) = 0$.

For the shift S_n from Lemma R0.4.1, define

$$\begin{aligned} \gamma_M(t)^2 &= \max_{Q \in \mathbb{R}^{n \times n}} a^\top Q a \\ \text{subject to } & Q = Q^\top \succeq 0, \quad Q_{00} = 1, \\ & (S_n^k)^\top Q S_n^k \preceq M^2 Q \quad (1 \leq k < n). \end{aligned} \quad (\text{R0.5.3})$$

All constraints and the objective are affine in Q .

Theorem R0.5.1 (Constant-factor finite characterization). *For every $M > 1$ and $t \geq 1$, the maximum in (R0.5.3) exists and*

$$\gamma_M(t) \leq G_M(t) \leq 2e \gamma_M(t). \quad (\text{R0.5.4})$$

The prescribed dimension satisfies $n = O_M(t \log t)$.

R0.5.1 Why a Gram matrix is the correct variable

Suppose $N^n = 0$ and $\sup_k \|N^k\| \leq M$. For a unit vector v , let K have columns $v, Nv, \dots, N^{n-1}v$. Then $KS_n = NK$, so $Q = K^*K$ obeys the Hermitian versions of the constraints in (R0.5.3). Moreover,

$$a^*Qa = \|f_t(N)v\|^2.$$

Because a and S_n are real, replacing Q by its entrywise real part preserves feasibility and this value.

Conversely, a feasible Q defines a Hilbert space by quotienting $\mathbb{C}^n$ by $\ker Q$, with $\|[u]\|_Q^2 = u^*Qu$. The $k = 1$ constraint makes the kernel S_n -invariant. The induced operator $N[u] = [S_n u]$ is nilpotent and power bounded by M , while $[e_0]$ is a unit vector and

$$\|f_t(N)[e_0]\|_Q^2 = a^T Q a.$$

Thus the program exactly captures all nilpotent tests of order at most n . The maximum exists: $Q_{ii} \leq M^2$ follows by testing the $k = i$ constraint on e_0 , and positivity bounds every entry. The feasible set is compact and contains I .

Singular Q is not a defect in this argument; the quotient is essential. Alternatively, $(1 - \varepsilon)Q + \varepsilon I$ is positive definite and feasible. Its models are similar to a single Jordan block and approach the same objective. This also explains why real matrices suffice in (R0.4.4).

R0.5.2 Why the cutoff is sufficient

On $|z| \leq R$ one has $\operatorname{Re}(r - z) > 0$ and $|f_t(z)| \leq 1$. Cauchy's estimate gives $|a_k| \leq R^{-k}$. If $p_j(z) = \sum_{k=0}^j a_k z^k$, then for $C \in \mathcal{P}_M$,

$$\|f_t(C) - p_j(C)\| \leq M \sum_{k>j} |a_k| \leq 2M\lambda R^{-j} \leq \delta := \frac{1}{20e} \quad (j \geq d). \quad (\text{R0.5.5})$$

Apply Lemma R0.4.1 with $n = 2(d + 1)$. At least half the terms in (R0.4.2) have index $j \geq d$. For every unit x this gives

$$\|f_t(N_n)v_n\| \geq \frac{1}{\sqrt{2}}(\|f_t(C)x\| - \delta)_+.$$

Taking suprema and using the exact Gram formulation yields

$$\gamma_M(t) \geq \frac{1}{\sqrt{2}}(\mathcal{F}_M(t) - \delta).$$

Since $C = 0$ is allowed, $\mathcal{F}_M(t) \geq e^{-c/r} \geq e^{-1}$. The last lower bound is at least $\mathcal{F}_M(t)/2$. The opposite inequality $\gamma_M(t) \leq \mathcal{F}_M(t)$ is immediate, so (R0.3.7) proves (R0.5.4). Finally, $\log R \asymp_M t^{-1}$ in (R0.5.1), giving the stated size.

R0.5.3 What a genuine upper certificate would look like

The finite formulation makes the missing task precise. Set $E_{00} = e_0 e_0^T$. If $Y_k \succeq 0$ and a real number μ satisfy

$$\mu E_{00} + \sum_{k=1}^{n-1} (S_n^k Y_k (S_n^k)^T - M^2 Y_k) \succeq aa^T, \quad (\text{R0.5.6})$$

then every feasible Q obeys

$$a^T Q a \leq \mu + \sum_{k=1}^{n-1} \operatorname{tr} \left[Y_k ((S_n^k)^T Q S_n^k - M^2 Q) \right] \leq \mu.$$

A feasible family Q_t provides lower bounds; matrices satisfying (R0.5.6) provide upper bounds. The attempts produced neither family with matching asymptotics.

R0.5.4 Why the program is not a completed asymptotic solution

The program has $n(n+1)/2$ scalar variables and $n-1$ power inequalities of matrix size n , in addition to positivity. The sufficient cutoff is already large: for $M=2$, it gives $n=1066$ at $t=10$ and $n=13318$ at $t=100$. The latter entails 88,691,221 symmetric matrix variables. These are sizes of this sufficient formulation, not lower bounds on necessary dimension or on the complexity of every possible algorithm.

Smaller- n computations can supply lower witnesses, but they do not inherit the constant-factor upper guarantee without an independent truncation argument. Near-singular Gram matrices also make numerical feasibility delicate: tiny negative eigenvalues of constraint residuals are not proof of feasibility, and small perturbations can change the norm in the quotient model substantially.

What this establishes. There is an explicit finite proxy with a proved uniform approximation factor, and a concrete certificate language for both sides of the remaining optimization.

What remains difficult. Knowing that a finite optimization equals the answer up to constants is different from evaluating its growth. Calling $\gamma_M(t)$ the desired rate merely relocates the unevaluated extremal problem. No analysis of its optimal values yet separates logarithmic, iterated-logarithmic, or intermediate behavior.

R0.6 A direct sum removes one possible quantifier obstruction

Proposition R0.6.1. *There is a single admissible generator A_* on a separable complex Hilbert space such that*

$$\left\| e^{tA_*^{-1}} \right\| = G_M(t) \quad (t \geq 0). \quad (\text{R0.6.1})$$

Proof. For each nonnegative rational q and integer $m \geq 1$, choose a finite model from (R0.4.4) within $1/m$ of $G_M(q)$. Form their Hilbert direct sum. The common bound Me^{-s} implies strong continuity, first on finitely supported vectors and then by approximation. Its generator has the natural direct-sum domain. The block inverses have norms at most M , so their direct sum is the bounded inverse of this generator. Equality holds at every rational time.

For all admissible A and $s, t \in [0, K]$,

$$\left\| e^{tA^{-1}} - e^{sA^{-1}} \right\| \leq Me^{MK}|t-s|.$$

The envelope is therefore locally Lipschitz as well. Equality extends from rational times to every $t \geq 0$. $\square$

This result shows that separate near-extremizers can be assembled into one example. It does *not* amplify a lower bound, because

$$\left\| \bigoplus_j B_j \right\| = \sup_j \|B_j\|,$$

not a sum of the block norms. Combining many favorable scales by direct sum selects the largest contribution; it does not make their contributions add. That distinction becomes central in the functional-calculus approach below.

R0.7 Foguel–Hankel models: a proved obstruction to a natural construction

R0.7.1 The tempting route and the external input

One route was to seek a lower bound among the classical scalar-symbol operators

$$R_h = \begin{pmatrix} S^* & \Gamma_h \\ 0 & S \end{pmatrix} \quad \text{on } \ell^2(\mathbb{N}_0) \oplus \ell^2(\mathbb{N}_0), \quad (\text{R0.7.1})$$

where S is the unilateral shift and Γ_h is a bounded scalar Hankel operator. This family contains power-bounded operators that are not polynomially bounded [Peller, Section 6]. However, failure of a uniform polynomial calculus does not imply large values for the particular functions in MF-17.

The needed quantitative input from Peller [Peller, Lemma 6.1 and Theorem 6.2, pp. 364–366] is

$$\sup_{k \geq 0} \|R_h^k\| \leq M \implies \|F(R_h)\| \leq C(1 + M) \|F\|_X, \quad (\text{R0.7.2})$$

for F analytic near $\overline{\mathbb{D}}$, where

$$\|F\|_X = \|F\|_{H^\infty} + \sup_{\|g\|_{H^1} \leq 1} \int_{\mathbb{D}} |F'(z)g(z)| dA(z). \quad (\text{R0.7.3})$$

Here C is absolute. Uniformity in h follows from the power formula R_h^k with upper-right block $k\Gamma_{(S^*)^{k-1}h}$ and the norm estimates in those results; radial regularization extends the estimate from smooth symbols. Equivalently, the derivative term in (R0.7.3) is controlled by the Carleson-measure norm of $|F'| dA$ [Peller, p. 365].

R0.7.2 The function family is uniformly controlled in this calculus

The affine generator construction gives

$$A_{\lambda,h} = \lambda R_h - (\lambda + 1)I, \quad e^{tA_{\lambda,h}^{-1}} = F_{t/\lambda, 1+1/\lambda}(R_h),$$

where

$$F_{c,r}(z) = \exp\left(-\frac{c}{r-z}\right), \quad c > 0, \quad r > 1. \quad (\text{R0.7.4})$$

The key calculation in the second attempt was the following uniform estimate.

Lemma R0.7.1 (Radial variation). *For every $c > 0$ and $r > 1$,*

$$\boxed{\sup_{\theta \in \mathbb{R}} \int_0^1 |F'_{c,r}(\rho e^{i\theta})| d\rho \leq \pi.} \quad (\text{R0.7.5})$$

Proof. Fix θ and set $w(\rho) = c/(r - \rho e^{i\theta})$. Then

$$\int_0^1 |F'_{c,r}(\rho e^{i\theta})| d\rho = \int_{w([0,1])} e^{-\operatorname{Re} w} |dw|. \quad (\text{R0.7.6})$$

For $\sin \theta = 0$, the curve is a monotone interval on the positive real axis and this integral is at most one. Otherwise, writing $w = u + iv$, the image lies on

$$\left(u - \frac{c}{2r}\right)^2 + \left(v - \frac{c \cot \theta}{2r}\right)^2 = \frac{c^2}{4r^2 \sin^2 \theta}.$$

Furthermore,

$$u - \frac{c}{2r} = \frac{c(r^2 - \rho^2)}{2r|r - \rho e^{i\theta}|^2} > 0.$$

The image is therefore an injectively parametrized subarc of the right semicircle. Let $a = c/(2r)$ and $R_0 = c/(2r|\sin \theta|)$. The weighted length of the full right semicircle is

$$R_0 e^{-a} \int_{-\pi/2}^{\pi/2} e^{-R_0 \cos \psi} d\psi.$$

Since $\cos \psi \geq 1 - 2|\psi|/\pi$ on this interval, this is at most

$$2R_0 e^{-a} \int_0^{\pi/2} e^{-R_0(1-2\psi/\pi)} d\psi = \pi e^{-a} (1 - e^{-R_0}) \leq \pi.$$

This proves (R0.7.5). $\square$

For every boundary arc I and its Carleson box $S(I)$, polar coordinates and Lemma R0.7.1 give

$$\int_{S(I)} |F'_{c,r}(z)| dA(z) \leq \int_I \int_0^1 |F'_{c,r}(\rho e^{i\theta})| d\rho d\theta \leq \pi |I|.$$

Together with $\|F_{c,r}\|_\infty \leq 1$, the characterization in (R0.7.3) implies

$$\sup_{c>0, r>1} \|F_{c,r}\|_X < \infty. \quad (\text{R0.7.7})$$

Combining (R0.7.2) and (R0.7.7) proves

$$\boxed{\sup_{h: \sup_k \|R_h^k\| \leq M} \sup_{\lambda>0, t \geq 0} \|e^{tA_{\lambda,h}^{-1}}\| \leq C_M < \infty.} \quad (\text{R0.7.8})$$

The case $t = 0$ simply gives the identity operator.

What this establishes. The classical scalar Foguel–Hankel route fails for a proved reason, not merely because numerical examples were too small. Varying the symbol and λ with time still cannot produce unbounded growth at fixed power bound.

What remains difficult. This is a family-specific obstruction. It does not exclude arbitrary upper-triangular operators, general nilpotent matrices, operator-valued Hankel variants, or other constructions not covered by (R0.7.2). It also does not improve the upper bound for the full envelope G_M .

R0.8 Why the logarithmic Besov norm did not turn into a lower bound

The functions in (R0.7.4) have a very different behavior in the radial analytic Besov norm

$$\mathfrak{B}(F) = \|F\|_\infty + \int_0^1 \sup_\theta |F'(\rho e^{i\theta})| d\rho. \quad (\text{R0.8.1})$$

A direct calculation gives the exact leading logarithmic coefficient.

Proposition R0.8.1. *For every fixed $c > 0$,*

$$\mathfrak{B}(F_{c,1+\varepsilon}) = e^{-1-c/2} \log(1/\varepsilon) + O_c(1) \quad (\varepsilon \downarrow 0). \quad (\text{R0.8.2})$$

Proof. Set $q = |r - \rho e^{i\theta}|^2$ and $D = r^2 - \rho^2$. Since $r - \rho \cos \theta = (D + q)/(2r)$,

$$|F'_{c,r}(\rho e^{i\theta})| = \frac{c}{q} e^{-c/(2r)} \exp\left(-\frac{cD}{2rq}\right).$$

At fixed ρ , the allowed interval is $q \in [(r - \rho)^2, (r + \rho)^2]$. The unconstrained maximum occurs at $q = cD/(2r)$ and equals

$$e^{-1-c/(2r)} \frac{2r}{r^2 - \rho^2}. \quad (\text{R0.8.3})$$

For fixed c , this maximizing q lies in the allowed interval when $r - \rho$ is sufficiently small. Thus (R0.8.3) is exact in a fixed neighborhood of $\rho = 1$ for all sufficiently small $r - 1$. Away from that neighborhood the derivative integral is uniformly bounded in terms of c . Consequently,

$$\int_0^1 \sup_{\theta} |F'_{c,r}(\rho e^{i\theta})| d\rho = e^{-1-c/(2r)} \log \frac{r+1}{r-1} + O_c(1).$$

Setting $r = 1 + \varepsilon$ and using $\|F_{c,r}\|_{\infty} \leq 1$ proves (R0.8.2). $\square$

The mismatch is therefore explicit:

$$\sup_{\theta} \int_0^1 |F'_{c,r}(\rho e^{i\theta})| d\rho \leq \pi, \quad (\text{R0.8.4})$$

$$\int_0^1 \sup_{\theta} |F'_{c,r}(\rho e^{i\theta})| d\rho \asymp_c \log \frac{1}{r-1} \quad (r \downarrow 1). \quad (\text{R0.8.5})$$

The maximizing angle changes with the radial scale. An integral of separate maxima can collect logarithmically many favorable contributions without providing a single operator, input vector, and output test that realizes them coherently.

This identifies the failed implication in the approach. An upper calculus estimate of the form $\|F(C)\| \leq C_M \mathfrak{B}(F)$, even with $\mathfrak{B}(F_t) \asymp \log t$, does not give a reverse estimate. In fact, (R0.7.8) and (R0.8.2) show that no such reverse bound holds for this function family when one restricts the operator supremum to the classical scalar Foguel–Hankel models.

The main unresolved structural issue is coherent accumulation across scales under a fixed forward bound. The computations above establish this as a difficulty for the explored routes, not as a characterization theorem for all possible extremizers. A general proof could instead show that the Besov upper estimate loses cancellations and is not sharp.

R0.9 Other exploratory routes and their specific limitations

R0.9.1 Changing the spectral spacing in conditional-basis constructions

The cited lower construction uses uniformly bounded partial-sum projections, conditionality growing like n^α , and doubly exponential eigenvalue scales [LMV, Theorem 1.1 and Section 2]. The conversion between dimension and observation time is therefore crucial: a gain polynomial in n need only be a power of $\log \log t$ after normalization.

The attempted improvement was to obtain comparable inverse-phase separation with less expensive spectral spacing. The difficulty was not simply to choose phases at a single time. The same choices must keep *every* forward semigroup operator within Me^{-s} . No replacement spacing with both properties was proved. In particular, the appearance of double logarithms in that construction is not itself an upper bound for all constructions.

R0.9.2 Triangular commutators

Another route used nonnormality through a block similarity. For finite matrices D and K , put

$$L = \begin{pmatrix} I & K \\ 0 & I \end{pmatrix}, \quad A = L \begin{pmatrix} D & 0 \\ 0 & D \end{pmatrix} L^{-1}.$$

For every function analytic near the spectrum,

$$f(A) = \begin{pmatrix} f(D) & Kf(D) - f(D)K \\ 0 & f(D) \end{pmatrix}. \quad (\text{R0.9.1})$$

If $D = \text{diag}(z_j)$, the (i, j) entry of the upper-right block is $K_{ij}(f(z_j) - f(z_i))$. The hope was to keep this small for $f(z) = e^{sz}$ at all forward times, while making it large for $f(z) = e^{t/z}$ at a chosen inverse time.

Spectral separation, oscillation, and dimension-dependent commutator norms have to be balanced simultaneously. Entrywise cancellation estimates did not produce the required operator-norm control. Increasing K to enhance the inverse block can also enhance the forward block or the similarity constant in (R0.2.2). No dimension-uniform family resolving this tradeoff was obtained. This remains an unsuccessful exploration, not a theorem excluding triangular constructions.

R0.9.3 Sparse shifts, dyadic Hankel coefficients, and square-function heuristics

Sparse shift witnesses and dyadically separated Hankel interactions were tested to see whether many favorable bands could accumulate. A proposed gain such as $\sqrt{\log t}$ would require a genuine orthogonality or square-function estimate; the Hilbert-space setting alone does not make different scale contributions orthogonal. On the lower-bound side, constructive alignment must also respect all the power inequalities.

The simple scalar Hankel route is now covered by (R0.7.8). Other sparse or more general triangular models were not excluded, but the experiments did not yield an asymptotic family with verified fixed- M bounds. No logarithmic or square-root-logarithmic rate should be inferred from these heuristics.

R0.9.4 Coefficientwise estimates and direct numerical optimization

For $C \in \mathcal{P}_M$, the bound

$$\|f_t(C)\| \leq M \sum_{k \geq 0} |a_k|$$

is immediate. It forgets the geometry of the vectors $C^k x$, however, and it provides no lower witness. Trying to align coefficients independently ignores the coupled constraints $\|C^k\| \leq M$ for all k . The Gram formulation was introduced precisely to retain these correlations.

Numerical optimization did not remove the asymptotic burden. A feasible matrix at one time is a lower witness at that time, not a growth law. A finite time-grid check of (R0.1.1) is not a proof for all $s \geq 0$. And numerical agreement between an objective and a guessed rate is not a substitute for matching feasible primal and upper-certificate families.

R0.10 What was actually checked computationally

The retained archives contain two reproducible diagnostic scripts. They test algebraic and scalar analytic calculations, not the universal sharp-growth claim. Their recorded outcomes are as follows.

Check	Recorded result and scope
Resolvent inverse identity	Error approximately 2.24×10^{-16} on the sample matrix.
Coefficient recurrence and exponential	Sample nilpotent errors at most 1.23×10^{-14} ; the general-parameter check was about 5.72×10^{-16} .
Finite shift model	For shift orders 8, 16, 32, 64, the test-vector values approached the chosen target norm, about 0.62276.
Radial variation	80 sampled parameter combinations were below π ; the maximum recorded value was about 1.88214.
Besov leading term	12 evaluations: three fixed c values and four ε values, comparing the integral with its logarithmic leading term.

The two retained scripts were rerun when preparing this report. Their sample assertions passed. Floating-point quadrature can miss narrow peaks, and floating-point matrix inequalities require certified residual control before being used as exact feasibility statements. The analytic proofs, not these samples, support the universal propositions above.

No semidefinite program was solved in the retained reproducible work. The earlier attempts found no installed semidefinite solver, and an installation attempt failed. An informal exploratory record mentioned a dimension-24 candidate at $M = 2$ with inverse-exponential norm above 2.01, but neither its matrix nor a complete feasibility certificate is present in the retained archives. That report is *not* treated here as a certified result or as evidence of an asymptotic rate. No independent referee review or Lean verification of the deductions in this report has taken place.

R0.11 The precise unfinished task

The reductions locate, but do not solve, the remaining mathematical problem. With $c = 1/(M+1)$ and $r_t = 1 + 1/((M+1)t)$, one must determine the asymptotic order of

$$\sup_{\sup_k \|C^k\| \leq M} \left\| \exp(-c(r_t I - C)^{-1}) \right\|. \quad (\text{R0.11.1})$$

By (R0.3.7), this is equivalent to the original target up to an absolute factor. It is also equivalent, up to $2e$, to estimating the finite program (R0.5.3) with the explicit cutoff (R0.5.1).

A completion through the finite program requires a family of feasible Gram matrices whose objectives have a quantified lower growth rate and an upper bound of the same order, for example through (R0.5.6). A completion through functional calculus requires either a construction that realizes the relevant scale contributions or a sharper general upper theorem exploiting cancellation that the Besov norm discards. Neither has been supplied.

The two attempts do establish that several apparent difficulties are not fundamental: the generator need not be unbounded in each finite witness; eigenvalues need not be spread across a complicated set; the supremum can be represented in finite dimension at each time; and separate-time witnesses can be combined into one admissible generator. What remains is the uniformly power-bounded, nonnormal geometry that these reductions preserve.

Bottom line. There is a global finite-dimensional reduction and a proved exclusion of a natural lower-bound family. There is no matching large-time rate. The most important obstruction encountered was the gap between estimating the size of a function in an upper-bound calculus and realizing that size by an admissible operator with a fixed forward bound.

Part III

Damping, a finite witness, and variation

Source and scope. Full mathematical body of the first follow-up report from the supplied results archive. The finite 48-dimensional data and exact verifier are included. The numerical example and variation obstruction are not asymptotic lower and upper bounds for the full class.

R1.1 The target and the outcome

For fixed $M > 1$, let $\mathcal{A}_M$ be the class of pairs (H, A) , where H is a complex Hilbert space and A generates a strongly continuous semigroup satisfying

$$\|e^{sA}\| \leq Me^{-s} \quad (s \geq 0).$$

The inverse is bounded, with $\|A^{-1}\| \leq M$. The target in the supplied problem is the matching asymptotic order of

$$G_M(t) = \sup_{(H,A) \in \mathcal{A}_M} \|e^{tA^{-1}}\|.$$

The constants in a matching estimate must be uniform over the entire class, while M remains fixed.

The lower estimate of Lorist–Meyries–Veraar [LMV], with its parameters chosen to respect this fixed bound, gives

$$\frac{1}{250} (\log \log(t + e^e))^{\alpha_M} \leq G_M(t), \quad \alpha_M = \frac{M-1}{M+49}. \quad (\text{R1.1.1})$$

The cited construction yields the envelope lower bound (R1.1.1). The previously supplied report records the upper estimate $2M^2(2 - e^{-1} + e^{-1} \log t)$ from [BGT, Corollary 5.7]. Section R1.2 gives the following explicit alternative, using the continuous-parameter form of the argument in [Zwart]:

$$G_M(t) \leq 1 + M^2 + \frac{M^2}{2} \log \frac{t}{2M^2} \quad (t \geq 2M^2). \quad (\text{R1.1.2})$$

This sharpens the displayed logarithmic estimate, but *does not change its asymptotic order*. The gap between (R1.1.1) and (R1.1.2) remains.

The exact numerical construction establishes

$$G_2(100) > \frac{5}{2}. \quad (\text{R1.1.3})$$

In particular, a proposed universal estimate $G_M(t) \leq M$ is false. This is a finite-time assertion, not a growth law.

The final proved result concerns a restricted method. For phase-lattice spectra $1/(-1 + iy_j)$ with $y_{j+1} - y_j \geq 1$, a uniform bound on the total variation of the forward exponential multipliers forces the number of coordinates to be $O(\log \log y_N)$. It is therefore not enough simply to pack more such eigenvalues into a smaller range while retaining the same bounded-variation certificate. A corollary also controls the inverse multiplier when arbitrarily dense clusters of eigenvalues are allowed. These statements are not upper bounds for arbitrary admissible generators.

R1.2 A continuous damping estimate

The following argument is based on the Lyapunov and damping comparison in Zwart [Zwart]. It is included in full to specify constants, domains, and uniformity. No priority claim is made for the continuous-parameter formulation.

Lemma R1.2.1 (An inverse-generator energy estimate). *Suppose D generates a semigroup with $\|e^{sD}\| \leq Me^{-\varepsilon s}$, where $\varepsilon > 0$. Put $R = D^{-1}$. Then, for all $x, y \in H$,*

$$\int_0^\infty \|Re^{uR}x\|^2 du \leq \frac{M^2}{2\varepsilon} \|x\|^2, \quad (\text{R1.2.1})$$

$$\int_0^\infty \|R^*e^{uR^*}y\|^2 du \leq \frac{M^2}{2\varepsilon} \|y\|^2. \quad (\text{R1.2.2})$$

No boundedness assumption on the inverse semigroup is used.

Proof. Define the bounded positive operator P by its quadratic form

$$\langle Px, x \rangle = \int_0^\infty \|e^{sD}x\|^2 ds.$$

Then $0 \leq P \leq M^2(2\varepsilon)^{-1}I$. Differentiating on $D(D)$ and integrating gives the Lyapunov identity

$$D^*P + PD = -I$$

in the sense of forms on $D(D)$. Since R maps H into $D(D)$, substitution of Rx and Ry in the form identity gives the bounded-operator identity

$$R^*P + PR = -R^*R.$$

Consequently

$$\frac{d}{du} \langle Pe^{uR}x, e^{uR}x \rangle = -\|Re^{uR}x\|^2.$$

Integrate to a finite upper endpoint and discard the nonnegative terminal energy, then let the endpoint tend to infinity. This proves (R1.2.1). Apply the same argument to D^* for (R1.2.2); the adjoint semigroup is strongly continuous on a Hilbert space and has the same bound. $\square$

Lemma R1.2.2 (Lipschitz dependence on logarithmic damping). *Let C generate a bounded semigroup, $\|e^{sC}\| \leq M$. For $\varepsilon > 0$ write*

$$R_\varepsilon = (C - \varepsilon I)^{-1}, \quad S_\varepsilon(t) = e^{tR_\varepsilon}.$$

For $0 < a \leq b$ and $t \geq 0$,

$$\|S_a(t) - S_b(t)\| \leq \frac{M^2}{2} \log \frac{b}{a}. \quad (\text{R1.2.3})$$

Proof. The resolvent is norm differentiable in ε , and

$$\frac{dR_\varepsilon}{d\varepsilon} = R_\varepsilon^2, \quad \frac{\partial S_\varepsilon(t)}{\partial \varepsilon} = tR_\varepsilon^2 S_\varepsilon(t).$$

The commutation of all these bounded operators implies

$$t \langle R_\varepsilon^2 S_\varepsilon(t)x, y \rangle = \int_0^t \langle R_\varepsilon S_\varepsilon(u)x, R_\varepsilon^* S_\varepsilon(t-u)^* y \rangle du.$$

Cauchy–Schwarz and Lemma R1.2.1 give

$$\|\partial_\varepsilon S_\varepsilon(t)\| \leq \frac{M^2}{2\varepsilon}.$$

Integrate this norm bound from a to b . $\square$

Theorem R1.2.3 (Uniform logarithmic-time modulus). *For $0 < t_1 \leq t_2$,*

$$0 \leq G_M(t_2) - G_M(t_1) \leq \frac{M^2}{2} \log \frac{t_2}{t_1}. \quad (\text{R1.2.4})$$

Moreover,

$$G_M(t) \leq 1 + M\sqrt{t/2} \quad (t \geq 0), \quad (\text{R1.2.5})$$

and (R1.1.2) follows for $t \geq 2M^2$.

Proof. For monotonicity, set $b = t_2/t_1 \geq 1$. If $A \in \mathcal{A}_M$, then bA also belongs to the class, and $e^{t_2(bA)^{-1}} = e^{t_1 A^{-1}}$.

For the other inequality, fix $A \in \mathcal{A}_M$ and set $C = A + I$. The semigroup of C is bounded by M . The operator

$$A_b = C/b - I$$

belongs to $\mathcal{A}_M$, because its semigroup is $e^{-s}e^{sC/b}$. Also $A_b^{-1} = bR_b$, in the notation of Lemma R1.2.2. Hence

$$\|e^{t_2 A^{-1}}\| \leq \|e^{t_2 R_b}\| + \frac{M^2}{2} \log b = \|e^{t_1 A_b^{-1}}\| + \frac{M^2}{2} \log b \leq G_M(t_1) + \frac{M^2}{2} \log b.$$

Take the supremum over A .

For (R1.2.5), integrate the derivative of $e^{uA^{-1}}x$ and apply (R1.2.1) with $\varepsilon = 1$:

$$\|e^{tA^{-1}}x\| \leq \|x\| + \sqrt{t} \left(\int_0^t \|A^{-1}e^{uA^{-1}}x\|^2 du \right)^{1/2} \leq (1 + M\sqrt{t/2})\|x\|.$$

Finally, use $t_1 = 2M^2$ in (R1.2.4) and $G_M(2M^2) \leq 1 + M^2$ from (R1.2.5). $\square$

R1.3 An exact 48-dimensional witness

The matrices are supplied in the accompanying `certificate.json` and `Q_integer.txt`. The certificate, including positive-definiteness verification, can be checked by the standard-library-only script `verify_certificate.py`. The numerical search is separate from this verifier.

Let $n = 48$, and let S be the right nilpotent shift on $\mathbb{C}^{48}$: $Se_j = e_{j+1}$ for $0 \leq j < 47$, and $Se_{47} = 0$. Let Q_{int} denote the explicit symmetric integer matrix in the certificate, and put

$$Q = 10^{-6}Q_{\text{int}}, \quad \|x\|_Q^2 = x^*Qx, \quad A = 400S - 401I.$$

The certificate establishes

$$Q \succ 0, \quad Q_{00} = 1, \quad 4Q - S^{*k}QS^k \succ 0 \quad (1 \leq k < 48). \quad (\text{R1.3.1})$$

Thus $H = (\mathbb{C}^{48}, \langle \cdot, \cdot \rangle_Q)$ is a complex Hilbert space and $\|S^k\|_Q \leq 2$ for every $k \geq 0$. For $k \geq 48$ this follows from nilpotence, and for $k = 0$ it is immediate. Therefore, for every real $s \geq 0$,

$$\|e^{sA}\|_Q \leq e^{-401s} \sum_{k=0}^{47} \frac{(400s)^k}{k!} \|S^k\|_Q \leq 2e^{-s}. \quad (\text{R1.3.2})$$

This proves the forward bound for *all times*, not merely on a sampled grid.

R1.3.1 Exact verification of positive definiteness

For $k = 0$ let $L_0 = Q_{\text{int}}$, and for $1 \leq k < 48$ let

$$L_k = 4Q_{\text{int}} - S^{*k}Q_{\text{int}}S^k.$$

For each k , the certificate contains an upper triangular integer matrix R_k with nonzero diagonal. The verifier multiplies integers to form $Z_k = R_k^T L_k R_k$, and checks

$$(Z_k)_{ii} > \sum_{j \neq i} |(Z_k)_{ij}| \quad (0 \leq i < 48). \quad (\text{R1.3.3})$$

A real symmetric matrix satisfying (R1.3.3) is positive definite. Since R_k is invertible, L_k is positive definite by congruence. No floating-point eigenvalue calculation is used in this verification. Across all 48 cones, the smallest checked integer margin is

$$999\,856\,920\,846\,456\,106 > 0.$$

The scale originally used to generate the R_k is irrelevant to this exact positivity argument.

R1.3.2 Exact verification of the inverse-exponential lower bound

Put $r = 401/400$ and $c = 1/4$. Then

$$e^{100A^{-1}} = F(S), \quad F(z) = \exp\left(-\frac{c}{r-z}\right).$$

Write

$$F(z) = e^{-100/401} \sum_{k \geq 0} b_k z^k.$$

All b_k are rational. They are obtained without evaluating a transcendental function, from

$$b_{-1} = 0, \quad b_0 = 1, \quad b_{k+1} = \frac{(2rk - c)b_k - (k-1)b_{k-1}}{r^2(k+1)}. \quad (\text{R1.3.4})$$

Because $S^{48} = 0$, only $b = (b_0, \dots, b_{47})^T$ matters. The unit vector e_0 satisfies

$$\|e^{100A^{-1}}e_0\|_Q^2 = e^{-200/401} b^T Q b.$$

For $x = 200/401 \in (0, 1)$, the odd alternating partial sum

$$L_{17} = \sum_{j=0}^{17} \frac{(-x)^j}{j!}$$

is a rigorous positive lower bound on e^{-x} . The verifier checks the rational inequalities

$$L_{17} b^T Q b > \frac{6348}{1000} > \frac{25}{4}. \quad (\text{R1.3.5})$$

For orientation only, its exact rational lower bound has decimal value approximately 6.348773015777, giving a norm lower bound approximately 2.5196771650. The assertions in the verifier use rational comparisons, not these rounded decimals. Equations (R1.3.2) and (R1.3.5) prove (R1.1.3).

Remark R1.3.1 (Scope of this computation). The search used a finite Gram semidefinite formulation from the supplied prior report. No global optimality of the numerical result is asserted. The exact certificate proves one admissible example and one lower bound. It does not provide an asymptotic family with a proved growth rate, a finite-dimensional upper bound for the full problem, or matching asymptotics.

R1.4 Why a bounded-variation phase-lattice construction cannot be packed arbitrarily densely

Here is a precise obstruction to a natural modification of the known conditional-basis construction. It concerns the multiplier certificate, not all bounded semigroups.

Let $2 \leq y_1 < \dots < y_N = Y$, with $y_{j+1} - y_j \geq 1$, and put

$$\lambda(y) = \frac{1}{-1 + iy}, \quad V = \sup_{u \geq 0} \sum_{j=1}^{N-1} |e^{u\lambda(y_{j+1})} - e^{u\lambda(y_j)}|.$$

The separation assumption holds, after an inessential fixed scaling, when inverse phases are chosen from a fixed lattice to produce alternating signs.

Theorem R1.4.1 (A bounded-variation spacing obstruction). *With*

$$L(Y) = 1 + \log \log(Y + e^e),$$

one has

$$\boxed{N \leq 4096 + (1 + 10\pi eV)(2 + 10eVL(Y))}. \quad (\text{R1.4.1})$$

In particular, for a fixed variation budget V , the number of points is $O_V(\log \log Y)$.

Proof. Write $h(y) = y/(1 + y^2)$, so $\lambda(y) = -1/(1 + y^2) - ih(y)$.

Step 1: a dyadic interval contains only $O(1 + V)$ points. Fix $a \geq 2$ and consider consecutive points $y < z$ in $[a, 2a)$. Since $z - y \geq 1$,

$$\omega = h(y) - h(z) = \int_y^z \frac{v^2 - 1}{(1 + v^2)^2} dv \geq \frac{1}{10a^2}.$$

For $0 \leq u \leq a^2$, both exponential moduli are at least e^{-1} . Therefore

$$|e^{u\lambda(y)} - e^{u\lambda(z)}| \geq 2e^{-1} |\sin(\omega u/2)|.$$

For every $q \geq 1/20$,

$$\frac{1}{q} \int_0^q |\sin v| dv \geq \frac{1}{20\pi}.$$

For $q \leq \pi/2$ this follows from $\sin v \geq 2v/\pi$; for larger q it follows by including a quarter-period or complete periods. Consequently

$$\frac{1}{a^2} \int_0^{a^2} |e^{u\lambda(y)} - e^{u\lambda(z)}| du \geq \frac{1}{10\pi e}.$$

If there are m points in the interval, summing over their $m - 1$ consecutive pairs and using the definition of V gives $m \leq 1 + 10\pi eV$.

Step 2: only $O(1 + VL(Y))$ well-separated dyadic intervals can occur. Suppose $4096 \leq y < z$ and $z \geq 2y$. Monotonicity of h and an elementary calculation give

$$\omega = h(y) - h(z) \geq h(y) - h(2y) \geq \frac{1}{4y}.$$

On $[y^{3/2}, y^2]$, the same modulus argument gives a lower bound by $2e^{-1} |\sin(\omega u/2)|$. Let $w(u) = 1/(u \log u)$. The function $|\sin(\omega u/2)| - 2/\pi$ has mean zero and period $2\pi/\omega$. Its primitive starting

at zero is bounded in modulus by $2\pi/\omega$. Integration by parts against the positive decreasing function w gives

$$\begin{aligned} \int_{y^{3/2}}^{y^2} |\sin(\omega u/2)| w(u) du &\geq \frac{2}{\pi} \log \frac{4}{3} - \frac{4\pi}{\omega} w(y^{3/2}) \\ &\geq \frac{2}{\pi} \log \frac{4}{3} - \frac{32\pi}{3\sqrt{y} \log y} > \frac{1}{10}. \end{aligned}$$

For the last numerical inequality it suffices to use $y \geq 4096$, $\log 4096 > 8$, $\log(4/3) > 2/7$, and $\pi < 22/7$. Hence

$$\int_{y^{3/2}}^{y^2} |e^{u\lambda(y)} - e^{u\lambda(z)}| \frac{du}{u \log u} \geq \frac{1}{5e}. \quad (\text{R1.4.2})$$

Choose one representative from each nonempty dyadic interval $[2^k, 2^{k+1})$ above 4096, first for even k , then for odd k . Within either subsequence, consecutive representatives have ratio at least two. Removing points cannot increase total variation, by the triangle inequality. If one subsequence has K representatives, sum (R1.4.2) over its $K-1$ edges. All its integration intervals lie inside $[e, Y^2]$. Thus

$$\frac{K-1}{5e} \leq V \int_e^{Y^2} \frac{du}{u \log u} = V \log \log(Y^2) \leq VL(Y).$$

There are therefore at most $2 + 10eVL(Y)$ occupied dyadic intervals above 4096. Multiply by the occupancy bound from Step 1 and add the at most 4096 points below the threshold. This proves (R1.4.1). $\square$

Proposition R1.4.2 (Dense clusters do not remove this obstruction). *The minimum-spacing assumption can be dropped for the corresponding multiplier bound. Let $2 \leq y_1 < \dots < y_N \leq Y$ be arbitrary, and define V by the same formula as above. Suppose the basis carrying these diagonal multipliers has partial-sum projection constant at most K . Then the multiplier D_Y with coefficients e^{-1+iy_j} satisfies*

$$\|D_Y\| \leq C_V K [1 + \log \log(Y + e^e)], \quad (\text{R1.4.3})$$

where C_V is independent of N , Y , and the spacings of the points.

Proof. Partition $[2, Y]$ into half-open unit intervals $[m, m+1)$, and let B be the number of occupied intervals. Choose one representative from each occupied interval. Split these representatives into the subsequences with even and odd integer interval index. Consecutive representatives in either subsequence are separated by more than one. Removing points cannot increase the total variation at any fixed time. Applying Theorem R1.4.1 separately to the two subsequences therefore gives

$$B \leq C_V [1 + \log \log(Y + e^e)].$$

Within one unit interval, the sum of consecutive differences of e^{iy_j} is at most the sum of the corresponding y -gaps, and hence at most one. The jump between consecutive occupied intervals is at most two. Thus the total variation of $d_j = e^{-1+iy_j}$ is at most $3B/e$. Abel summation with the partial-sum projections P_j gives

$$D_Y = d_N P_N + \sum_{j=1}^{N-1} (d_j - d_{j+1}) P_j, \quad \|D_Y\| \leq \frac{K}{e} (1 + 3B).$$

This proves (R1.4.3). $\square$

R1.4.1 What this does and does not exclude

The commonly used multiplier argument bounds a diagonal semigroup on a Schauder basis using its partial-sum projection constant and the scalar total variation of its coefficients. Theorem R1.4.1 shows that, for the phase-lattice family just specified, a fixed variation budget already enforces an iterated-logarithmic coordinate count. Global rescaling of all original eigenvalues does not change V . Any stronger exponentially weighted variation certificate also supplies such an unweighted bound; only this necessary condition is used here.

For example, normalize the least damping of the finite spectrum to one by multiplying $\lambda(y_j)$ by $1 + Y^2$. At an inverse observation time $t = 1 + Y^2$, the inverse multipliers are e^{-1+iy_j} . Then $N = O_V(\log \log t)$. If the basis has partial-sum constant K uniformly bounded, its individual coordinate projections have norm at most $2K$, and any diagonal multiplier with coefficient modulus at most one has norm at most $2KN$. Consequently this particular uniformly bounded-variation scheme can yield at most $O_{K,V}(\log \log t)$ at those observation times. Proposition R1.4.2 extends that conclusion to arbitrarily dense clusters of the y_j , without a minimum-spacing assumption.

The crucial limitation is that a bounded semigroup can have scalar multiplier variation much larger than its actual operator norm. The theorem does not assert that a fixed semigroup bound M forces a fixed V . It therefore does not imply $G_M(t) = O(\log \log t)$, and cannot be used as such an upper bound.

R1.5 Reproducibility and the remaining gap

The accompanying archive contains the integer certificate, a pure-Python verifier, the explicit integer Gram matrix, the numerical search code and its retained log, this report and its source, and SHA-256 checksums. To verify the finite-time claim, run

```
python verify_certificate.py certificate.json
```

The search code requires NumPy and SciPy; the verifier does not. The generator is specified in the Q -inner product. Passing to the ordinary Euclidean inner product by a square root of Q gives an equivalent conventional matrix example, but computing that square root is unnecessary for the proof.

A resolution still requires a general upper bound and an admissible lower construction of the same asymptotic order. The new finite-time certificate cannot be extrapolated into such a construction. The variation obstruction only excludes a specific certification strategy. The continuous damping estimate improves an explicit upper bound but remains logarithmic. Thus the claimed outcome of this report is exactly these proved statements, not a solution of MF-17.

Part IV

The first doubly logarithmic upper proof

Source and scope. Full mathematical body of the saved double-logarithmic upper-bound report.

This historical upper route is reproduced even where source S later reuses its window lemma. It provides an independent location for the full Fourier-window calculation, not an independent external verification.

U.1 Statement and the remaining gap

For $M \geq 1$, let $\mathcal{A}_M$ consist of all complex Hilbert spaces H and generators A of strongly continuous semigroups satisfying

$$\|e^{sA}\| \leq Me^{-s} \quad (s \geq 0).$$

As usual, A can be unbounded; exponential stability implies that A^{-1} is bounded. Define

$$G_M(t) = \sup_{(H,A) \in \mathcal{A}_M} \|e^{tA^{-1}}\| \quad (t \geq 0).$$

Theorem U.1.1. *There is an absolute constant C such that, for every $M \geq 1$ and $t > 0$,*

$$G_M(t) \leq 1 + CM^3[1 + \log \log(t + e^e)]. \quad (\text{U.1.1})$$

All constants are independent of the Hilbert space, the generator, and t .

The proof is based on a shifted-window estimate involving three semigroup factors. A window of width N controls an entire oscillatory range from $2N$ to a constant multiple of $N^{4/3}$. Iterating these ranges requires only $O(\log \log t)$ pieces. The factorization underlying the shifted estimate is the continuous-semigroup counterpart of the M_3 shifted-multiplier argument in Pisier [Pisier, Lemma 2.3]; a direct proof is given below.

The lower bound supplied by Loris–Meyries–Veraar [LMV, Theorem 1.2(2)], with their permissible constant 50, gives for each fixed $M > 1$

$$\frac{1}{250} [\log \log(t + e^e)]^{\alpha_M} \leq G_M(t), \quad \alpha_M = \frac{M-1}{M+49}. \quad (\text{U.1.2})$$

Thus (U.1.1) is not a full resolution of the fixed- M question: $\alpha_M < 1$, and α_M cannot tend to one while M remains fixed. The earlier logarithmic upper estimates are due to Zwart [Zwart] and, with explicit uniform constants, Batty–Gomilko–Tomilov [BGT].

U.2 A three-factor shifted-window estimate

All Hilbert inner products in this note are linear in the first argument. For $N > 0$, set

$$g_N(s) = N^{-1/2} \mathbf{1}_{[N/2, 3N/2]}(s), \quad h_N = g_N * g_N. \quad (\text{U.2.1})$$

Then $\|g_N\|_{L^2(\mathbb{R})} = 1$ and

$$h_N(v) = \left(1 - \left|\frac{v}{N} - 2\right|\right)_+.$$

In particular, h_N is supported on $[N, 3N]$ and

$$\sum_{j=0}^{\infty} h_N(u - jN) = 1 \quad (u \geq 2N). \quad (\text{U.2.2})$$

At most two terms are nonzero except at endpoints.

Lemma U.2.1 (Shifted-window estimate). *Let T be a strongly continuous semigroup on a complex Hilbert space with $\sup_{u \geq 0} \|T(u)\| \leq M$. For $x, y \in H$ put $\phi(u) = \langle T(u)x, y \rangle$ and*

$$F_k(\theta) = \int_0^\infty h_N(v) \phi(v+k) e^{-i\theta v} dv, \quad k \geq 0.$$

For every countable set $E \subset [0, \infty)$,

$$\int_{\mathbb{R}} \sup_{k \in E} |F_k(\theta)| \frac{d\theta}{2\pi} \leq M^3 \|x\| \|y\|. \quad (\text{U.2.3})$$

Proof. Define, as a strong operator integral,

$$U(\theta) = \int_0^\infty g_N(s) e^{-i\theta s} T(s) ds.$$

The semigroup law, Fubini's theorem on a compact integration region, and commutativity with $T(k)$ give

$$F_k(\theta) = \langle T(k)U(\theta)^2 x, y \rangle = \langle T(k)U(\theta)x, U(\theta)^* y \rangle.$$

Consequently

$$\sup_{k \in E} |F_k(\theta)| \leq M \|U(\theta)x\| \|U(\theta)^* y\|.$$

The Hilbert-valued Plancherel theorem gives

$$\begin{aligned} \int_{\mathbb{R}} \|U(\theta)x\|^2 \frac{d\theta}{2\pi} &= \int_0^\infty |g_N(s)|^2 \|T(s)x\|^2 ds \leq M^2 \|x\|^2, \\ \int_{\mathbb{R}} \|U(\theta)^* y\|^2 \frac{d\theta}{2\pi} &= \int_0^\infty |g_N(s)|^2 \|T(s)^* y\|^2 ds \leq M^2 \|y\|^2. \end{aligned}$$

For the second line, the Fourier sign is reversed. The adjoint semigroup is strongly continuous on a Hilbert space. Cauchy–Schwarz now proves (U.2.3). Countability makes the displayed supremum measurable. $\square$

Fix a smooth function ψ supported in $(1/2, 7/2)$ and equal to one on $[1, 3]$. It will be fixed throughout the proof.

Lemma U.2.2 (Window synthesis). *Let $K \in C_c^\infty((0, \infty))$ vanish on $(0, 2N)$. Put*

$$b_j(\theta) = \int_{\mathbb{R}} K(v+jN) \psi(v/N) e^{i\theta v} dv, \quad j \in \mathbb{N}_0.$$

Only finitely many b_j are nonzero. Under the assumptions of Lemma U.2.1,

$$\left\| \int_0^\infty K(u) T(u) du \right\| \leq M^3 \sup_{\theta \in \mathbb{R}} \sum_{j \geq 0} |b_j(\theta)|. \quad (\text{U.2.4})$$

The operator integral is a strong integral.

Proof. Using (U.2.2), followed by Fourier inversion with the indicated opposite signs, gives for unit x, y

$$\begin{aligned} \int_0^\infty K(u) \phi(u) du &= \sum_{j \geq 0} \int_{\mathbb{R}} K(v+jN) h_N(v) \phi(v+jN) dv \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{j \geq 0} b_j(\theta) F_{jN}(\theta) d\theta. \end{aligned}$$

Here $\psi(v/N) = 1$ on $\text{supp } h_N$. The sums are finite and each integrand is integrable: b_j is rapidly decreasing, and the F_{jN} satisfy Lemma U.2.1. Taking absolute values and applying that lemma proves (U.2.4). $\square$

U.3 The scalar oscillatory estimate

The key point is that the Fourier transforms of the translated windows are localized at different frequencies. Their absolute values have a uniformly bounded sum, even though the number of windows tends to infinity.

Lemma U.3.1 (Summable frequency tails). *Set*

$$J_j = [(j + 7/2)^{-1/2}, (j + 1/2)^{-1/2}], \quad j \in \mathbb{N}_0.$$

There is an absolute constant C such that

$$\sup_{\eta \in \mathbb{R}} \sum_{j \geq 0} \left(1 + (j + 2)^{3/2} \text{dist}(\eta, J_j) \right)^{-2} \leq C. \quad (\text{U.3.1})$$

Proof. For $\eta \leq 0$, the quantity multiplying the distance is bounded below by $c(j + 2)$, so comparison with $\sum (j + 2)^{-2}$ suffices. For $\eta > 0$, put $L = \eta^{-2}$.

First suppose $L \geq 16$. If $j + 2 \leq L/8$, then $j + 7/2 \leq L/4$, and hence

$$\text{dist}(\eta, J_j) \geq \frac{1}{2}(j + 7/2)^{-1/2}.$$

These terms again have a summable $C(j + 2)^{-2}$ bound. If $j + 2 \geq 8L$, then $j + 1/2 \geq 4L$ and $\text{dist}(\eta, J_j) \geq \eta/2$. Their sum is bounded by

$$C \sum_{j+2 \geq 8L} \frac{L}{(j + 2)^3} \leq C.$$

For the remaining indices, $j + 2 \in [L/8, 8L]$. The mean value theorem for $r \mapsto r^{-1/2}$, on an interval contained in $[cL, CL]$, gives

$$(j + 2)^{3/2} \text{dist}(\eta, J_j) \geq c \text{dist}(L, [j + 1/2, j + 7/2]).$$

The corresponding inverse-square tails have uniformly bounded sum over the integer translates of an interval of fixed length.

If $0 < L < 16$, there are only a fixed number of indices with $j + 2 < 128$; each contributes at most one. The remaining indices have $\text{dist}(\eta, J_j) \geq \eta/2$, and the same $CL(j + 2)^{-3}$ bound applies. This proves (U.3.1). $\square$

Lemma U.3.2 (Power-annular oscillatory bound). *Fix $C_0 \geq 1$ and constants A_0, A_1, A_2 . Suppose $N \geq 2$ and $a \in C_c^\infty((0, \infty))$ satisfies*

$$\text{supp } a \subset [2N, C_0 N^{4/3}], \quad |a^{(\ell)}(u)| \leq A_\ell u^{-3/4-\ell}, \quad \ell = 0, 1, 2. \quad (\text{U.3.2})$$

For either sign, put $K(u) = a(u)e^{\pm 2i\sqrt{u}}$ and form the b_j in Lemma U.2.2. Then

$$\sup_{\theta} \sum_{j \geq 0} |b_j(\theta)| \leq C, \quad (\text{U.3.3})$$

where C depends only on C_0, A_0, A_1, A_2 and the fixed cutoff ψ , not on N . Consequently

$$\left\| \int K(u) T(u) du \right\| \leq CM^3. \quad (\text{U.3.4})$$

Proof. It suffices to handle the positive phase; the negative phase reverses the frequency axis. Set

$$U_j = (j+2)N, \quad \lambda_j = N^2 U_j^{-3/2}, \quad w_j = N U_j^{-3/2} = \lambda_j / N.$$

On the j th window, $u = v + jN$ lies in $[(j+1/2)N, (j+7/2)N]$, so $u \asymp U_j$, with absolute comparison constants. If b_j is nonzero, then

$$U_j \leq C_0 N^{4/3} + \frac{3}{2}N \leq (C_0 + 2)N^{4/3},$$

whence

$$\lambda_j \geq (C_0 + 2)^{-3/2} > 0. \quad (\text{U.3.5})$$

After $v = Nx$, write

$$b_j(\theta) = \int B_j(x) e^{iP_{j,\theta}(x)} dx,$$

where

$$B_j(x) = Na(N(j+x))\psi(x), \quad P_{j,\theta}(x) = 2\sqrt{N(j+x)} + N\theta x.$$

The integration is over $[1/2, 7/2]$. From (U.3.2),

$$\max_{\ell=0,1,2} \|B_j^{(\ell)}\|_\infty \leq C N U_j^{-3/4} = C \sqrt{\lambda_j}. \quad (\text{U.3.6})$$

Also

$$|P_{j,\theta}''(x)| \asymp \lambda_j, \quad |P_{j,\theta}'''(x)| \leq C \lambda_j,$$

and the second derivative has a fixed sign. The second-derivative van der Corput estimate therefore gives

$$|b_j(\theta)| \leq C \quad (\text{U.3.7})$$

for every θ . One elementary proof of the estimate used here splits the integration into $|P'| \leq \sqrt{\lambda_j}$ and its complement. The first region has length $O(\lambda_j^{-1/2})$; integration by parts on the at most two complementary intervals, and the monotonicity of P' , gives the same bound times $\|B_j\|_\infty + \|B_j'\|_1$.

The stationary-frequency interval is

$$I_j = -N^{-1/2} J_j.$$

Put $D = \text{dist}(\theta, I_j)$. If $D \geq w_j$, then throughout the window

$$|P'_{j,\theta}| \geq ND =: q \geq \lambda_j.$$

Two integrations by parts have no endpoint terms, because B_j vanishes near the endpoints. More explicitly, the resulting amplitude is, up to a constant of modulus one,

$$\frac{B_j''}{(P')^2} - \frac{3B_j'P'' + B_jP'''}{(P')^3} + \frac{3B_j(P'')^2}{(P')^4}.$$

Equations (U.3.6) and (U.3.5) imply

$$\begin{aligned} |b_j(\theta)| &\leq C \sqrt{\lambda_j} (q^{-2} + \lambda_j q^{-3} + \lambda_j^2 q^{-4}) \\ &\leq C \lambda_j^{-3/2} (D/w_j)^{-2} \leq C (D/w_j)^{-2}. \end{aligned}$$

Together with (U.3.7), this yields

$$|b_j(\theta)| \leq C \left(1 + \frac{\text{dist}(\theta, I_j)}{w_j} \right)^{-2}. \quad (\text{U.3.8})$$

With $\eta = -\theta\sqrt{N}$, the ratio in (U.3.8) equals $(j+2)^{3/2} \text{dist}(\eta, J_j)$. Lemma U.3.1 proves (U.3.3); Lemma U.2.2 gives (U.3.4). $\square$

U.4 Applying the estimate to inverse semigroups

U.4.1 The exact Bessel representation

Write

$$\mathcal{K}(u) = \frac{J_1(2\sqrt{u})}{\sqrt{u}} \quad (u > 0), \quad \mathcal{K}(0) = 1.$$

Lemma U.4.1. *For every admissible A and every $t > 0$,*

$$e^{tA^{-1}} = I - \int_0^\infty \sqrt{t/s} J_1(2\sqrt{ts}) e^{sA} ds \quad (\text{U.4.1})$$

$$= I - \int_0^\infty \mathcal{K}(u) e^{-u/t} T_t(u) du, \quad T_t(u) = e^{(u/t)(A+I)}. \quad (\text{U.4.2})$$

The integrals converge in the strong sense and define bounded operators; T_t is a strongly continuous semigroup bounded by M .

Proof. For $n \geq 1$, the semigroup resolvent formula gives

$$A^{-n} = \frac{(-1)^n}{(n-1)!} \int_0^\infty s^{n-1} e^{sA} ds.$$

Insert this into the norm-convergent exponential series. Absolute interchange is justified by

$$M \sum_{n \geq 1} \frac{t^n}{n!(n-1)!} \int_0^\infty s^{n-1} e^{-s} ds = M(e^t - 1) < \infty.$$

The power series for J_1 gives (U.4.1). Substituting $u = ts$ proves (U.4.2); the claimed bound on T_t follows directly from the hypothesis on A . $\square$

Only the leading large-argument Bessel asymptotic is needed. For fixed order one, the standard expansion [DLMF, §10.17(i)] says

$$\mathcal{K}(u) = \pi^{-1/2} u^{-3/4} \cos(2\sqrt{u} - 3\pi/4) + E(u), \quad |E(u)| \leq C u^{-5/4} \quad (u \geq 1). \quad (\text{U.4.3})$$

In particular, the error is integrable. No differentiated asymptotic expansion or exact symbol representation of a Hankel function is needed.

U.4.2 A partition into power-annular intervals

Fix a smooth nonincreasing cutoff $\chi : [0, \infty) \rightarrow [0, 1]$, equal to one on $[0, 1]$ and zero on $[2, \infty)$. Set

$$N_0 = 16, \quad N_{q+1} = N_q^{4/3} \quad (q \geq 0).$$

The part of (U.4.2) containing $\chi(u/(2N_0))\mathcal{K}(u)$ has norm at most CM , uniformly in t . On its complement, (U.4.3) shows that replacing $\mathcal{K}$ by its leading cosine changes the operator by at most

$$M \int_1^\infty |E(u)| du \leq CM.$$

It remains to estimate the two complex exponentials in the leading term. For each sign, the scalar kernel is a sum of terms

$$a_{q,t}(u) e^{\pm 2i\sqrt{u}}, \quad a_{q,t}(u) = u^{-3/4} e^{-u/t} \left[\chi\left(\frac{u}{2N_{q+1}}\right) - \chi\left(\frac{u}{2N_q}\right) \right]. \quad (\text{U.4.4})$$

The sum telescopes to $u^{-3/4}e^{-u/t}(1 - \chi(u/(2N_0)))$. Moreover,

$$\text{supp } a_{q,t} \subset [2N_q, 4N_q^{4/3}].$$

For $\ell = 0, 1, 2$, differentiation of (U.4.4) gives

$$|a_{q,t}^{(\ell)}(u)| \leq C_\ell e^{-N_q/t} u^{-3/4-\ell}. \quad (\text{U.4.5})$$

Indeed, derivatives of either cutoff have their natural scale comparable to u wherever they are nonzero, and

$$t^{-m}e^{-u/t} \leq C_m u^{-m} e^{-u/(2t)} \leq C_m u^{-m} e^{-N_q/t} \quad (u \geq 2N_q).$$

Apply Lemma U.3.2 with $N = N_q$, $C_0 = 4$, and semigroup T_t . Each annular term costs at most $CM^3 e^{-N_q/t}$.

Finally,

$$\sum_{q \geq 0} e^{-N_q/t} \leq C[1 + \log \log(t + e^e)]. \quad (\text{U.4.6})$$

To see this, note that $N_q = \exp((4/3)^q \log 16)$. The number of indices with $N_q \leq t$ is $O(1 + \log \log(t + e^e))$. Starting with the first $N_q > t$, the remaining ratios N_{q+m}/t are bounded below by $16^{(4/3)^{m-1}}$, and their exponentials have uniformly summable tails. The same argument handles $t < 16$ by starting with $q = 0$.

Combining (U.4.2)–(U.4.6), and using $M \geq 1$, proves Theorem U.1.1. All scalar error estimates and all operator constants are uniform in H, A, t .

U.5 Two consequences with different quantifiers

Corollary U.5.1 (No smaller power works for all forward bounds). *For every real $\beta < 1$, there exists a fixed $M > 1$ such that*

$$\lim_{t \rightarrow \infty} \frac{G_M(t)}{[\log \log(t + e^e)]^\beta} = \infty.$$

Proof. Choose $\alpha \in (\max\{\beta, 0\}, 1)$ and set $M = 1 + 50\alpha/(1 - \alpha)$. The cited lower construction in (U.1.2), equivalently Theorem 1.2(ii) of [LMV], gives $G_M(t) \geq (1/250)[\log \log(t + e^e)]^\alpha$. Dividing by the displayed smaller power proves the assertion. $\square$

This is sharpness only within the family of fixed powers with one exponent required to work for every M . It does not establish a matching lower bound for an individually prescribed M . It also does not rule out an improvement such as $\log \log t / \log \log \log t$.

Corollary U.5.2 (Power-bounded operator formulation). *Let B be a bounded operator on a complex Hilbert space with $\sup_{n \geq 0} \|B^n\| \leq M$. For $r > 1$ and $c > 0$,*

$$\left\| \exp[-c(rI - B)^{-1}] \right\| \leq 1 + CM^3 \left[1 + \log \log \left(\frac{c}{r-1} + e^e \right) \right].$$

The constant is the same absolute constant as in Theorem U.1.1.

Proof. The spectral-radius formula gives $\sigma(B) \subset \overline{\mathbb{D}}$, so $rI - B$ is invertible. Set $A = (B - rI)/(r - 1)$. Its exponential series gives

$$\left\| e^{sA} \right\| \leq e^{-rs/(r-1)} \sum_{n \geq 0} \frac{(s/(r-1))^n}{n!} \|B^n\| \leq Me^{-s}.$$

At $t = c/(r - 1)$, one has $tA^{-1} = -c(rI - B)^{-1}$. Apply Theorem U.1.1. $\square$

U.6 What this proves, and what it does not

The proof reduces the upper rate from a logarithm to a double logarithm. Its operative distinction from the earlier two-factor estimates is the additional shift $T(jN)$ in Lemma U.2.1. This permits all windows in $[2N, CN^{4/3}]$ to be estimated together rather than separately. Their different stationary frequencies keep the pointwise sum of their Fourier magnitudes bounded.

For the requested fixed- M envelope, the bounds justified here are

$$\frac{1}{250} [\log \log(t + e^e)]^{(M-1)/(M+49)} \leq G_M(t) \leq 1 + CM^3 [1 + \log \log(t + e^e)].$$

The ratio between the displayed upper and lower rates can still tend to infinity. Thus a matching fixed- M lower construction, or a sharper upper bound with matching dependence in the exponent, is still needed.

The proof above is analytic. The accompanying Fourier-window computations are diagnostics only; they are not used to infer any universal operator inequality, optimality, or fixed- M lower asymptotic.

Part V

The binary-cut intermediate route

Source and scope. Reconstruction from the displayed conversation; no separate report file survives.

The rate here is superseded by the proposed all-cut exponent in S. The proof is included to preserve what was tried and where its geometric loss occurs.

B.1 The intermediate binary-cut proof

This part reconstructs the intermediate argument stated in the conversation between reports U and S. There is no separate mounted source file for it. It is retained as a superseded method: it explains the first fixed- M power saving and the precise geometric information later discarded. It is not used as a premise of report S.

B.1.1 From oscillatory blocks to a bounded separator

Use the Bessel blocks of report U and the scales

$$N_0 = 16, \quad N_{q+1} = N_q^{4/3}, \quad B_q^\pm = \int a_{q,t}(u) e^{\pm 2i\sqrt{u}} T(u) du, \quad \sup_u \|T(u)\| \leq M.$$

The amplitude is supported in $[2N_q, 4N_{q+1}]$ and satisfies

$$|a_{q,t}^{(\ell)}(u)| \leq C_\ell e^{-N_q/t} u^{-3/4-\ell}, \quad \|B_q^\pm\| \leq CM^3 e^{-N_q/t}.$$

Let C be the generator of T , and write $R_r = r(r - C)^{-1}$, so $\|R_r\| \leq M$. The integration-by-parts estimates later retained in S are

$$\begin{aligned} \|R_r B_q^\pm\| &\leq CM^4 e^{-N_q/t} r \sqrt{N_{q+1}} + CM^2 e^{-N_q/t} N_q^{-1/4}, \\ \|(I - R_r) B_q^\pm\| &\leq CM^4 e^{-N_q/t} \frac{1}{r \sqrt{N_q}}. \end{aligned}$$

At the split between k and $k + 1$, choose $r = N_{k+1}^{-1/2}$. The ratio sums converge, uniformly in the split, and give

$$\sum_{q \leq k} \|R_r B_q^\pm\| + \sum_{q > k} \|(I - R_r) B_q^\pm\| \leq CM^4.$$

The resolvent is not an idempotent. This argument uses only its bounded norm and its small separation errors.

B.1.2 Abstract approximate-separation lemma

Suppose a sequence of operators satisfies $\|B_j\| \leq B$, and for each cut k there exists R_k with $\|R_k\| \leq M$ and

$$\sum_{j \leq k} \|R_k B_j\| \leq D, \quad \sum_{j > k} \|(I - R_k) B_j\| \leq D.$$

Let $F(n)$ be the largest norm of an interval sum with at most n terms. For a unit vector x , split an interval of at most $2n$ terms into two halves and put

$$u = \sum_{\text{first}} B_j x, \quad v = \sum_{\text{second}} B_j x.$$

Then $\|u\|, \|v\| \leq F(n)$, $\|Ru\| \leq D$, and $\|Rv - v\| \leq D$.

If one of $\|u\|, \|v\|$ is at most $4D$, the sum is bounded by $F(n) + 4D$. Otherwise write $a = \|u\|$, $b = \|v\|$, and choose a scalar z minimizing $\|v + zu\|$. It satisfies $|z| \leq b/a$ and

$$\|v + zu\|^2 = b^2 - \frac{|\langle v, u \rangle|^2}{a^2}.$$

The separator gives

$$M\|v + zu\| \geq \|R(v + zu)\| \geq b - D - |z|D \geq b/2.$$

Thus

$$|\langle u, v \rangle| \leq \tilde{\rho}_M ab, \quad \tilde{\rho}_M = \sqrt{1 - \frac{1}{4M^2}},$$

and

$$F(2n) \leq q_M F(n) + 4D, \quad q_M = \sqrt{2(1 + \tilde{\rho}_M)} < 2.$$

Iteration gives

$$F(n) \leq C(B + D)n^{b(M)}, \quad b(M) = \log_2 q_M < 1.$$

Since $M \geq 1$, the factor $q_M - 1$ stays bounded away from zero, so this particular recurrence has an absolute iteration constant. Moreover

$$\begin{aligned} 1 - b(M) &= \frac{1}{2 \log 2} \log \frac{2}{1 + \sqrt{1 - 1/(4M^2)}} \\ &\geq \frac{1}{32(\log 2)M^2}. \end{aligned}$$

Applying this to the blocks, summing the exponentially damped tail, and keeping the bounded initial Bessel remainder yields the recorded weaker estimate

$$G_M(t) \leq CM^4 L(t)^{1 - [32(\log 2)M^2]^{-1}}.$$

The main loss is now transparent: the recurrence uses one split and one safety factor at a time. It does not use the full ordered projection geometry that produces $a(M)$ in S.

B.1.3 What the power saving did and did not imply

With $c = (32 \log 2)^{-1}$, the preceding estimate implies, for each fixed $k > 0$,

$$G_M(t) \leq C_k M^{4+2k} \frac{L(t)}{[\log L(t)]^k}, \quad C_k = C \left(\frac{k}{ec} \right)^k,$$

by maximizing $s^k e^{-cs/M^2}$ for $s = \log L(t)$. For every fixed finite M , it also implies little- o relative to every such logarithmically reduced rate. No one fixed saving $L(t)^{1-\varepsilon}$ follows uniformly over all M ; the lower exponents approach one as $M \rightarrow \infty$.

The conversation compared the old lower exponent $1 - 50/(M + 49)$ to this upper exponent. Their equality would require

$$1600(\log 2)M^2 - M - 49 = 0.$$

Its positive root is below one, outside the admissible class. Adjusting a positive prefactor could make two displayed expressions intersect at a chosen time, but could not make their ratio bounded as $t \rightarrow \infty$ for fixed admissible M . This was an algebraic obstruction to resolving the gap by constants alone, not an obstruction to improving the proof's exponent.

The later proposed exponent $a(M)$ supersedes both exponents. The conclusions about extra logarithms relative to $L(t)$ remain compatible with it because $0 < a(M) < 1$ for every finite $M > 1$.

B.1.4 Other regime checks recorded in the conversation

At fixed $t > 0$, the discussion established a different order, $G_M(t) = \Theta_t(M)$ as $M \rightarrow \infty$. An explicit lower witness is

$$A_M = -2I + (M-1)E, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Indeed

$$e^s \|e^{sA_M}\| \leq e^{-s} + (M-1)se^{-s} \leq M,$$

while

$$\|e^{tA_M^{-1}}\| \geq e^{-t/2} \frac{t(M-1)}{4}.$$

The energy bound $G_M(t) \leq 1 + M\sqrt{t/2}$ gives the matching fixed-time upper order.

Near $M = 1$, a resolvent calculation gives

$$1 \leq G_M(t) \leq e^{(M-1)t/2},$$

so $G_M(t) \rightarrow 1$ for fixed t as $M \downarrow 1$. To see the upper bound, set $B = A^{-1}$ and $r = 1/(M-1)$. The resolvent bound $\|(r-A)^{-1}\| \leq 1/r$ applied to $(r-A)Bx$ gives

$$\|rBx - x\| \geq r\|Bx\|, \quad \Re\langle Bx, x \rangle \leq \frac{M-1}{2}\|x\|^2.$$

Differentiate $\|e^{tB}x\|^2$. These are useful normalization checks, not estimates of a large- t leading coefficient at fixed M .

Part VI

Complete proposed sharp-rate proof

Source and scope. Full mathematical body of MF-17-sharp-fixed-bound-rate.tex, including all its appendices.

The source calls this a research proof without independent peer review. That status is unchanged. Its diagnostic-archive references are historical: the complete diagnostic archive is not mounted, whereas the source PDF and TeX are. The original argument is reproduced without silently replacing its lemmas by later optimizations.

S.1 Statement

For $M \geq 1$, let $\mathcal{A}_M$ consist of pairs (H, A) , where H is a complex Hilbert space and A generates a strongly continuous semigroup satisfying

$$\|e^{sA}\| \leq Me^{-s} \quad (s \geq 0).$$

Exponential stability implies $0 \in \rho(A)$ and $A^{-1} \in \mathcal{L}(H)$. Put

$$G_M(t) = \sup_{(H,A) \in \mathcal{A}_M} \|e^{tA^{-1}}\|, \quad L(t) = \log \log(t + e^e), \quad a(M) = \frac{2}{\pi} \arccos \frac{1}{M}.$$

All inner products below are linear in their first argument; all logarithms are natural. Constants denoted C_M, c_M depend only on the fixed number M ; they may change between displays.

Theorem S.1.1. *For every fixed $M > 1$, there are constants $0 < c_M \leq C_M < \infty$ and $t_M < \infty$ such that*

$$\boxed{c_M L(t)^{a(M)} \leq G_M(t) \leq C_M L(t)^{a(M)} \quad (t \geq t_M).} \quad (\text{S.1.1})$$

Moreover $G_1(t) = 1$ for every $t \geq 0$. Finite-dimensional generators suffice for the lower envelope in (S.1.1).

The original problem asks for matching multiplicative bounds with M fixed, not merely an exponent that becomes sharp as $M \rightarrow \infty$. The preceding continuations had an upper exponent of the form $1 - c/M^2$ and used the lower exponent $(M - 1)/(M + 49)$ supplied by Loris–Meyries–Veraar [LMV]. Neither exponent is retained here. The exact parameter $a(M)$ comes from the angle between the two sides of a Hilbert-space coordinate cut.

The proof has three main parts. First, all coordinate cuts together give a sharp growth estimate through a scalar Hardy-space argument. Second, sparse resolvent localization lets that estimate be applied to both factors of the inverse-semigroup oscillatory blocks without losing the endpoint exponent. Third, an analytic modification of power weights makes a conditional basis robust under a single complex phase. Orthogonal replication then gives the lower construction within the exact prescribed bound M .

S.2 Sharp geometry of all coordinate cuts

We use normalized Lebesgue measure m on $\mathbb{T}$. The Hilbert transform is normalized by $\mathcal{H}(e^{ik\theta}) = -i \operatorname{sgn}(k)e^{ik\theta}$. Thus the imaginary part of an analytic function is the Hilbert transform of its real part, up to its constant term.

Lemma S.2.1 (The cut angle). *If P is a projection on a Hilbert space and $\|P\| \leq K$, $K \geq 1$, then*

$$|\langle u, v \rangle| \leq \rho_K \|u\| \|v\|, \quad u \in \operatorname{ran} P, \quad v \in \ker P, \quad \rho_K = \sqrt{1 - K^{-2}}.$$

Conversely this angle bound implies $\|P\| \leq K$.

Proof. Minimize $\|u + zv\|^2$ over $z \in \mathbb{C}$ and use $\|u\| = \|P(u + zv)\| \leq K \|u + zv\|$. The minimum is $\|u\|^2 - |\langle u, v \rangle|^2 / \|v\|^2$. The same computation proves the converse. $\square$

Lemma S.2.2 (A scalar concentration bound). *Let w be a strictly positive trigonometric polynomial. If the Riesz projection P_+ onto nonnegative Fourier coefficients satisfies $\|P_+\|_{L^2(w) \rightarrow L^2(w)} \leq K$, then*

$$(P_r * w)(\theta) \leq 4K^3 \left(\int_{\mathbb{T}} w \, dm \right) \left(\frac{1+r}{1-r} \right)^{a(K)} \quad (0 < r < 1). \quad (\text{S.2.1})$$

Consequently, if w has degree at most $n-1$, then

$$\|w\|_{\infty} \leq CK^3 n^{a(K)} \int_{\mathbb{T}} w \, dm. \quad (\text{S.2.2})$$

Proof. Normalize $\int w \, dm = 1$. By Fejér–Riesz factorization, write $w = |h|^2$, where h is a polynomial with no zeros in $\overline{\mathbb{D}}$. Both h and $1/h$ are bounded analytic functions. Put $g = \bar{h}/h$ on $\mathbb{T}$ and $\rho = \sqrt{1 - K^{-2}}$. Multiplication by h maps the nonnegative weighted Hardy subspace onto H^2 , while multiplication by $\bar{h}$ maps the negative subspace onto H_-^2 . Lemma S.2.1 therefore gives

$$\|H_g\| \leq \rho, \quad H_g f = P_-(gf), \quad f \in H^2. \quad (\text{S.2.3})$$

The elementary form of Nehari’s theorem [Nehari] gives $f \in H^\infty$ with $\|g - f\|_{\infty} \leq \rho$. Here is the needed argument: factor any analytic polynomial p with $p(0) = 0$ as $p = ab$, $a \in H^2$, $b \in zH^2$, with $\|a\|_2 \|b\|_2 = \|p\|_1$. This follows by splitting its outer factor into two equal square roots and placing its inner factor in b . Equation (S.2.3) implies $|\int gp \, dm| \leq \rho \|p\|_1$. Hahn–Banach and L^1 duality give a function ψ with $\|\psi\|_{\infty} \leq \rho$ and $\int (g - \psi)p \, dm = 0$. The vanishing negative Fourier coefficients imply $f = g - \psi \in H^\infty$.

Fix ρ' with $\rho < \rho' < 1$. For $s < 1$ sufficiently close to one, $f_s(z) = f(sz)$ satisfies $\|g - f_s\|_{\infty} \leq \rho'$: use uniform convergence of the Poisson means of the continuous function g , and contractivity of Poisson convolution on L^∞ . The boundary function g has winding number zero. Since f_s is uniformly within distance $\rho' < 1$ of g , it has the same winding number and hence no zeros in $\overline{\mathbb{D}}$. A continuous logarithm is therefore available.

Set $b = \arcsin \rho'$. The argument difference $v = \arg g - \arg f_s$ can be chosen continuously with $\|v\|_{\infty} \leq b$. With $u = -\log |f_s|$ and $c = \int (\log w + \log |f_s|) \, dm$, the conjugate-function identities give

$$\log w = u + \mathcal{H}v + c. \quad (\text{S.2.4})$$

Jensen’s inequality and $1 - \rho' \leq |f_s| \leq 1 + \rho'$ imply

$$e^{u+c} \leq \frac{1 + \rho'}{1 - \rho'}.$$

There is an analytic function F with boundary logarithm $\log F = \mathcal{H}v - iv$. It lies in the sector $|\arg F| \leq b$. Thus $w \leq [(1 + \rho')/(1 - \rho')] \sec b \operatorname{Re} F$ on the boundary. Taking Poisson means and using positivity gives

$$P_r * w \leq \frac{1 + \rho'}{1 - \rho'} \sec b |F(re^{i\theta})|.$$

The normalized L^1 norm of the conjugate Poisson kernel is exactly

$$\int_{\mathbb{T}} |Q_r| dm = \frac{2}{\pi} \log \frac{1+r}{1-r}.$$

It follows that $|F(re^{i\theta})| \leq [(1+r)/(1-r)]^{2b/\pi}$. Let $\rho' \downarrow \rho$. Then $2b/\pi \rightarrow a(K)$ and

$$\frac{1+\rho}{1-\rho} \frac{1}{\sqrt{1-\rho^2}} = K^3(1+\rho)^2 \leq 4K^3.$$

This proves (S.2.1).

For (S.2.2), the case $n = 1$ is immediate. For $n \geq 2$ take $r = 1 - 1/n$. Bernstein's inequality gives $\|w'\|_\infty \leq (n-1)\|w\|_\infty$. On an interval of length $1/n$ around a maximum, $w \geq \|w\|_\infty/2$, and P_r is bounded below there by cn . Thus $\|w\|_\infty \leq C\|P_r * w\|_\infty$. Apply (S.2.1) and $(1+r)/(1-r) \leq 2n$. $\square$

Theorem S.2.3 (All-cut synthesis). *Let $u_1, \dots, u_n$ be linearly independent vectors in a Hilbert space, $\|u_j\| \leq B$. Assume that each initial coordinate projection on their span has norm at most $K \geq 1$. Then*

$$\left\| \sum_{j=1}^n u_j \right\| \leq CK^{3/2} B n^{(1+a(K))/2}. \quad (\text{S.2.5})$$

Proof. Define the positive trigonometric polynomial

$$w(e^{i\theta}) = \frac{1}{n} \left\| \sum_{j=1}^n u_j e^{ij\theta} \right\|^2.$$

It is strictly positive because the vectors are independent, and $\int w dm = n^{-1} \sum_j \|u_j\|^2 \leq B^2$. For a Laurent polynomial $p = \sum_m a_m z^m$, vector-valued Parseval gives

$$\int |p|^2 w dm = \frac{1}{n} \sum_m \left\| \sum_{j=1}^n a_{m-j} u_j \right\|^2.$$

Replacing p by $P_+ p$ replaces the sum in the m th term by its initial cut $j \leq m$. Hence $\|P_+\|_{L^2(w)} \leq K$. Lemma S.2.2, evaluated at $\theta = 0$, proves (S.2.5). $\square$

S.3 Approximate cuts and two-sided summation

The previous one-cut argument discarded most of the coordinate-cut structure. The following form retains it and will be used twice, once on each side of an operator product.

Lemma S.3.1 (Orthogonal augmentation). *Suppose vectors $u_1, \dots, u_n$, $\|u_j\| \leq B$, admit operators R_k with $\|R_k\| \leq M > 1$ and*

$$\sum_{j \leq k} \|R_k u_j\| + \sum_{j > k} \|R_k u_j - u_j\| \leq D \quad (1 \leq k < n).$$

For every $0 < \varepsilon \leq 1$,

$$\left\| \sum_{j=1}^n u_j \right\| \leq C_M \left(B + \frac{D}{\sqrt{\varepsilon}} \right) n^{(1+a(M+\varepsilon))/2}. \quad (\text{S.3.1})$$

The constants are uniform in ε, n and the vectors.

Proof. Put $e_{kj} = R_k u_j - \mathbf{1}_{\{j>k\}} u_j$ and let $E_k : \ell_2^n \rightarrow H$ have columns e_{kj} . Then $\|E_k\| \leq D$. For $c > 0$ augment u_j to $w_j = (u_j, c\delta_j)$ in $H \oplus \ell_2^n$. The operator

$$\tilde{R}_k = \begin{pmatrix} R_k & -E_k/c \\ 0 & Q_k \end{pmatrix}, \quad Q_k \delta_j = \mathbf{1}_{\{j>k\}} \delta_j,$$

acts as the exact tail coordinate projection on $\text{span}\{w_j\}$. Its norm is at most the norm of the scalar matrix $\begin{pmatrix} M & D/c \\ 0 & 1 \end{pmatrix}$, namely

$$\frac{\sqrt{(M+1)^2 + (D/c)^2} + \sqrt{(M-1)^2 + (D/c)^2}}{2} \leq M + \frac{MD^2}{2(M^2-1)c^2}. \quad (\text{S.3.2})$$

Take $c = D\sqrt{M/[2(M^2-1)\varepsilon]}$ (and pass to a limit if $D = 0$). The complementary initial projections have the same norm as the nontrivial tail projections on a Hilbert space. Apply Theorem S.2.3 to the independent w_j and discard their orthogonal components. $\square$

Lemma S.3.2 (Two-sided summation). *Let $(X_j)_{j \in I}$ and $(Y_j)_{j \in I}$ be finite ordered families, $|I| \leq n$. Suppose the members of each family and its adjoint family have uniformly bounded norms. More precisely, for each of $Z = X, Y, X^*, Y^*$ and each cut of its ordered index set there is an operator R with $\|R\| \leq M > 1$ such that the operator-norm error satisfies*

$$\sum_{j \text{ before cut}} \|RZ_j\| + \sum_{j \text{ after cut}} \|(I-R)Z_j\| \leq D \leq \sqrt{\varepsilon}, \quad 0 < \varepsilon \leq 1.$$

The same bounds are assumed for every ordered subfamily; its cuts may be different from the cuts used for the full family. Then

$$\left\| \sum_{j \in I} Y_j X_j \right\| \leq C_M n^{a(M+\varepsilon)}. \quad (\text{S.3.3})$$

Here the constant may also depend on the uniform member bounds, but not on n, ε or the Hilbert space.

Proof. Apply Lemma S.3.1 to $u_j = Z_j x_j$, $\|x_j\| \leq 1$, for any ordered subset and any of the four families Z . For $\beta = (1 + a(M + \varepsilon))/2$, this gives

$$\left\| \sum_{j \in F} Z_j x_j \right\| \leq C_M |F|^\beta.$$

By duality, for each vector x , $\sum_{j \in F} \|Z_j^* x\| \leq C_M |F|^\beta \|x\|$. The decreasing rearrangement therefore obeys $(\|Z_j^* x\|)_\ell^* \leq C_M \ell^{\beta-1} \|x\|$. Apply this to X_j^* and Y_j , and use the rearrangement inequality:

$$\left| \left\langle \sum_j Y_j X_j x, y \right\rangle \right| \leq \sum_j \|X_j x\| \|Y_j^* y\| \leq C_M \|x\| \|y\| \sum_{\ell=1}^n \ell^{a(M+\varepsilon)-1}.$$

Since $a(M + \varepsilon) \geq a(M) > 0$, the last sum is at most $[1 + a(M)^{-1}] n^{a(M+\varepsilon)}$. $\square$

S.4 Oscillatory blocks and sparse resolvent cuts

All integrals of a semigroup below are strong integrals on vectors; the displayed norm estimates make them bounded operators. No operator-norm continuity of an unbounded generator's semigroup is assumed. On a Hilbert space the adjoint semigroup is strongly continuous as well.

The analytic input from the previous continuation is the following window estimate. Its proof is reproduced in Appendix S.A; in particular, no unproved operator-valued oscillatory integral theorem is being assumed.

Lemma S.4.1 (Power-sized oscillatory window). *Let T be a strongly continuous Hilbert-space semigroup with $\sup_{u \geq 0} \|T(u)\| \leq M$. Suppose $N \geq 16$ and a smooth scalar amplitude a is supported in $[2N, 4N^{4/3}]$ and satisfies*

$$|a^{(\ell)}(u)| \leq Au^{-3/4-\ell}, \quad \ell = 0, 1, 2.$$

Then

$$\left\| \int_0^\infty a(u) e^{\pm 2i\sqrt{u}T(u)} du \right\| \leq CM^3 A. \quad (\text{S.4.1})$$

Fix once and for all a nonincreasing C^∞ function χ on $[0, \infty)$ which is one on $[0, 1]$ and zero on $[2, \infty)$. Put

$$N_0 = 16, \quad N_{q+1} = N_q^{4/3}, \quad \lambda_q = N_q^{-1/2}.$$

For $t > 0$ set

$$a_{q,t}(u) = u^{-3/4} e^{-u/t} \left[\chi\left(\frac{u}{2N_{q+1}}\right) - \chi\left(\frac{u}{2N_q}\right) \right], \quad (\text{S.4.2})$$

$$B_q^\pm = \int_0^\infty a_{q,t}(u) e^{\pm 2i\sqrt{u}T(u)} du. \quad (\text{S.4.3})$$

For every fixed derivative order,

$$|a_{q,t}^{(\ell)}(u)| \leq C_\ell e^{-N_q/t} u^{-3/4-\ell}, \quad \text{supp } a_{q,t} \subset [2N_q, 4N_{q+1}]. \quad (\text{S.4.4})$$

The constants in (S.4.4) are uniform in q, t : factors $(u/t)^j$ are absorbed by half of the exponential. Lemma S.4.1 yields

$$\|B_q^\pm\| \leq CM^3 e^{-N_q/t}. \quad (\text{S.4.5})$$

Let C be the generator of T , and define

$$R_r = r(rI - C)^{-1} = \int_0^\infty r e^{-rs} T(s) ds, \quad r > 0.$$

Thus $\|R_r\| \leq M$. All the operators here commute.

Lemma S.4.2 (High- and low-frequency errors). *For all $r > 0$ and $q \geq 0$,*

$$\|R_r B_q^\pm\| \leq CM^4 e^{-N_q/t} \frac{r}{\lambda_{q+1}} + CM^2 e^{-N_q/t} N_q^{-1/4}, \quad (\text{S.4.6})$$

$$\|(I - R_r) B_q^\pm\| \leq CM^4 e^{-N_q/t} \frac{\lambda_q}{r}. \quad (\text{S.4.7})$$

For $q \geq 1$ let

$$W_q = R_{\lambda_{q-1}} - R_{\lambda_{q+2}}. \quad (\text{S.4.8})$$

Then $\|W_q\| \leq 2M$ and

$$\sum_{q \geq 1} \|B_q^\pm - W_q B_q^\pm\| \leq CM^4. \quad (\text{S.4.9})$$

Proof. We give the positive-phase proof. Write $a = a_{q,t}$, $K(u) = a(u)e^{2i\sqrt{u}}$, and

$$h(u) = -i\sqrt{u}a(u)e^{2i\sqrt{u}}, \quad d(u) = (-i\sqrt{u}a(u))'e^{2i\sqrt{u}}.$$

Then $K = h' - d$. Lemma S.4.1 applied to h , and direct integration applied to d , give

$$\left\| \int hT \right\| \leq CM^3 e^{-N_q/t} \lambda_{q+1}^{-1}, \quad \left\| \int dT \right\| \leq CM e^{-N_q/t} N_q^{-1/4}.$$

For $h \in C_c^1((0, \infty))$, the strong integral $H_h = \int h(u)T(u)du$ maps H into $D(C)$ and $CH_h = -\int h'(u)T(u)du$. This follows directly from the difference quotient for $T(s)H_h$ and integration by parts. Since $R_r C = r(R_r - I)$, the preceding bounds prove (S.4.6).

Also $CB_q^+ = -\int K'(u)T(u)du$ and $I - R_r = -r^{-1}R_r C$ on $D(C)$. The amplitude of K' and its first two amplitude derivatives satisfy (S.4.4) with the additional factor $C\lambda_q$. Lemma S.4.1 proves (S.4.7). The negative phase has the identical estimates.

Finally,

$$B_q^\pm - W_q B_q^\pm = (I - R_{\lambda_{q-1}})B_q^\pm + R_{\lambda_{q+2}}B_q^\pm.$$

Apply (S.4.6)–(S.4.7) and sum the convergent series $\sum_q \lambda_q/\lambda_{q-1}$, $\sum_q \lambda_{q+2}/\lambda_{q+1}$, and $\sum_q N_q^{-1/4}$. $\square$

Lemma S.4.3 (Errors vanish on sparse tails). *Fix $J = 8$. There are absolute $b, c > 0$ such that the following holds. Restrict q to any one residue class modulo J and to $q \geq Q \geq 1$. Every cut of this ordered family admits an operator R_r of norm at most M such that, for $Z_q = B_q^+$, B_q^- , or W_q ,*

$$\sum_{q \text{ before cut}} \|R_r Z_q\| + \sum_{q \text{ after cut}} \|(I - R_r)Z_q\| \leq C_M \exp[-c(4/3)^Q]. \quad (\text{S.4.10})$$

The statement remains true for adjoints and for ordered subfamilies.

Proof. For a cut between $q = k$ and $q = k + J$, choose

$$r = \sqrt{\lambda_{k+2}\lambda_{k+J-1}}.$$

For $0 < r < b < a$ with $r \leq b/2$, the resolvent identity gives

$$\|R_r(R_a - R_b)\| \leq CM \frac{r}{b}. \quad (\text{S.4.11})$$

For $0 < b < a < r$ with $a \leq r/2$, it gives

$$\|(I - R_r)(R_a - R_b)\| \leq CM \frac{a}{r}. \quad (\text{S.4.12})$$

Indeed $R_r R_a = R_r + [r/(a-r)](R_r - R_a)$, so the leading R_r cancels in (S.4.11); and $(I - R_r)R_a = [a/(r-a)](R_r - R_a)$. For the stated scales and $J = 8$, the required ratios are bounded away from one already at $k \geq 1$.

For W_q , apply (S.4.11) to earlier indices and (S.4.12) to later ones. The two sums are bounded by a constant times

$$M \sqrt{\lambda_{k+J-1}/\lambda_{k+2}}.$$

For $B_q^\pm$, use (S.4.6)–(S.4.7); the extra sum of $N_q^{-1/4}$ over $q \geq Q$ is equally small. Since $\lambda_q = \exp[-(\log 16)(4/3)^q/2]$, all these quantities are bounded by the right side of (S.4.10), uniformly in the cut. Commutativity permits taking adjoints, and deleting indices only decreases the error sums. A cut in a subfamily may be placed at the corresponding cut of the full residue class. $\square$

Proposition S.4.4 (Endpoint sum of the blocks). *For every fixed $M > 1$ and $n \geq 1$,*

$$\left\| \sum_{q=0}^{n-1} B_q^\pm \right\| \leq C_M n^{a(M)}. \quad (\text{S.4.13})$$

Proof. Put $\varepsilon = 1/\log(n + e)$. Choose $Q \geq 1$ so that the right side of (S.4.10) is at most $\sqrt{\varepsilon}$. One may choose

$$Q \leq C_M [1 + \log \log(n + e)].$$

This deliberately coarse estimate is enough: $Q \leq C_M n^{a(M)}$. If $Q \geq n$, use (S.4.5) directly. Otherwise discard the first Q blocks at cost $C_M^3 Q$, replace the remaining blocks by $W_q B_q^\pm$ using (S.4.9), and split into the J residue classes. Lemma S.3.2 applies to each class by Lemma S.4.3. It yields $C_M n^{a(M+\varepsilon)}$ in total. Finally

$$a'(x) = \frac{2}{\pi x \sqrt{x^2 - 1}}, \quad x > 1,$$

is decreasing. Therefore $[a(M + \varepsilon) - a(M)] \log n \leq a'(M)$, and $n^{a(M+\varepsilon)} \leq e^{a'(M)} n^{a(M)}$. All discarded terms are absorbed into (S.4.13). $\square$

S.5 Completion of the upper bound

For $A \in \mathcal{A}_M$, the Bessel identity is

$$e^{tA^{-1}} = I - \int_0^\infty \sqrt{\frac{t}{s}} J_1(2\sqrt{ts}) e^{sA} ds. \quad (\text{S.5.1})$$

It follows by the series of J_1 and $\int_0^\infty s^k e^{sA} ds = (-1)^{k+1} k! A^{-k-1}$. Absolute integration using $\|e^{sA}\| \leq M e^{-s}$ justifies the series interchange. In particular, unbounded generators create no domain issue here. With $u = ts$ and $T_t(u) = e^{(u/t)(A+I)}$, this becomes

$$e^{tA^{-1}} = I - \int_0^\infty \frac{J_1(2\sqrt{u})}{\sqrt{u}} e^{-u/t} T_t(u) du, \quad \sup_{u \geq 0} \|T_t(u)\| \leq M. \quad (\text{S.5.2})$$

The standard Bessel asymptotic [DLMF, §10.17] is

$$\frac{J_1(2\sqrt{u})}{\sqrt{u}} = \frac{1}{\sqrt{\pi}} u^{-3/4} \cos(2\sqrt{u} - 3\pi/4) + O(u^{-5/4}).$$

The bounded initial interval and the integrable remainder contribute at most C_M . The leading amplitude decomposes into the blocks (S.4.2) and a bounded initial piece.

Let n be one plus the largest index with $N_q \leq t$, or $n = 1$ if there is no such index. Then $n \leq CL(t)$. Proposition S.4.4 bounds the initial block sum by $C_M L(t)^{a(M)}$. The remaining tail is bounded using (S.4.5) and $\sum_{N_q > t} e^{-N_q/t} \leq C$. This proves the upper bound in Theorem S.1.1.

Remark S.5.1 (Why no logarithmic endpoint loss remains). Using all adjacent cuts with a fixed nonzero error would give an extra factor $\log L(t)$ after orthogonal augmentation. Sparse residue classes make the cut errors super-exponentially small in the first retained block index. Consequently the exact triangular estimate (S.3.2) can be used with $\varepsilon \asymp 1/\log L(t)$ while keeping the augmented vector lengths bounded. This is why the endpoint $a(M)$, rather than every exponent strictly above it, is obtained.

S.6 A replication lemma for one active phase

The lower construction needs more than bounded coordinate cuts: while a semigroup passes from one cut to the next, one coordinate can make a full complex rotation. The following lemma controls that transition at the exact same bound K , not at $K + \varepsilon$.

Lemma S.6.1 (Phase-robust replication). *Fix $K > 1$ and $B < \infty$. Let $e_1, \dots, e_m$ be a normalized basis of a finite-dimensional Hilbert space. Write β_i for its coordinate functionals and Q_i for the tail projection onto $e_i, \dots, e_m$, with $Q_{m+1} = 0$. Suppose, for some $0 < \delta \leq 1$, that every i and $\sigma \in \{0, 1\}$ satisfy*

$$D_{i,\sigma}(x) := K^2 \|x\|^2 - \|Q_{i+1-\sigma}x\|^2 \geq \delta \|x\|^2, \quad (\text{S.6.1})$$

$$|\beta_i(x)|^2 + |\langle Q_{i+1-\sigma}x, e_i \rangle|^2 \leq BD_{i,\sigma}(x). \quad (\text{S.6.2})$$

There is an integer $r = r(K, B)$ and a normalized basis of dimension mr , ordered as r consecutive copies of each original coordinate, with these properties:

(a) Every multiplier which is zero before one coordinate, has an arbitrary value z with $|z| \leq 1$ on that coordinate, and is one afterwards, has norm at most $\sqrt{K^2 - c_K \delta}$, where $c_K > 0$ is independent of m, δ .

(b) Repeating a diagonal multiplier's value on all r copies preserves its original operator norm as a lower bound.

Proof. The original tail projections have norm at most K , and hence the rank-one coordinate projection $E_i x = \beta_i(x) e_i$ has norm at most $2K$. Fix i , abbreviate $e = e_i$, and write $x = v + ce$, with $\beta_i(v) = 0$. Put $a = Q_i v = Q_{i+1} v$, $u = \langle v, e \rangle$, $w = \langle a, e \rangle$, and

$$d_z(v, c) = K^2 \|v + ce\|^2 - \|a + zce\|^2, \quad |z| \leq 1.$$

Minimizing over $c \in \mathbb{C}$ gives

$$d_z(v, c) = m_z(v) + (K^2 - |z|^2)|c - c_z(v)|^2, \quad (\text{S.6.3})$$

$$c_z(v) = -\frac{K^2 u - \bar{z} w}{K^2 - |z|^2},$$

$$m_z(v) = K^2 \|v\|^2 - \|a\|^2 - \frac{|K^2 u - \bar{z} w|^2}{K^2 - |z|^2}.$$

Let $h(v) = |u|^2 + |w|^2$. For $\sigma = 0$, (S.6.2) at the minimizer says $h(v) \leq Bm_0(v)$. For $\sigma = 1$, it says $|c_1|^2 + |w + c_1|^2 \leq Bm_1(v)$. Using $K^2 u = w - (K^2 - 1)c_1$ then gives $h(v) \leq 8Bm_1(v)$. Thus, with $\kappa = (8B)^{-1}$,

$$m_0(v), m_1(v) \geq \kappa h(v). \quad (\text{S.6.4})$$

Also $v = (I - E_i)(v + ce)$ and $\|I - E_i\| \leq 1 + 2K$, so (S.6.1) implies

$$m_0(v), m_1(v) \geq \frac{\delta}{(1 + 2K)^2} \|v\|^2. \quad (\text{S.6.5})$$

On the other hand, $K^2 \|v\|^2 - \|a\|^2 \geq 0$ and

$$m_z(v) \geq -C_0 h(v), \quad C_0 = \frac{2K^4}{K^2 - 1}, \quad |z| \leq 1. \quad (\text{S.6.6})$$

Choose an integer $r \geq 2$ with $(r - 1)\kappa/2 \geq C_0$; explicitly,

$$r = 2 + \left\lceil \frac{32BK^4}{K^2 - 1} \right\rceil \quad (\text{S.6.7})$$

is sufficient. Combining half of (S.6.4) and half of (S.6.5) for each of $r - 1$ terms shows that, for arbitrary $\sigma_1, \dots, \sigma_{r-1} \in \{0, 1\}$,

$$m_z(v) + \sum_{a=1}^{r-1} m_{\sigma_a}(v) \geq \frac{(r-1)\delta}{2(1+2K)^2} \|v\|^2. \quad (\text{S.6.8})$$

Here is the precise replication. For each i , take an auxiliary copy of $\{\xi \in \mathbb{C}^r : \sum_a \xi_a = 0\}$. In the orthogonal sum of the original Hilbert space and these auxiliary spaces, define

$$\widehat{e}_{i,a} = r^{-1/2} e_i \oplus (\delta_a - r^{-1} \mathbf{1})_i, \quad 1 \leq a \leq r. \quad (\text{S.6.9})$$

These vectors form a basis, each has norm one, and the r copies of a fixed i are orthonormal. Order them first by i , then by a .

For a cut within the copies of i , there is an isometric realization in r orthogonal copies of the original space, together with the unaffected auxiliary spaces for $j \neq i$. In this realization, the components of a vector are

$$X_a = r^{-1/2}(v + c_a e_i), \quad \beta_i(v) = 0.$$

The same v occurs in every copy. This can be checked directly from (S.6.9): all coordinates $j \neq i$ enter the r original spaces equally, whereas the a th copy of i enters only the a th space. The zero/one/one-phase multiplier acts in these components as $r^{-1/2}(a + z_a c_a e_i)$, where one z_a is the active value z and every other z_a is zero or one. The remaining auxiliary spaces are multiplied by zero or one.

Its norm deficit is therefore the sum of $r^{-1} d_{z_a}(v, c_a)$ and the nonnegative auxiliary deficits. By (S.6.3) and (S.6.8), this is bounded below by

$$\frac{(r-1)\delta}{2r(1+2K)^2} \|v\|^2 + \frac{K^2 - 1}{r} \sum_a |c_a - c_{z_a}(v)|^2 + (K^2 - 1) \|\text{auxiliary part}\|^2.$$

Since $\|a\| \leq K \|v\|$, $|c_z(v)| \leq [K/(K-1)] \|v\|$ uniformly for $|z| \leq 1$. It follows that the norm squared of the input is at most a constant depending only on K times $\|v\|^2 + r^{-1} \sum_a |c_a - c_{z_a}(v)|^2 + \|\text{auxiliary part}\|^2$. Using $r \geq 2$ and $\delta \leq 1$ proves (a). One explicit value is

$$c_K = \min \left\{ \frac{K^2 - 1}{4}, \frac{1}{4(1+2K)^2(2+4K^2/(K-1)^2)} \right\}. \quad (\text{S.6.10})$$

Indeed the input norm squared is at most $(2+4K^2/(K-1)^2) \|v\|^2 + 4r^{-1} \sum_a |c_a - c_{z_a}(v)|^2$ plus the squared auxiliary norm.

Finally $r^{-1/2} \sum_a \widehat{e}_{i,a} = e_i \oplus 0$. A multiplier repeated on all copies leaves the original summand invariant and acts there as the original multiplier. This proves (b). $\square$

S.7 Rounded sector weights at the endpoint

Power-weighted trigonometric systems provide conditional Hilbert-space bases; see, for example, Ansorena [Ansorena]. The quantitative sector bound and the phase-margin modification needed here are proved directly below. Fix $K > 1$ for the remainder of the lower construction, and set

$$\alpha = a(K), \quad b = \frac{\pi\alpha}{2} = \arccos(1/K), \quad \nu = \frac{1-\alpha}{2}, \quad \eta = \frac{\alpha}{4}. \quad (\text{S.7.1})$$

On $\mathbb{D}$, use the principal powers of $1 - z$ and define the zero-free analytic function

$$h(z) = (1 - z)^{-\alpha/2} \exp\{\eta[(1 - z)^\nu - 1]\}. \quad (\text{S.7.2})$$

In particular $h(0) = 1$. The exponential factor and its reciprocal are bounded on $\mathbb{D}$, so $|h|^{\pm 2}$ are comparable, with K -dependent constants, to $|1 - z|^{\mp \alpha}$ on $\mathbb{T}$.

Lemma S.7.1 (Sector margin). *The boundary values of h satisfy $|2 \arg h| \leq b$. With $c = \cos b = 1/K$ and $\rho = \sin b$, the function*

$$d(\theta) = 2c(\cos(2 \arg h(e^{i\theta})) - c) \quad (\text{S.7.3})$$

is positive away from $\theta = 0$ and satisfies

$$d(\theta) \geq c_K |1 - e^{i\theta}|^\nu. \quad (\text{S.7.4})$$

Moreover d^{-1} , $|h|^2/d$, and $|h|^{-2}/d$ are integrable on $\mathbb{T}$.

Proof. For $0 < \theta < \pi$, write $\phi = (\pi - \theta)/2$ and $R = 2 \cos \phi = |1 - e^{i\theta}|$. Then

$$\arg h(e^{i\theta}) = \frac{\alpha}{2} \phi - \eta R^\nu \sin(\nu \phi).$$

The second term is nonnegative and is at most $\eta 2^\nu \nu \phi \leq \alpha \phi/4$, since $0 < \nu < 1/2$. Hence the argument lies between $\alpha \phi/4$ and $\alpha \phi/2$, strictly below $b/2$ away from zero. The other half-circle is obtained by complex conjugation. Near zero,

$$b - 2|\arg h(e^{i\theta})| = 2\eta \sin(\nu \pi/2) |\theta|^\nu + o(|\theta|^\nu).$$

Positivity and compactness away from zero now imply (S.7.4). The three integrability assertions follow from $\nu < 1$ and $\alpha + \nu = (1 + \alpha)/2 < 1$.

An explicit lower constant in (S.7.4) is available:

$$d_* := \frac{4c(1-c)\eta \sin(\nu \pi/4)}{b} > 0, \quad d(\theta) \geq d_* |1 - e^{i\theta}|^\nu. \quad (\text{S.7.5})$$

Indeed, with $m = b - 2|\arg h|$, concavity gives $\cos(b - m) - \cos b \geq (1 - c)m/b$. For $\phi \geq \pi/4$, $m \geq 2\eta R^\nu \sin(\nu \pi/4)$; for $\phi \leq \pi/4$, the term $\alpha(\pi/2 - \phi)$ gives the same lower bound. Thus compactness is not needed to specify the positive margin constant. $\square$

Let

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad d_0 = \frac{1}{2} \int_{\mathbb{T}} (|h|^2 + |h|^{-2}) dm,$$

and set

$$H(z) = d_0^{-1/2} U \begin{pmatrix} h(z) & 0 \\ 0 & h(z)^{-1} \end{pmatrix} U^*. \quad (\text{S.7.6})$$

For $n \geq 1$, let $\mathcal{H}_n$ be the space of $\mathbb{C}^2$ -valued polynomials of degree less than n , with Hilbert norm

$$\|p\|_{\mathcal{H}_n}^2 = \int_{\mathbb{T}} \|H(z)p(z)\|_{\mathbb{C}^2}^2 dm(z). \quad (\text{S.7.7})$$

Order its coefficient basis as $e_{0,1}, e_{0,2}, e_{1,1}, e_{1,2}, \dots, e_{n-1,1}, e_{n-1,2}$, where $e_{j,s}(z) = z^j \delta_s$. The choice of d_0 makes every basis vector have norm one.

Proposition S.7.2 (The endpoint basis). *The bases of $\mathcal{H}_n$ have the following properties, uniformly in n :*

- (i) *Every coefficient tail projection has norm at most K .*
- (ii) *The boundary-deficit estimate (S.6.2) holds with $B = B_K < \infty$.*
- (iii) *The strict deficit (S.6.1) holds with $\delta_n = c_K n^{-p}$, after decreasing c_K if necessary, where $p = 2\alpha + \nu$ and $0 < \delta_n \leq 1$.*
- (iv) *The diagonal multiplier which is $+1$ on $e_{j,1}$ and -1 on $e_{j,2}$ has norm at least $c_K n^\alpha$.*

Proof. We first prove the cut estimates. Shift a cut's boundary frequency to zero; multiplication by a power of z is an isometry in the weighted norm. Write $x = u + v$, where u is the tail part and v the initial part. The function u has positive Fourier coefficients and a specified subspace of its constant coefficient, while v has negative coefficients and the complementary constant subspace. Since $H(0) = d_0^{-1/2}I$, the functions

$$a = Hu, \quad b_- = \bar{H}v$$

are orthogonal in the ordinary vector-valued L^2 space: their nonconstant Fourier supports are disjoint, and their constant terms are orthogonal. Here $\bar{H}$ denotes the entrywise boundary conjugate; H is symmetric and diagonalized by the fixed real orthogonal matrix U .

The matrix $g = \bar{H}H^{-1}$ is unitary on the boundary. Its eigenvalues are $\bar{h}/h$ and $h/\bar{h}$. Therefore

$$(g - cI)^*(g - cI) = \rho^2 I - d(\theta)I. \quad (\text{S.7.8})$$

Also $\|u\|_{\mathcal{H}_n} = \|a\|_2$, $\|v\|_{\mathcal{H}_n} = \|b_-\|_2$, and $\langle u, v \rangle_{\mathcal{H}_n} = \langle ga, b_- \rangle_{L^2}$. Using $\langle a, b_- \rangle = 0$, completing the square gives the exact identity

$$K^2 \|x\|_{\mathcal{H}_n}^2 - \|u\|_{\mathcal{H}_n}^2 = K^2 \left[\|b_- + (g - cI)a\|_2^2 + \int_{\mathbb{T}} d(\theta) \|a(\theta)\|^2 dm \right]. \quad (\text{S.7.9})$$

This proves (i). It also implies

$$\int_{\mathbb{T}} d(\theta) (\|a\|^2 + \|b_-\|^2) dm \leq C_K \left(K^2 \|x\|_{\mathcal{H}_n}^2 - \|u\|_{\mathcal{H}_n}^2 \right), \quad (\text{S.7.10})$$

since $b_- = [b_- + (g - cI)a] - (g - cI)a$ and $d, g - cI$ are bounded.

For (ii), the boundary coefficient of x is a component of $\sqrt{d_0}(a_0 + (b_-)_0)$. Cauchy–Schwarz with weight d and $\int d^{-1} < \infty$ controls its square by (S.7.10). Also, for either boundary component e_s ,

$$|\langle u, e_s \rangle_{\mathcal{H}_n}|^2 = \left| \int_{\mathbb{T}} \langle a, He_s \rangle dm \right|^2 \leq \left(\int_{\mathbb{T}} d \|a\|^2 dm \right) \left(\int_{\mathbb{T}} \frac{\|He_s\|^2}{d} dm \right).$$

The last integral is finite by Lemma S.7.1. This proves (S.6.2) uniformly at either cut adjacent to the boundary coordinate. More explicitly, write

$$I_0 = \int_{\mathbb{T}} \frac{1}{d} dm, \quad I_1 = \int_{\mathbb{T}} \frac{|h|^2 + |h|^{-2}}{d} dm.$$

One may take

$$B_K = 1 + K^{-2} \left(8d_0 I_0 + \frac{I_1}{2d_0} \right). \quad (\text{S.7.11})$$

To check this constant, (S.7.9) implies $\int d(\|a\|^2 + \|b_-\|^2) dm \leq 4K^{-2}D$, where D denotes the cut deficit. Apply weighted Cauchy–Schwarz to $a_0 + (b_-)_0$ and use $\|He_s\|^2 = (|h|^2 + |h|^{-2})/(2d_0)$.

For (iii), we record an elementary finite-degree estimate. If a vector-valued Laurent polynomial q has at most n consecutive Fourier modes, then $\|q\|_{\infty}^2 \leq n \|q\|_2^2$. At least half of its L^2 mass is outside an interval of length c/n around zero. Consequently, for $s > 0$,

$$\int_{\mathbb{T}} |1 - z|^s \|q\|^2 dm \geq c_s n^{-s} \|q\|_2^2.$$

Splitting at $|\theta| = 1/n$ similarly gives

$$\int_{\mathbb{T}} |1 - z|^{-\alpha} \|q\|^2 dm \leq C_{\alpha} n^{\alpha} \|q\|_2^2.$$

The smallest squared singular value of H is at least $c_K|1 - z|^\alpha$, and the largest is at most $C_K|1 - z|^{-\alpha}$. Together with (S.7.4), these inequalities imply

$$\int d \|Hq\|^2 dm \geq c_K n^{-(2\alpha+\nu)} \|q\|_{\mathcal{H}_n}^2.$$

The same is true with $\bar{H}$. Apply it to u, v in (S.7.10), and use $\|x\|^2 \leq 2(\|u\|^2 + \|v\|^2)$. For example, set

$$C_\alpha = (\pi/2)^\alpha \left(1 + \frac{1}{\pi(1-\alpha)}\right), \quad \delta_n = \min \left(1, K^2 - 1, \frac{K^2 d_* e^{-4\eta} 2^{-4\alpha}}{16C_\alpha}\right) n^{-(2\alpha+\nu)}.$$

This is a valid explicit choice: the lower weighted moment estimate has constant $1/2$ by discarding $|\theta| < \pi/(2n)$, and the upper moment estimate follows by splitting at $|\theta| = 1/n$.

Finally, in the U -coordinates the multiplier in (iv) interchanges the two components. Let $D_n(z) = \sum_{j=0}^{n-1} z^j$. Applied to the vector with only its second U -component equal to D_n , its amplification is

$$\frac{\|hD_n\|_2}{\|h^{-1}D_n\|_2} \geq c_K n^\alpha.$$

Indeed $|h|^{\pm 2} \asymp_K |1 - z|^{\mp \alpha}$, $|D_n| \geq cn$ for $|\theta| \leq 1/n$, and $|D_n| \leq \min(n, C/|\theta|)$. These give the numerator squared at least $c_K n^{1+\alpha}$ and the denominator squared at most $C_K n^{1-\alpha}$. $\square$

Remark S.7.3 (Why the bounded analytic correction matters). Replacing (S.7.2) by the pure power $(1 - z)^{-\alpha/2}$ preserves the growth n^α , but the elementary sector deficit then vanishes only linearly at zero. The integral $\int |h|^2/d$ is no longer finite. The bounded analytic exponential in (S.7.2) changes that margin to $|\theta|^\nu$ without changing either the endpoint sector angle b or the power singularity α . This is the additional term that makes the exact fixed- K lower construction possible in this proof.

S.8 Realization by exponentially stable generators

We now set $K = M > 1$. Apply Lemma S.6.1 to Proposition S.7.2. The replication number $r = r(M)$ is fixed independently of n . We obtain a normalized basis $f_1, \dots, f_N$, $N = 2rn$, such that every one-active-phase multiplier has squared norm at most

$$M^2 - \gamma_n, \quad \gamma_n = c_M n^{-p}, \quad p = 2\alpha + \nu, \quad \alpha = a(M). \quad (\text{S.8.1})$$

We decrease c_M so that $0 < \gamma_n \leq \min(1, M^2 - 1)$. There is a sign multiplier S_n in this basis, with the sign repeated on all replicas of an original coordinate, such that

$$\|S_n\| \geq c_M n^\alpha. \quad (\text{S.8.2})$$

The coordinate rank-one projections E_i have norm at most $2M$, because ordinary tails are among the one-active-phase multipliers.

The matrices can also be stated in the usual Euclidean norm. Let G be the Gram matrix of the coefficient basis in (S.7.7). The Gram matrix of the replicated basis is explicitly

$$\hat{G}_{(i,a),(j,b)} = \delta_{ij} \delta_{ab} + \frac{G_{ij} - \delta_{ij}}{r}. \quad (\text{S.8.3})$$

Its positivity also follows from the orthogonal construction (S.6.9). If a generator has coordinate matrix D in that basis, its ordinary Euclidean realization is $\hat{G}^{1/2} D \hat{G}^{-1/2}$. Thus no change of the problem's Hilbert-space operator norm is hidden in the construction.

S.8.1 Quadratically separated scalar eigenvalues

Choose

$$\varepsilon_n = \frac{\gamma_n}{100M^2}, \quad \kappa_n = \max \left(2, \left\lceil \varepsilon_n^{-1} \log \frac{N}{\varepsilon_n} \right\rceil \right). \quad (\text{S.8.4})$$

Thus $N \exp(-\varepsilon_n \kappa_n) \leq \varepsilon_n$ and $\kappa_n \leq C_M n^p \log(n+e)$. Choose integers y_i with parity prescribed by the sign of S_n : $(-1)^{y_i}$ equals that sign, $2 \leq y_1 \leq 3$, and

$$\kappa_n(1 + y_i^2) \leq y_{i+1} \leq \kappa_n(1 + y_i^2) + 2. \quad (\text{S.8.5})$$

There is an integer of either parity in each indicated interval. Define

$$\mu_i = -1 + iy_i, \quad \lambda_i = \mu_i^{-1}, \quad C_n f_i = \lambda_i f_i. \quad (\text{S.8.6})$$

The quadratic separation reflects the decay time $1 + y_i^2$ of the i th scalar exponential; a cubic separation would work but is unnecessary.

Lemma S.8.1 (Uniform forward bound). *The finite-dimensional generator C_n satisfies*

$$\sup_{s \geq 0} \|e^{sC_n}\| \leq M. \quad (\text{S.8.7})$$

Proof. For $s < \varepsilon_n y_1$, every coefficient is close to one:

$$\sum_i |e^{s\lambda_i} - 1| \leq s \sum_i |\lambda_i| \leq 2\varepsilon_n.$$

For $s \geq \varepsilon_n y_1$, choose m so that $\varepsilon_n y_m \leq s < \varepsilon_n y_{m+1}$, with $m = N$ allowed. For $i < m$, use $\text{Re } \lambda_i = -(1 + y_i^2)^{-1}$ and $y_m \geq y_{i+1} \geq \kappa_n(1 + y_i^2)$ to get

$$\sum_{i < m} |e^{s\lambda_i}| \leq N \exp(-\varepsilon_n \kappa_n) \leq \varepsilon_n.$$

For $i > m$, $\text{Re } \lambda_i < 0$ and the fundamental theorem of calculus give $|e^{s\lambda_i} - 1| \leq s|\lambda_i|$. By (S.8.5), $y_{i+1} \geq 2y_i$, and hence

$$\sum_{i > m} |e^{s\lambda_i} - 1| \leq \varepsilon_n y_{m+1} \sum_{i > m} y_i^{-1} \leq 2\varepsilon_n.$$

Thus e^{sC_n} differs, in the sum of the absolute diagonal coefficients, by at most $3\varepsilon_n$ from the multiplier which is zero before m , is $z = e^{s\lambda_m}$ at m , and one after m . Its operator norm error is at most $6M\varepsilon_n$.

By (S.8.1), the model's norm is at most $\sqrt{M^2 - \gamma_n} \leq M - \gamma_n/(2M)$. The chosen ε_n makes $6M\varepsilon_n$ smaller than $\gamma_n/(2M)$. The initial model is I , whose deficit is at least as large because $\gamma_n \leq M^2 - 1$. The result follows for all $s \geq 0$. $\square$

S.8.2 Normalization and the inverse observation time

Set

$$\omega_n = \frac{1}{100\pi MN(1 + y_N^2)}, \quad A_n = \omega_n^{-1} C_n - I, \quad t_n = \frac{\pi}{\omega_n}. \quad (\text{S.8.8})$$

Then $\|e^{sA_n}\| \leq M e^{-s}$, so $A_n \in \mathcal{A}_M$. Moreover

$$t_n A_n^{-1} = \pi(C_n - \omega_n I)^{-1}.$$

For each eigenvalue,

$$\left| \frac{1}{\lambda_i - \omega_n} - \mu_i \right| = \left| \frac{\omega_n \mu_i^2}{1 - \omega_n \mu_i} \right| \leq \omega_n(1 + y_N^2),$$

because $|1 - \omega_n \mu_i| \geq 1$. Both numbers being exponentiated have negative real part, so integration along their joining segment gives

$$\left| \exp\left(\frac{\pi}{\lambda_i - \omega_n}\right) - e^{\pi \mu_i} \right| \leq \pi \omega_n (1 + y_N^2).$$

Summing with $\|E_i\| \leq 2M$ yields

$$\left\| e^{t_n A_n^{-1}} - e^{-\pi} S_n \right\| \leq \frac{1}{50}. \quad (\text{S.8.9})$$

Thus, for all sufficiently large n ,

$$\left\| e^{t_n A_n^{-1}} \right\| \geq c_M n^\alpha. \quad (\text{S.8.10})$$

The time cost is exactly of the required double-exponential size. Because $y_i \geq 2$ and $\kappa_n \geq 2$, (S.8.5) gives $\kappa_n y_i^2 \leq y_{i+1} \leq 2\kappa_n y_i^2$. Consequently,

$$\log y_N \leq 2^{N-1} (\log y_1 + \log(2\kappa_n)).$$

Since $N = 2rn$, r is fixed, and κ_n is polynomial in n up to a logarithm, it follows that

$$L(t_n) \leq C_M n. \quad (\text{S.8.11})$$

In fact the same inequalities from above and below show the more precise relation

$$L(t_n) = 2rn \log 2 + \log \log n + \log p + o_M(1) \quad (n \rightarrow \infty). \quad (\text{S.8.12})$$

Indeed $\log \kappa_n = p \log n + \log \log n + O_M(1)$ and $\log y_N = 2^{N-1} (\log \kappa_n + O(1))$; the fixed $y_1 \in \{2, 3\}$ is absorbed into the latter $O(1)$ term. The coarse bound (S.8.11) alone is sufficient for the theorem.

For completeness, G_M is nondecreasing: if $c \geq 1$ and $A \in \mathcal{A}_M$, then $cA \in \mathcal{A}_M$ and $e^{t(cA)^{-1}} = e^{(t/c)A^{-1}}$. Given a large t , choose n proportional to $L(t)$ with a sufficiently small M -dependent proportionality constant. Equation (S.8.11) ensures $t_n \leq t$, and scaling A_n by t/t_n transfers (S.8.10) to time t . Therefore

$$G_M(t) \geq c_M L(t)^\alpha,$$

which completes the lower bound and the proof of Theorem S.1.1 for $M > 1$.

S.9 Endpoints, consequences, and scope

If $M = 1$, the generator A is dissipative, as is its bounded inverse: for $x = Ay$, $\text{Re}\langle A^{-1}x, x \rangle = \text{Re}\langle y, Ay \rangle \leq 0$. Hence $\left\| e^{tA^{-1}} \right\| \leq 1$. Scalar generators $A = -aI$, $a \geq 1$, show that the supremum equals one by letting $a \rightarrow \infty$.

The exponent in the main theorem has the limits and expansions

$$a(1) = 0, \quad a(M) \uparrow 1, \\ a(M) \sim \frac{2\sqrt{2}}{\pi} \sqrt{M-1} \quad (M \downarrow 1), \quad a(M) = 1 - \frac{2}{\pi M} - \frac{1}{3\pi M^3} + O(M^{-5}) \quad (M \rightarrow \infty).$$

For example,

$$G_{\sqrt{2}}(t) \asymp \sqrt{L(t)}, \quad G_2(t) \asymp L(t)^{2/3}.$$

These examples concern fixed forward-bound classes; the implicit constants are different for the different values of M .

Corollary S.9.1 (The underlying finite-dimensional profile). *For $K > 1$, let $\mathcal{U}_K(n)$ be the supremum of the norms of diagonal multipliers with coefficient moduli at most one, over all normalized n -dimensional Hilbert-space bases whose initial coordinate projections have norms at most K . Then*

$$\mathcal{U}_K(n) \asymp_K n^{a(K)} \quad (n \rightarrow \infty).$$

Proof. For such a basis let E_j be the rank-one coordinate projections; $\|E_j\| \leq 2K$. The families E_j and $\theta_j E_j$, $|\theta_j| \leq 1$, and their adjoints have exact cuts of norm at most K . Apply Lemma S.3.2 with zero error and let $\varepsilon \downarrow 0$. Since $E_j^2 = E_j$, the upper conclusion is $C_K n^{a(K)}$. Proposition S.7.2(iv) supplies the lower bound in even dimensions. Append an orthogonal unit vector to cover odd dimensions. $\square$

This profile explains the common exponent in the semigroup theorem. The upper proof reduces the oscillatory sum to almost-separated products; the lower proof realizes highly conditional multipliers through scalar semigroup transitions. The replication step is needed for the latter realization, not for the finite-dimensional profile by itself.

Corollary S.9.2 (A single generator realizing the lower rate). *For every $M > 1$, there is a separable Hilbert space and a single generator $A \in \mathcal{A}_M$ for which $\|e^{tA^{-1}}\| \asymp_M L(t)^{a(M)}$ for all sufficiently large t .*

Proof. Take the orthogonal direct sum of the finite blocks qA_n over all n and all rational $q \geq 1$. Their forward bounds are uniformly Me^{-s} , so the direct sum is a strongly continuous semigroup with that bound; strong continuity follows first on finite-support vectors and then by uniform boundedness. Its generator has a bounded inverse by exponential stability. For $t \geq t_n$, rational q can approximate t/t_n , and continuity in this scalar parameter makes the supremum of the inverse block norms at time t at least the norm in (S.8.10). The upper bound follows from Theorem S.1.1. $\square$

S.9.1 Audit of the principal losses in the earlier argument

Earlier step	Replacement or check	Effect
Leading Bessel asymptotic	The next remainder is already $O(u^{-5/4})$ and integrable.	Further Bessel terms change a bounded contribution, not the asymptotic exponent.
Triangle sum of oscillatory blocks	All-cut Hardy-space synthesis and localization on both sides of each block.	Produces the exact angle parameter $a(M)$ instead of exponent one.
One binary split and a safety factor in the angle	Stationarize the entire finite Gram matrix; use the exact cut angle $\sqrt{1 - M^{-2}}$.	Avoids an artificial M^{-2} exponent deficit.
Triangle estimate on an augmented block matrix	Use the exact scalar triangular norm (S.3.2).	The loss is quadratic in the augmentation error, not linear.
Fixed error at every adjacent cut	Split the block indices into eight residue classes and discard a very short initial segment.	Removes the otherwise surviving endpoint logarithmic factor.
Crude lower-bound basis constants	Use sector weights with $K = \sec(\pi\alpha/2)$.	Matches the upper parameter exactly.
Pure power lower weight	Multiply by the bounded analytic exponential in (S.7.2).	Creates an integrable boundary deficit without changing the power growth.

Earlier step	Replacement or check	Effect
A one-coordinate complex phase	Orthogonal replication and the exact quadratic minimization (S.6.3).	Maintains the exact forward bound M through the transition.
Asymptotic scalar semigroup approximation	Retain a polynomial finite-size norm margin and choose the frequency separation accordingly.	Gives a genuine all-times semigroup estimate, not a sampled-time check.

Constants and verification. The theorem asserts M -dependent multiplicative constants, as permitted by MF-17. They are not claimed to have the polynomial dependence appearing in the earlier, weaker upper bounds. All uniformities needed for the fixed- M conclusion are proved above. The independent mathematical verification of this research proof remains a separate task; floating-point tests are not being substituted for any lemma.

S.A Proof of the oscillatory-window estimate

This appendix reproduces the analytic input of the previous report in a self-contained form. It is the continuous shifted-window factorization related to the three-factor multiplier argument in Pisier [Pisier].

For $N > 0$ put

$$g_N(s) = N^{-1/2} \mathbf{1}_{[N/2, 3N/2]}(s), \quad h_N = g_N * g_N.$$

Then $\|g_N\|_2 = 1$, $h_N(v) = (1 - |v/N - 2|)_+$, and

$$\sum_{j \geq 0} h_N(u - jN) = 1 \quad (u \geq 2N). \quad (\text{S.A.1})$$

For $\phi(u) = \langle T(u)x, y \rangle$, define

$$F_k(\theta) = \int_0^\infty h_N(v) \phi(v + k) e^{-i\theta v} dv, \quad k \geq 0.$$

With $U(\theta) = \int_0^\infty g_N(s) e^{-i\theta s} T(s) ds$, the semigroup law gives

$$F_k(\theta) = \langle T(k)U(\theta)x, U(\theta)^*y \rangle.$$

Hilbert-valued Plancherel and Cauchy–Schwarz imply

$$\int_{\mathbb{R}} \sup_{k \in E} |F_k(\theta)| \frac{d\theta}{2\pi} \leq M^3 \|x\| \|y\| \quad (\text{S.A.2})$$

for every countable $E \subset [0, \infty)$. Indeed each of $\int \|U(\theta)x\|^2 d\theta/(2\pi)$ and its adjoint analogue is at most M^2 times the squared norm of the corresponding vector.

Fix $\psi \in C_c^\infty((1/2, 7/2))$ equal to one on $[1, 3]$. For a smooth compactly supported kernel K vanishing below $2N$, put

$$b_j(\theta) = \int_{\mathbb{R}} K(v + jN) \psi(v/N) e^{i\theta v} dv.$$

Fourier inversion, (S.A.1), and (S.A.2) give

$$\left\| \int_0^\infty K(u) T(u) du \right\| \leq M^3 \sup_{\theta \in \mathbb{R}} \sum_{j \geq 0} |b_j(\theta)|. \quad (\text{S.A.3})$$

Only finitely many terms occur; their Fourier transforms are rapidly decreasing.

Take $K(u) = a(u)e^{2i\sqrt{u}}$ as in Lemma S.4.1 and first normalize $A = 1$. The j th window has $U_j = (j+2)N$ and dimensionless curvature

$$\Lambda_j = N^2 U_j^{-3/2}.$$

For every nonzero window, $U_j \leq CN^{4/3}$, hence $\Lambda_j \geq c > 0$. After $v = Nw$, the phase is $2\sqrt{N(j+w)} + N\theta w$, whose second derivative has absolute value comparable to Λ_j and constant sign on $1/2 \leq w \leq 7/2$. The amplitude and its first two derivatives are bounded by $CNU_j^{-3/4}$. For completeness, the second-derivative estimate follows by splitting at $|\phi'| \leq \sqrt{\Lambda_j}$. Monotonicity of ϕ' makes that interval have length at most $C\Lambda_j^{-1/2}$. On its complement, one integration by parts bounds the integral by $C\Lambda_j^{-1/2}$ times the amplitude's supremum plus its total variation. In the latter estimate, $\int |\phi''|/|\phi'|^2$ is bounded by $C\Lambda_j^{-1/2}$ by the substitution $\xi = \phi'$. Thus

$$|b_j(\theta)| \leq CNU_j^{-3/4} \Lambda_j^{-1/2} = C.$$

The stationary-frequency interval is

$$I_j = -N^{-1/2}[(j+7/2)^{-1/2}, (j+1/2)^{-1/2}].$$

Away from this interval, two integrations by parts, using $|\phi'| \geq N \text{dist}(\theta, I_j)$ and the corresponding second and third phase derivative bounds, give

$$|b_j(\theta)| \leq C \left(1 + \frac{\text{dist}(\theta, I_j)}{NU_j^{-3/2}} \right)^{-2}. \quad (\text{S.A.4})$$

For explicit scale bookkeeping, if A is the scaled amplitude and $p = \phi'$, the twice-integrated amplitude, apart from a factor of modulus one, is

$$-\frac{A''}{p^2} + \frac{3A'p'}{p^3} + \frac{Ap''}{p^3} - \frac{3A(p')^2}{p^4}.$$

There are no boundary terms because the cutoff is smooth and compactly supported. Put $s = \text{dist}(\theta, I_j)/(NU_j^{-3/2})$. When $s \geq 1$, the twice-integrated terms are bounded by $C\Lambda_j^{1/2}\Lambda_j^{-2}(s^{-2} + s^{-3} + s^{-4})$, which is at most Cs^{-2} because $\Lambda_j \geq c$.

These tails have uniformly bounded overlap. After scaling $\zeta = -\theta\sqrt{N}$, their sum is bounded by

$$C \sum_{j \geq 0} \left(1 + (j+2)^{3/2} \text{dist}(\zeta, [(j+7/2)^{-1/2}, (j+1/2)^{-1/2}]) \right)^{-2}.$$

For $\zeta \leq 0$, this is bounded by $C \sum (j+2)^{-2}$. For $\zeta > 0$, put $R = \zeta^{-2}$. Indices $j+2 \leq R/8$ contribute $C(j+2)^{-2}$; those with $j+2 \geq 8R$ contribute $CR(j+2)^{-3}$. For $R/8 < j+2 < 8R$, the mean value theorem identifies the normalized distance with the distance of R from an interval of fixed length centered at j ; the resulting inverse-square lattice sum is uniformly bounded. When R is bounded, the finitely many small indices are separated off and the same tail estimate applies. Thus $\sup_{\theta} \sum_j |b_j(\theta)| \leq C$. Equation (S.A.3) proves Lemma S.4.1. The negative phase is identical with the frequency sign reversed.

S.A.1 Arbitrarily high-order primitives

The early-frequency estimate (S.4.6) can be sharpened to any fixed order. For the positive phase let $\varphi(u) = 2\sqrt{u}$ and define amplitudes

$$h_0(u) = \frac{a_{q,t}(u)}{i\varphi'(u)}, \quad h_{j+1}(u) = \frac{h'_j(u)}{i\varphi'(u)}.$$

For each integer $P \geq 1$, direct differentiation gives the exact identity

$$a_{q,t}(u)e^{i\varphi(u)} = \left[e^{i\varphi(u)} \sum_{j=0}^{P-1} (-1)^j h_j(u) \right]' + (-1)^P h'_{P-1}(u)e^{i\varphi(u)}.$$

The symbol estimates (S.4.4) imply $|h_j^{(\ell)}(u)| \leq C_{j,\ell} e^{-N_q/t} u^{-1/4-j/2-\ell}$. Applying the same proof as Lemma S.4.2 therefore yields

$$\|R_r B_q^\pm\| \leq C_P M^4 e^{-N_q/t} \frac{r}{\lambda_{q+1}} + C_P M^2 e^{-N_q/t} N_q^{1/4-P/2}. \quad (\text{S.A.5})$$

Thus the residual $N_q^{-1/4}$ in the first-order calculation is not an intrinsic limitation. However, it is already summable and already decays super-exponentially in the block index q . Higher orders improve constants and how soon a tail has a specified error, not the final exponent $a(M)$.

S.A.2 Why a single stationary window cannot yield a power saving

The scalar bound on an individual b_j is of the correct scale. Choose $\xi \in C_c^\infty((2,3))$, $\xi(w_0) > 0$, and set $a_N(u) = u^{-3/4} \xi(u/N)$, $\theta_N = -N^{-1/2} w_0^{-1/2}$. For $j = 0$, the cutoff ψ is identically one on the support, so

$$b_0(\theta_N) = N^{1/4} \int w^{-3/4} \xi(w) \exp\{i\sqrt{N}(2\sqrt{w} - w/\sqrt{w_0})\} dw.$$

There is one nondegenerate stationary point at w_0 , with second derivative $-\frac{1}{2}w_0^{-3/2}$. The change of scale $w - w_0 = N^{-1/4}v$, Taylor expansion of the phase there, and integration by parts outside a shrinking neighborhood give the Fresnel limit

$$|b_0(\theta_N)| \longrightarrow 2\sqrt{\pi} \xi(w_0) > 0.$$

Equivalently this is the leading nondegenerate stationary-phase calculation. Consequently the uniform $O(1)$ window bound cannot be replaced by $o(1)$. The power saving in the theorem genuinely comes from joint Hilbert-space geometry across the windows, not from a smaller estimate for each window.

S.B Higher-order checks on the lower construction

The leading exponent is determined by the cut angle, but higher-order scalar terms can also be retained. They confirm that the analytic rounding does not introduce a hidden logarithmic loss into the lower construction.

For $-1 < s < 1$, let

$$F_n(s) = \int_{\mathbb{T}} |D_n(z)|^2 |1-z|^s dm(z), \quad c_0(s) = \frac{\Gamma(1+s)}{\Gamma(1+s/2)^2}, \quad I_s = \frac{1}{\Gamma(2-s) \cos(\pi s/2)}.$$

There is an exact expression

$$F_n(s) = I_s \frac{\Gamma(n+1-s/2)}{\Gamma(n+s/2)} - \frac{s c_0(s)}{2(1-s)}. \quad (\text{S.B.1})$$

To verify it, the Fourier coefficients of $|1-z|^s$ are $c_k = c_0(s)(-s/2)_k/(1+s/2)_k$ for $k \geq 0$, with $c_{-k} = c_k$. Their formula follows from the beta integral for c_0 , integration by parts, and the recurrence

$$2kc_k = (k-1-s/2)c_{k-1} + (k+1+s/2)c_{k+1}, \quad c_1 = -\frac{s}{s+2}c_0.$$

Integration by parts is legitimate for all $s > -1$ after multiplication by $1 - \cos \theta$, which makes its endpoint boundary term vanish. Now $F_n = nc_0 + 2 \sum_{k=1}^{n-1} (n-k)c_k$ and

$$F_{n+1} - F_n = c_0 \frac{(1-s/2)_n}{(1+s/2)_n}.$$

The right side of (S.B.1) has this difference and vanishes at $n = 0$ (with the removable case $s = 0$ read by continuity). The reflection formula for Γ identifies its prefactor as I_s .

The gamma quotient has the expansion

$$\begin{aligned} \frac{\Gamma(n+1-s/2)}{\Gamma(n+s/2)} &= n^{1-s} \left[1 + \frac{s(1-s)(2-s)}{24n^2} \right. \\ &\quad \left. + \frac{s(s-2)(s-1)(s+1)(s+2)(5s-12)}{5760n^4} + O_s(n^{-6}) \right]. \end{aligned} \quad (\text{S.B.2})$$

There is no n^{-1} term: the two gamma parameters sum to one. The coefficients follow by subtracting the Stirling expansions for the logarithms; the odd inverse powers vanish by the symmetry of the Bernoulli polynomials. Notice that the separate constant term in (S.B.1) must still be retained. At the borderline $s = 1$, direct summation instead gives

$$F_n(1) = \frac{4}{\pi} \sum_{j=0}^{n-1} \frac{1}{2j+1} = \frac{2}{\pi} \log n + O(1).$$

For the rounded weight in (S.7.2), put $q_\nu = \cos(\pi\nu/2)$. As $z \rightarrow 1$ along $\mathbb{T}$,

$$|h(z)|^{\pm 2} = e^{\mp 2\eta} |1-z|^{\mp \alpha} [1 \pm 2\eta q_\nu |1-z|^\nu + O_M(|1-z|^{2\nu})].$$

The same remainder bound holds globally on $\mathbb{T}$ after increasing its constant. Use (S.B.1) on the first two terms. The numerator remainder is bounded by $C_M F_n(-\alpha + 2\nu)$, and the denominator remainder by $C_M F_n(\alpha + 2\nu) = C_M F_n(1)$. It follows that the explicit Dirichlet witness satisfies

$$\begin{aligned} \frac{\|hD_n\|_2}{\|h^{-1}D_n\|_2} &= A_M n^\alpha \left[1 + B_M n^{-\nu} + O_M(n^{-2\nu} \log n) \right], \\ A_M &= e^{-2\eta} \sqrt{\frac{\Gamma(2-\alpha)}{\Gamma(2+\alpha)}}, \\ B_M &= \eta \cos(\pi\nu/2) \left(\frac{I_{-\alpha+\nu}}{I_{-\alpha}} + \frac{I_{\alpha+\nu}}{I_\alpha} \right) > 0. \end{aligned} \quad (\text{S.B.3})$$

This is a higher-order formula for one *specific witness*, not a claim that the leading multiplicative constant in G_M itself has been determined. In particular, the correction does not alter the sharp exponent $\alpha = a(M)$. For α close to one, ν is small and this convergence can be very slow; finite-dimensional experiments should not be fitted to the leading term alone.

S.B.1 The inverse-normalization error can be made arbitrarily small

The constant error in (S.8.9) is not a limitation. For any fixed $q > 0$, replace ω_n in (S.8.8) by $n^{-q}\omega_n$. The same proof gives an error at most $n^{-q}/50$. This changes $\log t_n$ by $q \log n$, which is negligible compared with $\log t_n \asymp_M 2^{2rn} \log n$; even the expansion (S.8.12) is unchanged.

The attenuation factor $e^{-\pi}$ can also be removed asymptotically. In the lower construction take $\mu_i = -d_n + iy_i$, $d_n = n^{-1}$, and choose

$$\frac{\kappa_n}{d_n} (d_n^2 + y_i^2) \leq y_{i+1} \leq \frac{\kappa_n}{d_n} (d_n^2 + y_i^2) + 2$$

with the same parity prescription. The earlier-coordinate decay estimate is unchanged, because the decay time is now $(d_n^2 + y_i^2)/d_n$. The future-coordinate estimate still uses $|\mu_i^{-1}| \leq y_i^{-1}$. Choose

$$\omega_n = [100\pi MNn^2(d_n^2 + y_N^2)]^{-1}, \quad A_n = \omega_n^{-1}C_n - I, \quad t_n = \pi/\omega_n.$$

Then the uniform forward bound remains Me^{-s} and

$$\left\| e^{t_n A_n^{-1}} - e^{-\pi/n} S_n \right\| \leq \frac{1}{50n^2}.$$

The frequency prefactor now has logarithm $(p+1)\log n + \log \log n + O_M(1)$, so

$$L(t_n) = 2rn \log 2 + \log \log n + \log(p+1) + o_M(1).$$

Combining this with (S.B.3) and monotonicity of G_M gives the explicit, non-optimal prefactor bound

$$\liminf_{t \rightarrow \infty} \frac{G_M(t)}{L(t)^{a(M)}} \geq \frac{A_M}{(2r \log 2)^{a(M)}} > 0. \quad (\text{S.B.4})$$

Indeed the successive values of $L(t_n)$ differ by $2r \log 2 + o_M(1)$, so the largest n with $t_n \leq t$ is asymptotic to $L(t)/(2r \log 2)$. This bound is for the explicit witness family only and does not identify the optimal leading constant of the envelope.

S.C An exact check for real-Gram phase models

The following observation is useful when checking the finite examples. It does not replace the dimension-independent replication proof.

Lemma S.C.1. *Let $e_1, \dots, e_m$ have a real positive-definite Gram matrix, and let $T_i(z)$ be the diagonal multiplier with values zero before i , z at i , and one after i . On the complexified Hilbert space,*

$$\sup_{|z| \leq 1} \|T_i(z)\| = \max\{\|T_i(1)\|, \|T_i(-1)\|\}.$$

Proof. Minimize the squared input norm over the coordinates preceding i , which make no contribution to the output. The resulting input Gram matrix is the real Schur complement S on the remaining coordinates, whereas the output uses the corresponding principal submatrix H of the original Gram matrix. Thus the squared multiplier norm is the largest generalized eigenvalue of $(D_z^* H D_z, S)$, where $D_z = \text{diag}(z, I)$.

Take q strictly larger than both endpoint squared norms. For $|z| = 1$, the lower-right block of $qS - D_z^* H D_z$ is the positive-definite real matrix $B = qS_{FF} - H_{FF}$. Its scalar Schur complement is

$$qS_{ii} - H_{ii} - (qS_{iF} - \bar{z}H_{iF})B^{-1}(qS_{Fi} - zH_{Fi}).$$

Since every matrix entry is real, this is affine in $\text{Re } z$. It is positive at $z = 1$ and $z = -1$, hence throughout the circle. This proves the bound for $|z| = 1$; convexity of the norm of the affine map $T_i(z)$ proves it inside the disk. Let q decrease to the maximum endpoint squared norm. If the future index set is empty, the same assertion is immediate without a Schur complement. $\square$

S.D Numerical diagnostics and reproducibility

The companion archive contains scripts testing finite Gram matrices, coordinate cuts, and one-active-phase models for the rounded-sector weight with the final choice $\eta = \alpha/4$. Further scripts check the replication isometry, the exact quadratic and resolvent identities, stationarization by Parseval, and the higher-order moments in Appendix S.B. They use floating-point linear algebra

and numerical quadrature and are explicitly not interval certificates. The proof above does not depend on their output.

For larger Fourier-Gram matrices, a single adaptive endpoint quadrature produced round-off warnings and inaccurate high-mode coefficients. Those large-size outputs are not used. The final composite rule resolves the oscillations interval by interval and uses $\theta = (\pi/n)u^{1/\nu}$ at the singular endpoint. Its results are compared at two quadrature orders and against the separately evaluated Dirichlet moments. Agreement of these floating-point calculations is a diagnostic, not a certified error bound.

The archive retains the algebraic checks and the source of this report. Previously generated finite-matrix certificates are separate results; they are not used to extrapolate an asymptotic lower bound here.

Part VII

Constants and threshold bookkeeping

Source and scope. Transcription and synthesis of the constants PDF, JSON ledger, and threshold discussion.

The complete original Q PDF is bundled; its full LaTeX source is not mounted. This part preserves the formulas and their distinct roles. The sharper eventual constructions in O and T do not retain these all-time constants.

Q.1 All-time constants: dependency recipe from source Q

This part transcribes and organizes the quantitative formulas in source Q and its JSON ledger. The complete original 20-page PDF and the ledger are bundled. The present part is not a verbatim reproduction of that PDF, and it does not supply a new independent audit. Its purpose is to prevent the multiple occurrences of c_M, C_M, t_M from being conflated. Later parts O and T improve eventual coefficients and do not need to retain the old enormous numerical package.

Q.1.1 First occurrences and distinct roles

In source S, the convention permitting c_M, C_M to change appears before Theorem 1.1. The theorem first introduces the final t_M . The first proof occurrence of C_M is the augmentation estimate (3.1), on page 4, source TeX line 230. It is not yet the final upper coefficient.

The first literal c_M in the lower construction occurs in (8.1), page 12, source line 883: it is a phase-margin coefficient. The next display (8.2), source line 890, uses the same symbol for sign amplification. The earlier c_K in the replication lemma is a different auxiliary coefficient. The first threshold-producing phrase is “for all sufficiently large n ” before (8.10), page 14, source line 1003. Source S does not then define a separate numerical t_M .

The ledger gives an explicit choice for each of these quantities. Its recorded final all-time bounds, conditional on the analytic proof, are

$$2^{-32}M^{-3}L(t)^\alpha \leq G_M(t) \leq 2^{154} \max\{M^7, (M-1)^{-3/2}\}L(t)^\alpha \quad (M > 1, t \geq 0).$$

At $M = 2$, a tighter admissible rounded pair is 10^{-6} and $2 \cdot 10^{45}$, still with threshold zero. None of these is presented as an optimal prefactor.

Q.1.2 Upper dependency chain

Set

$$g_M^2 = \frac{M}{2(M^2 - 1)}.$$

Q first replaces the unspecified synthesis constant by

$$\left\| \sum_{j=1}^n u_j \right\| \leq 12K^{3/2}B n^{(1+a(K))/2}.$$

It comes from the scalar Poisson coefficient $4K^3$, the Bernstein comparison $40\pi K^3$, and $\sqrt{40\pi} < 12$. The exact triangular augmentation then gives

$$\left\| \sum_j u_j \right\| \leq 12(M+1)^{3/2} \sqrt{B^2 + g_M^2 D^2 / \varepsilon} n^{(1+a(M+\varepsilon))/2}.$$

Thus the first proof C_M may be chosen as

$$C_M^{\text{aug}} = 12(M+1)^{3/2} \max\{1, g_M\}.$$

The unrounded square-root form is used subsequently. Two-sided summation gives the coefficient

$$\frac{144(M+1)^3}{\alpha} \sqrt{B_X^2 + g_M^2} \sqrt{B_Y^2 + g_M^2}.$$

Q fixes a universal package $U_0 = 2^{128}$. It is a recorded upper bound derived there from fixed cutoff derivatives, a scalar window constant 2^{60} , a block-symbol constant 2^{40} , and ratio sums. It is not a mathematically optimized number. The precise fixed cutoffs and their elementary derivative estimates are in Q Section 5. Their role is to make every hidden absolute constant finite and specified; O later removes this package from the asymptotic coefficient instead of optimizing its decimal value.

With

$$\begin{aligned} P_M &= 4 \log(U_0 M^4) + 4, \\ q_M &= 1 + \frac{\log P_M}{\log(4/3)} + \frac{2}{P_M \alpha e \log(4/3)}, \\ \mathcal{S}_M &= \frac{144(M+1)^3}{\alpha} \sqrt{4M^2 + g_M^2} \sqrt{U_0^2 M^6 + g_M^2}, \\ \mathcal{H}_M &= U_0 M^3 q_M + U_0 M^4 + 8e^{a'(M)} \mathcal{S}_M, \end{aligned}$$

a specified upper coefficient is

$$\hat{C}_M = 1 + 32M + U_0 M^3 + 5^\alpha \mathcal{H}_M.$$

The terms respectively cover the identity, bounded initial and remainder terms, the terminal tail, and the block sum. Here

$$a'(M) = \frac{2}{\pi M \sqrt{M^2 - 1}}.$$

A capped variant avoids an artificial exponential blow-up near $M = 1$. Use

$$\tau_M = \frac{1}{\max\{1, a'(M)\}}, \quad \varepsilon = \frac{\tau_M}{\log(n+e)}.$$

Replace P_M by $P_M + 2 \log \max\{1, a'(M)\}$ and $e^{a'(M)}$ by $e^{\min\{1, a'(M)\}} \leq e$. Q then bounds the resulting coefficient by

$$\begin{cases} 2^{154} (M-1)^{-3/2}, & 1 < M \leq 2, \\ 2^{146} M^7, & M \geq 2. \end{cases}$$

These are estimates on a selected proof recipe, not asymptotics of the best upper constant.

Q.1.3 Lower dependency chain

Put

$$\nu = \frac{1-\alpha}{2}, \quad \eta = \frac{\alpha}{4}, \quad p = 2\alpha + \nu, \quad J(s) = \frac{\Gamma(1+s)}{\Gamma(1+s/2)^2}.$$

The source's sector-margin constant is

$$d_* = \frac{2}{\pi M} \left(1 - \frac{1}{M}\right) \sin \frac{\pi \nu}{4} > 0.$$

A specified boundary constant is

$$\mathcal{B}_M = 1 + \frac{e^{2\eta}}{M^2 d_*} \left[4\{J(-\alpha) + J(\alpha)\}J(-\nu) + \frac{J(-\alpha - \nu) + J(\alpha - \nu)}{2} \right].$$

All arguments of J exceed -1 . Define

$$A_\alpha = (\pi/2)^\alpha \left(1 + \frac{1}{\pi(1-\alpha)} \right),$$

$$\delta_* = \min \left\{ 1, M^2 - 1, \frac{M^2 d_* e^{-\alpha} 2^{-4\alpha}}{16A_\alpha} \right\}, \quad r = 2 + \left\lceil \frac{32\mathcal{B}_M M^4}{M^2 - 1} \right\rceil,$$

$$\zeta_M = \min \left\{ \frac{M^2 - 1}{4}, \frac{1}{4(1+2M)^2[2 + 4M^2/(M-1)^2]} \right\}.$$

The first literal lower-proof constant in S.8.1 is therefore

$$c_M^{\text{phase}} = \zeta_M \delta_*, \quad \gamma_n = c_M^{\text{phase}} n^{-p}.$$

It must not be substituted for the sign-amplification coefficient. That coefficient can instead be taken to be

$$s_M = \frac{2e^{-\alpha/2}}{\pi} \sqrt{\frac{1+\alpha}{1-\alpha+\pi^2(1+\alpha)}}.$$

The retained Dirichlet estimates give $\|S_n\| \geq s_M n^\alpha$ for every $n \geq 1$.

The inverse comparison in S has error $1/50$. Since $S_n^2 = I$, $\|S_n\| \geq 1$, and hence

$$\|e^{t_n A_n^{-1}}\| \geq e^{-\pi} \|S_n\| - 1/50 \geq h_M n^\alpha, \quad h_M = (e^{-\pi} - 1/50) s_M > 0.$$

This removes the source's need to require n sufficiently large at that step. Using only $s_M n^\alpha$ and a half-factor instead would require

$$n_0 = \max \left\{ 1, \left\lceil \left(\frac{e^\pi}{25s_M} \right)^{1/\alpha} \right\rceil \right\}.$$

Q.1.4 A time slope is not a threshold

Define

$$E = \frac{\zeta_M \delta_*}{100M^2}, \quad K_0 = \frac{2}{E} \left[\log \frac{2r}{E} + p + 1 \right],$$

$$\mathcal{T}_M = \log(12K_0) + p + \log(400\pi^2 M r) + 3, \quad T_M = 2r \log 2 + 1 + \log \mathcal{T}_M.$$

The eigenvalue recursion and normalization give

$$\kappa_n \leq K_0 n^p \log(n+e), \quad L(t_n) \leq T_M n \quad (n \geq 1).$$

In particular, T_M is a slope in doubly logarithmic time, not the physical threshold t_M , and not the upper prefactor C_M .

One all-time lower coefficient is

$$\hat{c}_M = \frac{h_M}{(2T_M)^\alpha}.$$

For $L(t) \geq 2T_M$, take $n = \lfloor L(t)/T_M \rfloor \geq L(t)/(2T_M)$ and rescale the witness from t_n to t . Below that range, the bound $G_M(t) \geq 1$ covers the proposed lower expression, which is less than $h_M < 1$. Thus $t_M = 0$ is justified without requiring small constructed observation times.

The ledger's conservative simplification is $\hat{c}_M \geq 2^{-32}M^{-3}$. Q also notes an improved zero-threshold floor conversion:

$$c_M^{\text{prefix}} = \frac{h_M}{T_M^\alpha (1 + h_M^{1/\alpha})^\alpha}.$$

This is a larger admissible selected coefficient, not the optimal lower constant.

Q.1.5 Numerical checkpoints and their scope

The unrounded original recipe at $M = 2$ records

$$\mathcal{B}_2 \approx 206.55148, \quad r = 35254, \quad \delta_* \approx 1.58965829 \cdot 10^{-4}, \quad \zeta_2 = 1/1800,$$

$$T_2 \approx 48877.3905, \quad \hat{c}_2 \approx 1.57252975 \cdot 10^{-6}, \quad \hat{C}_2 \approx 1.80390308 \cdot 10^{45}.$$

A separate elementary rational check instead takes $\mathcal{B}_2 = 300$, $r = 51202$, $\delta_* = 10^{-4}$, $\zeta_2 = 1/1800$ and certifies the rounded pair 10^{-6} and $2 \cdot 10^{45}$. This script is mounted, bundled, and was rerun successfully while preparing this handoff. The source's general arbitrary-precision calculator and all of its larger finite-Gram scripts are not mounted; their reported outputs are not represented as new reruns.

Q.2 Threshold trade-offs and the true tail constants

Q.2.1 Optimizing after a chosen threshold

Set $R_M(t) = G_M(t)/L(t)^\alpha$. The best constants valid for all $t \geq \tau$ are

$$c_M^*(\tau) = \inf_{t \geq \tau} R_M(t), \quad C_M^*(\tau) = \sup_{t \geq \tau} R_M(t).$$

The first is nondecreasing in τ and the second nonincreasing. Their limits are $\underline{K}_M$ and $\overline{K}_M$. Therefore their gap tends to one exactly when the normalized envelope has a finite positive limit. Allowing a threshold depending on M can be absorbed into the notation c_M, C_M ; it never permits the coefficient to depend on the running observation time.

Larger thresholds leave previously admissible constants valid but do not improve them automatically. Nor must the optimal improvement be strict. Matching multiplicative order is weaker than a leading asymptotic equivalent.

Q.2.2 Explicit improvements inside the old recipe

Keeping the additive inverse error and the floor rather than immediately absorbing them gives, for $L(\tau) > T_M$,

$$c_M^{\text{tail}}(\tau) = \frac{e^{-\pi} s_M}{T_M^\alpha} \left(1 - \frac{T_M}{L(\tau)}\right)^\alpha - \frac{1}{50L(\tau)^\alpha}.$$

Once positive, this works for all $t \geq \tau$. Its maximum with the all-time coefficient is never worse, and it increases to $e^{-\pi} s_M/T_M^\alpha$. At $M = 2$ the original nested recipe approaches about $4.64687 \cdot 10^{-6}$ through this improvement alone. This is not the optimal tail coefficient.

Likewise, before absorbing bounded terms, the upper recipe reads

$$G_M(t) \leq 5^\alpha \mathcal{H}_M L(t)^\alpha + (1 + 32M + U_0 M^3).$$

Thus

$$C_M^{\text{tail}}(\tau) = 5^\alpha \mathcal{H}_M + \frac{1 + 32M + U_0 M^3}{L(\tau)^\alpha}$$

is an admissible tail coefficient. Waiting longer removes this additive loss but not the large scale-independent factors in $\mathcal{H}_M$.

O and T go substantially further: they alter the proofs, not just the threshold. In particular, S Appendix B.1 already removes $e^{-\pi}$ asymptotically by using real parts $-1/n$ in the inverse spectral parameters, with inverse error at most $1/(50n^2)$. That refinement, not the small threshold adjustment above, is the basis of the later leading lower coefficients.

Q.2.3 Why monotonicity and growth order are insufficient

For $0 < \alpha < 1$ and sufficiently small $\varepsilon > 0$,

$$F(x) = x^\alpha [1 + \varepsilon \sin(\log x)]$$

is increasing but $F(x)/x^\alpha$ has unequal positive liminf and limsup. This is a logical counterexample to inferring convergence from order and monotonicity; it is not asserted to arise from a semigroup envelope. T supplies an additional logarithmic-time modulus and compactness of rescaled profiles, but still does not prove uniqueness of a profile.

Part VIII

Complete eventual-constant optimization

Source and scope. Full mathematical content of optimized-constants-proof.md, apart from its redundant source-map appendix, which is incorporated into this handoff.

Its formulas are optimized sufficient coefficients, not the true tail constants. The original Markdown and arithmetic companion are bundled. Bracketed source-local references retain their original meaning.

Status and scope

This is a continuation of the proposed research proof in **MF-17-sharp-fixed-bound-rate.pdf** (denoted [S]) and its quantitative audit **MF-17-explicit-constants.pdf** (denoted [Q]). It is an analytic research argument, not an independent peer review or a formal verification of those reports. The arithmetic checker supplied with this note checks only the stated arithmetic identities and inequalities.

The object is

$$G_M(t) = \sup\{\|e^{tA^{-1}}\| : \|e^{sA}\| \leq Me^{-s} \ (s \geq 0)\}, \quad L(t) = \log \log(t + e^e),$$

where the supremum includes all complex Hilbert spaces and strongly continuous semigroup generators satisfying the displayed bound. Fix

$$\alpha = \frac{2}{\pi} \arccos(M^{-1}), \quad \rho = \sqrt{1 - M^{-2}}, \quad \beta = \frac{1 + \alpha}{2}, \quad M > 1.$$

Write

$$\underline{K}_M = \liminf_{t \rightarrow \infty} \frac{G_M(t)}{L(t)^\alpha}, \quad \bar{K}_M = \limsup_{t \rightarrow \infty} \frac{G_M(t)}{L(t)^\alpha}.$$

This note optimizes bounds on these two quantities. It does **not** claim that they are equal, or that the resulting bounds are globally optimal among all possible proofs or constructions.

A concrete consequence of the optimized argument is

$$\frac{1}{200} L(t)^{2/3} \leq G_2(t) \leq 15000 L(t)^{2/3} \quad (t \geq t_2)$$

for some finite threshold t_2 . The threshold is not asserted to be zero. The ratio of these two coefficients is 3,000,000, rather than the ratio $2 \cdot 10^{51}$ of the previous rounded all-time coefficients. This comparison changes the threshold convention and must not be described as an all-time improvement.

The main symbolic upper bound is

$$\bar{K}_M \leq \mathcal{U}_M := \frac{\kappa_* 2^{1-\alpha} (1 + \alpha)^2 M^7 (1 + \rho)^2}{\alpha^{1+\alpha} (1 - \alpha)^{1-\alpha} [\log(4/3)]^\alpha}. \quad (\text{O.1})$$

Here κ_* is the fixed scalar variational constant

$$\kappa_* = \inf_{\substack{g \in C_c^\infty(0,1), \ g \geq 0 \\ \|g\|_2 = 1}} \frac{\|g\|_1^2}{\|g * g\|_2^2}, \quad 1 \leq \kappa_* < \sqrt{2}. \quad (\text{O.2})$$

Replacing κ_* by $\sqrt{2}$ gives a completely explicit upper bound. No unknown semigroup norm is hidden in (2); it is an optimization of ordinary scalar convolution integrals.

An explicit symbolic lower optimization is given in Sections 8–11. It uses only Gamma functions, three circle integrals, two 2×2 eigenvalue calculations, and low-dimensional parameter optimizations. A simpler formula without the matrix optimization is given in Section 11.

O.1 Which constants are threshold costs rather than asymptotic prefactors?

The old constants include costs for an initial block prefix, a bounded Bessel remainder, finite inverse-normalization error, integer rounding, and controlling every time down to zero. Those costs should not be absorbed into the coefficient of $L(t)^\alpha$ when the threshold can increase.

More substantially, the following losses can be removed or improved:

- the factor $e^{a'(M)}$ and nonvanishing auxiliary-coordinate lengths;
- the replacement of M by $M + 1$ in the synthesis constant;
- the Bernstein/Poisson constant 40π , followed by its rounding to 144;
- the pointwise weak-sequence estimate in place of the exact partial-sum constraint;
- eight residue classes, each charged as though it contained all blocks;
- the estimate $n \leq 5L(t)$ rather than the asymptotic block density;
- the universal oscillatory package 2^{128} ;
- the attenuation $e^{-\pi}$, a deliberately crude Dirichlet witness, and the original replication count.

The removal of the attenuation was already present in [S, Appendix B.1]. It is reused here, not claimed as a new discovery.

O.2 Exact finite-degree Poisson comparison

The scalar concentration argument in [S, Lemma 2.2] gives, before rounding its prefactor,

$$(P_r * w)(\theta) \leq A(K) \left(\frac{1+r}{1-r} \right)^{a(K)} \int w \, dm, \quad A(K) = K^3(1 + \sqrt{1 - K^{-2}})^2. \quad (\text{O.3})$$

Here w is a positive trigonometric polynomial whose Riesz projection has norm at most K , and circle measure is normalized. The factor is the exact limit of $(1 + \rho')/(1 - \rho') \cdot (1 - (\rho')^2)^{-1/2}$ in that proof. The endpoint passage does not contribute another constant.

Let $w = |p|^2$, with p of degree at most $n - 1$. The Poisson Toeplitz matrix on its coefficients is

$$R_n(r) = (r^{|i-j|})_{0 \leq i, j < n}.$$

For $0 < r < 1$, Cauchy–Schwarz in this matrix gives the exact comparison

$$|p(1)|^2 \leq [\mathbf{1}^* R_n(r)^{-1} \mathbf{1}] (P_r * |p|^2)(1) = \left[1 + (n-1) \frac{1-r}{1+r} \right] (P_r * w)(1). \quad (\text{O.4})$$

Indeed, $R_n(r)^{-1}$ is tridiagonal, with endpoint diagonal entries $1/(1-r^2)$, interior entries $(1+r^2)/(1-r^2)$, and off-diagonal entries $-r/(1-r^2)$. Summing its entries proves (4). Rotation gives the same inequality at every boundary point.

Put $R = (1+r)/(1-r)$. Combining (3) and (4) reduces the comparison to minimizing

$$R^a + (n-1)R^{a-1}, \quad R \geq 1, \quad a = a(K).$$

For sufficiently large n , the minimizer is $R = (1-a)(n-1)/a$, and its value is

$$\frac{(n-1)^a}{a^a(1-a)^{1-a}}.$$

For the remaining small degrees, $\|w\|_\infty \leq n \int w$ supplies the same bound with n^a in place of $(n-1)^a$. Thus define

$$D(K) = \frac{K^3(1 + \sqrt{1 - K^{-2}})^2}{a(K)^{a(K)}(1 - a(K))^{1-a(K)}}. \quad (\text{O.5})$$

The convention at $a = 0$ is by continuity. The all-cut synthesis estimate becomes

$$\left\| \sum_{j=1}^n u_j \right\| \leq \sqrt{D(K)} B n^{(1+a(K))/2} \quad (\text{O.6})$$

whenever $\|u_j\| \leq B$ and all initial-coordinate cuts have norm at most K . The stationarization proof in [S, Theorem 2.3] is unchanged.

This optimizes the Poisson radius exactly within comparison (4), rather than selecting a convenient radius and estimating an arc around the maximum.

O.3 Exact use of the partial-sum constraints

Suppose a decreasing nonnegative sequence $b_1, \dots, b_n$ satisfies

$$\sum_{j=1}^m b_j \leq A m^\beta \quad (1 \leq m \leq n), \quad \frac{1}{2} < \beta < 1.$$

It is weakly majorized by

$$A(j^\beta - (j-1)^\beta)_{j=1}^n.$$

Consequently,

$$\sum_{j=1}^n b_j^2 \leq A^2 \sum_{j=1}^n (j^\beta - (j-1)^\beta)^2. \quad (\text{O.7})$$

One elementary proof uses $\sum_j (b_j - u)_+ = \max_m (\sum_{j \leq m} b_j - mu)$ and integrates this inequality against $2 du$ after using the layer-cake formula for squares.

The exact sum in (7) has asymptotic

$$\sum_{j=1}^n (j^\beta - (j-1)^\beta)^2 = \frac{\beta^2}{2\beta-1} n^{2\beta-1} + o(n^{2\beta-1}). \quad (\text{O.8})$$

The estimate is locally uniform in $\beta \in (1/2, 1)$.

Apply (6) and duality to every ordered subfamily, as in [S, Lemma 3.2]. Formula (7), followed by Cauchy–Schwarz between the two operator families, replaces the old product prefactor by

$$D(K) B_X B_Y \sum_{j=1}^n (j^{(1+a(K))/2} - (j-1)^{(1+a(K))/2})^2. \quad (\text{O.9})$$

Its leading coefficient is

$$D(K) B_X B_Y \frac{(1+a(K))^2}{4a(K)}.$$

The original pointwise rearrangement argument had discarded the square of $(1+a(K))/2$.

O.4 Asymptotically free augmentation and two residue classes

The approximate-cut augmentation uses vectors $(u_j, c_n \delta_j)$. Its cut norm is bounded by

$$\left\| \begin{pmatrix} M & D_Q/c_n \\ 0 & 1 \end{pmatrix} \right\| \leq M + \frac{M(D_Q/c_n)^2}{2(M^2-1)}. \quad (\text{O.10})$$

For a fixed $M > 1$, take

$$c_n = \frac{1}{\log(n+e)}, \quad D_Q \leq [\log(n+e)]^{-3}.$$

Then the augmented vector lengths converge to their original bounds, the cut norm is $M + O_M((\log n)^{-4})$, and

$$n^{a(K_n)-a(M)} \longrightarrow 1. \quad (\text{O.11})$$

The super-exponentially decaying sparse-tail errors permit this choice with a discarded prefix of size $O_M(1 + \log \log \log(n + e^{e^e}))$. Its cost is $o(n^\alpha)$. There is no residual factor $e^{a'(M)}$, no $M + 1$, and no nonzero limiting augmentation cost.

To reduce the number of residue classes, choose any fixed

$$1 < q < 4/3, \quad N_j = N_0^{q^j}, \quad \lambda_s = N_0^{-q^s/2} \quad (s \in \mathbb{R}).$$

For $0 < \delta < 1/2$, use

$$W_j = R_{\lambda_{j-\delta}} - R_{\lambda_{j+1+\delta}}, \quad R_r = r(r - C)^{-1}. \quad (\text{O.12})$$

The high/low estimates in [S, Lemma 4.2] imply

$$\sum_j \|B_j - W_j B_j\| < \infty.$$

The error is uniform over admissible semigroups and the observation time. A cut between $j = k$ and $j = k + 2$ can be placed at

$$r = \sqrt{\lambda_{k+1+\delta} \lambda_{k+2-\delta}}.$$

There is a genuine gap because $1 + \delta < 2 - \delta$. The resolvent identities in [S, Lemma 4.3] therefore give errors $O_M(e^{-cq^Q})$ on each of the **two** parity classes. The same is true for adjoints and every ordered subfamily. Taking $\delta = 1/4$ is sufficient.

If the two class sizes are n_0, n_1 , their leading costs add as

$$n_0^\alpha + n_1^\alpha \leq 2^{1-\alpha} (n_0 + n_1)^\alpha, \quad (\text{O.13})$$

not as $2n^\alpha$. Also, the actual block count obeys

$$\frac{n(t)}{L(t)} \longrightarrow \frac{1}{\log q}. \quad (\text{O.14})$$

All these improvements preserve the exact fixed value of M ; they do not replace it by a larger forward bound.

0.5 Replacing the astronomical oscillatory constant

0.5.1 A flexible exact partition

Fix a nonnegative $g \in C_c^\infty(0, 1)$ with $\|g\|_2 = 1$, and let $h = g * g$. For sufficiently small $d > 0$,

$$S_d(v) = \sum_{k \in \mathbb{Z}} h(v - kd)^2 > 0, \quad c_d(v) = \frac{h(v)}{S_d(v)}.$$

The function c_d is smooth and supported in $(0, 2)$. Its translates satisfy

$$\sum_{j \geq 0} h(x - jd) c_d(x - jd) = 1 \quad (x \geq 2). \quad (\text{O.15})$$

Moreover

$$\sup_x \sum_j c_d(x - jd) = \sup_x \frac{\sum_j h(x - jd)}{\sum_j h(x - jd)^2} \longrightarrow \frac{\int h}{\int h^2} = \frac{\|g\|_1^2}{\|g * g\|_2^2} =: \kappa(g) \quad (\text{O.16})$$

as $d \downarrow 0$. Uniform convergence follows directly from uniform Riemann-sum estimates for the smooth compactly supported functions h, h^2 .

Let $g_N(s) = N^{-1/2}g(s/N)$, so $g_N * g_N(v) = h(v/N)$. For a bounded semigroup T , define

$$U_N(\xi) = \int_0^\infty g_N(s) e^{-i\xi s} T(s) ds.$$

The translated-window identity and vector-valued Plancherel give

$$\int_{\mathbb{R}} \sup_{k \in E} |\langle T(k) U_N(\xi)^2 x, y \rangle| \frac{d\xi}{2\pi} \leq M^3 \|x\| \|y\| \quad (\text{O.17})$$

for any countable $E \subset [0, \infty)$. This is the same three-factor semigroup argument as before; only the scalar partition has changed.

For a scalar kernel supported above $2N$, (15)–(17) imply

$$\left\| \int K(u) T(u) du \right\| \leq M^3 \sup_{\xi} \sum_{j \geq 0} |b_j(\xi)|,$$

where

$$b_j(\xi) = \int K(u) c_d(u/N - jd) e^{i\xi(u - jdN)} du. \quad (\text{O.18})$$

All semigroup integrals are strong integrals on vectors; no norm continuity of T is assumed.

0.5.2 Uniform stationary phase on strictly smaller power windows

Suppose

$$K_N(u) = u^{-3/4} f_N(u) e^{2i\sqrt{u}}, \quad \text{supp } f_N \subset [2N, 4N^q], \quad 1 < q < 4/3,$$

with $0 \leq f_N \leq 1$ and uniformly bounded logarithmic derivatives $\sup_u u^\ell |f_N^{(\ell)}(u)|$, for the finitely many derivative orders used below. Then

$$\limsup_{N \rightarrow \infty} \sup_{\xi} \sum_j |b_j(\xi)| \leq 2\sqrt{\pi} \sup_x \sum_j c_d(x - jd). \quad (\text{O.19})$$

The same holds for the negative phase. The statement is uniform in the amplitude family with these fixed derivative bounds.

Here are details of the uniformity needed in (19). For negative ξ , the unique stationary point is $U = \xi^{-2}$. On a translated window of width comparable to N , centered at U_j , the curvature parameter is

$$\Lambda_j = N^2 U_j^{-3/2} \geq c N^{2-3q/2} \rightarrow \infty.$$

Only a bounded number of windows (with the bound depending on the fixed grid spacing and fixed support of c_d) meet a fixed enlargement of the stationary point. Uniform stationary phase on these windows gives

$$b_j(\xi) = 2\sqrt{\pi} f_N(U) c_d(U/N - jd) \exp\{i(2\sqrt{U} + \xi(U - jdN)) - i\pi/4\} + o(1). \quad (\text{O.20})$$

The $o(1)$ is uniform. One direct proof rescales about U , first integrates over a fixed bounded interval in the stationary variable, and then bounds the two tails by integration by parts. The rescaled amplitudes are uniformly Lipschitz and the rescaled phases have uniformly controlled higher derivatives; the lower curvature bound tends to infinity.

For all other windows, two integrations by parts give

$$|b_j(\xi)| \leq C \Lambda_j^{-3/2} \left(\frac{\text{dist}(\xi, I_j)}{w_j} \right)^{-2},$$

where I_j is the stationary-frequency interval of that window, $w_j \asymp NU_j^{-3/2}$, and the normalized distance is bounded below on the windows under consideration. The intervals have centers proportional to $j^{-1/2}$ and widths proportional to $j^{-3/2}$. Their inverse-square tails have uniformly bounded sum: split the indices below, comparable to, and above the index corresponding to U . Consequently these nonstationary windows have total $o(1)$. For $\xi \geq 0$, all windows are nonstationary. This proves (19), including frequencies for which the stationary point is outside the amplitude support.

Combining (16), (18), and (19) gives the asymptotic block estimate

$$\|B_j^\pm\| \leq [2\sqrt{\pi} \kappa(g) + o(1)]M^3. \quad (\text{O.21})$$

The observation-time damping $e^{-u/t}$ in the actual block amplitude satisfies the required derivative bounds uniformly in t . The older finite bound is still used for finitely many initial blocks and for the summable terminal tail. Its large numerical constant therefore disappears from the asymptotic prefactor.

O.5.3 The scalar optimization is nontrivial but explicit

Young's inequality gives $\kappa(g) \geq 1$. For a Gaussian, before compact truncation and rescaling,

$$\kappa(e^{-x^2/2}) = \frac{(\int e^{-x^2/2})^2 \int e^{-x^2}}{\int (e^{-x^2/2} * e^{-x^2/2})^2} = \sqrt{2}.$$

Smooth positive truncation, followed by a translation and rescaling into $(0, 1)$, preserves the limiting ratio. Thus $\kappa_* \leq \sqrt{2}$.

There is even a strict improvement. With $f_\epsilon(x) = e^{-x^2/2 - \epsilon x^4}$, Gaussian moment calculations give the right derivative

$$\left. \frac{d}{d\epsilon} \log \kappa(f_\epsilon) \right|_{0+} = -6 - \frac{3}{2} + \frac{27}{4} = -\frac{3}{4}. \quad (\text{O.22})$$

The three contributions come from the squared L^1 norm, the squared L^2 norm, and the autoconvolution norm. Therefore $\kappa_* < \sqrt{2}$. This is a genuine scalar improvement, not a claim that the exact value of κ_* is known. Formula (2), or the smaller one-parameter infimum over $\epsilon > 0$, may be retained symbolically.

O.6 Combining the optimized upper constants

The two families in the product estimate have asymptotic member bounds

$$B_X = 2M, \quad B_Y = 2\sqrt{\pi} \kappa(g)M^3.$$

Equations (8)–(14) and (21) show that the sum of either phase has leading coefficient at most

$$4\sqrt{\pi} \kappa(g)D(M)M^4 \frac{\beta^2}{\alpha} \frac{2^{1-\alpha}}{(\log q)^\alpha}.$$

The two phases in the Bessel cosine have coefficients of modulus $1/(2\sqrt{\pi})$. Their total contribution is therefore the preceding expression divided by $\sqrt{\pi}$, not the expression itself. Bounded initial terms, the Bessel remainder, the discarded prefix, localization errors, and the terminal tail are $o(L(t)^\alpha)$.

First take the large-time limit with all auxiliary choices fixed. Then take the grid spacing to zero, optimize g , and let $q \uparrow 4/3$. This order of limits gives exactly (1). It does not require the uniform stationary-phase assertion at $q = 4/3$, where the minimal curvature need not diverge.

For every $\epsilon > 0$, the resulting consequence is an eventual upper bound with coefficient $\mathcal{U}_M + \epsilon$. A limsup bound by $\mathcal{U}_M$ alone does not justify asserting the coefficient $\mathcal{U}_M$ itself for every sufficiently large time.

Using the explicit Gaussian relaxation gives

$$\boxed{\bar{K}_M \leq U_M^G := \frac{2^{3/2-\alpha}(1+\alpha)^2 M^7 (1+\rho)^2}{\alpha^{1+\alpha}(1-\alpha)^{1-\alpha} [\log(4/3)]^\alpha}} \quad (\text{O.23})$$

At $M = 2$, this simplifies to

$$U_2^G = 25 \cdot 2^{25/6} (2 + \sqrt{3})^2 [\log(4/3)]^{-2/3} < 15000. \quad (\text{O.24})$$

The strict rational verification is in Section 12.

These are optimized upper **bounds**, not optimal constants for G_M . For example, the remaining sector factor in (3), the member bound $\|W_j\| \leq 2M$, and the passage through the coefficient-sequence relaxation can still contain genuine slack.

O.7 Lower normalization: preserve the leading coefficient

The lower construction in [S, Appendix B.1] replaces the spectral parameters by

$$\mu_i = -n^{-1} + iy_i, \quad \lambda_i = \mu_i^{-1}.$$

The separation is modified to

$$\kappa_n n(n^{-2} + y_i^2) \leq y_{i+1} \leq \kappa_n n(n^{-2} + y_i^2) + 2,$$

with the same parity prescription. With the stated smaller normalization parameter, the resulting admissible generator satisfies

$$\|e^{t_n A_n^{-1}} - e^{-\pi/n} S_n\| \leq \frac{1}{50n^2}, \quad L(t_n) = 2rn \log 2 + o(n). \quad (\text{O.25})$$

The replication number r is fixed independently of n . The polynomially small forward margin affects the lower-order time cost, not $2r \log 2$.

Consequently, any sign-multiplier witness with $\|S_n\| \geq (a + o(1))n^\alpha$ yields

$$\boxed{K_M \geq \frac{a}{(2r \log 2)^\alpha}} \quad (\text{O.26})$$

To justify the statement for every large time, fix $\varepsilon > 0$ and use $L(t_n) \leq (2r \log 2 + \varepsilon)n$ for sufficiently large n . Choose n by the floor of $L(t)/(2r \log 2 + \varepsilon)$, and use the monotonicity of G_M . Let $\varepsilon \downarrow 0$. No assumption about exact monotonicity of the constructed sequence t_n is needed.

Thus the old factors $e^{-\pi}$, $e^{-\pi} - 1/50$, 2^α from a floor bound, and the coarse coefficient T_M^α are all absent from the asymptotic lower coefficient.

O.8 Optimize the test polynomial instead of fixing a Dirichlet kernel

For a weight

$$h(z) = (1 - z)^{-\alpha/2} \exp\{\eta[(1 - z)^\nu - 1]\}, \quad (\text{O.27})$$

the coefficient at its singularity is

$$|h(e^{i\theta})|^2 \sim e^{-2\eta|\theta|^{-\alpha}}, \quad |h(e^{i\theta})|^{-2} \sim e^{2\eta|\theta|^\alpha}.$$

Let $f_\sigma(s) = [s(1-s)]^\sigma$ on $[0, 1]$, with $\sigma \geq 0$, and use

$$p_n(z) = \sum_{j=0}^{n-1} f_\sigma(j/n) z^j.$$

For $\sigma = 0$, read the endpoint value as one. A Riemann-sum argument after $\theta = x/n$ gives

$$\frac{\|hp_n\|_2}{\|h^{-1}p_n\|_2} = [e^{-2\eta} \mathcal{D}_\alpha(\sigma) + o(1)] n^\alpha. \quad (\text{O.28})$$

Indeed, the functions f_σ have bounded variation. Summation by parts gives a uniform bound on the rescaled polynomials by $C \min(1, |x|^{-1})$. This is integrable after squaring and multiplying by $|x|^{\pm\alpha}$, since $0 < \alpha < 1$. Dominated convergence therefore justifies both weighted limits.

The coefficient has the explicit Gamma-function formula

$$\begin{aligned} \mathcal{D}_\alpha(\sigma)^2 &= 2^{-2\alpha} \frac{\Gamma((1-\alpha)/2) \Gamma(\sigma + (1+\alpha)/2)}{\Gamma((1+\alpha)/2) \Gamma(\sigma + (1-\alpha)/2)} \\ &\quad \times \frac{\Gamma(\sigma + 1 - \alpha/2) \Gamma(2\sigma + 3/2 - \alpha/2)}{\Gamma(\sigma + 1 + \alpha/2) \Gamma(2\sigma + 3/2 + \alpha/2)}. \end{aligned} \quad (\text{O.29})$$

It follows, for example, by evaluating

$$\int_{\mathbb{R}} \left| \int_0^1 f_\sigma(s) e^{ixs} ds \right|^2 |x|^u dx$$

with the Bessel representation of the inner integral and the equal-order Bessel Mellin integral. A derivation is given in Appendix A.

The old Dirichlet constant is recovered at $\sigma = 0$:

$$\mathcal{D}_\alpha(0) = \sqrt{\Gamma(2-\alpha)/\Gamma(2+\alpha)}.$$

Define the optimized one-parameter coefficient

$$S_\alpha = \max_{\sigma \geq 0} \mathcal{D}_\alpha(\sigma). \quad (\text{O.30})$$

The maximum exists: the function is continuous and positive, and the Gamma quotient shows $\mathcal{D}_\alpha(\sigma) = O_\alpha(\sigma^{-\alpha/2})$ as $\sigma \rightarrow \infty$. This is a scalar optimization independent of the rounding parameters ν, η .

An interior maximizing parameter satisfies

$$\begin{aligned} 0 &= \psi(\sigma + (1+\alpha)/2) - \psi(\sigma + (1-\alpha)/2) \\ &\quad + \psi(\sigma + 1 - \alpha/2) - \psi(\sigma + 1 + \alpha/2) \\ &\quad + 2\psi(2\sigma + 3/2 - \alpha/2) - 2\psi(2\sigma + 3/2 + \alpha/2), \end{aligned} \quad (\text{O.31})$$

where $\psi = \Gamma'/\Gamma$. No numerical root finding is required to retain (30) symbolically.

A still larger variational family replaces f_σ by arbitrary smooth nonzero functions supported in $(0, 1)$, and maximizes the quotient of the $|x|^{-\alpha}$ and $|x|^\alpha$ Fourier energies. This is finite by elementary support/uncertainty estimates. Formula (30) is preferable when a completely explicit Gamma-function optimization is desired; no claim is made that the beta-profile family is globally optimal.

O.9 A sharper phase-replication certificate

O.9.1 Admissible rounded weights

Allow

$$0 < \nu < 1 - \alpha, \quad 0 < \eta < \eta_{\max}(\alpha, \nu),$$

where

$$\eta_{\max}(\alpha, \nu) = \inf_{0 < \phi < \pi/2} \frac{\alpha(\phi + \pi/2)}{2(2 \cos \phi)^\nu \sin(\nu \phi)}. \quad (\text{O.32})$$

This is an explicit sufficient-and-necessary strict interior condition for the lower side of the sector inequality for this family. In fact,

$$\arg h(e^{i\theta}) = \frac{\alpha}{2}\phi - \eta(2 \cos \phi)^\nu \sin(\nu \phi), \quad \phi = (\pi - \theta)/2,$$

on the upper half circle. The upper sector inequality is automatic for $\eta > 0$, and (32) ensures the lower one. In particular, $|2 \arg h| < \pi\alpha/2$ away from $\theta = 0$. The sector margin vanishes at zero as a positive constant times $|\theta|^\nu$.

Put

$$c = M^{-1}, \quad k = M^2, \quad \Delta(\theta) = 2c\{\cos(2 \arg h(e^{i\theta})) - c\},$$

$$d_0 = \frac{1}{2} \int_{\mathbb{T}} (|h|^2 + |h|^{-2}) dm, \quad I_0 = \int \Delta^{-1} dm, \quad I_1 = \int \frac{|h|^2 + |h|^{-2}}{\Delta} dm.$$

All these quantities are finite. The three scalar integrals used below are summarized by

$$A = d_0 I_0, \quad B = \frac{I_1}{2d_0}, \quad C = \frac{1}{2} \int_{\mathbb{T}} \frac{\operatorname{Re}(h + h^{-1})}{\Delta} dm, \quad D = \frac{d_0}{2c}. \quad (\text{O.33})$$

The letter B in (33) is not the old boundary constant, and D is not an approximate-cut error.

O.9.2 The exact deficit and a three-functional Gram matrix

Use the paired weighted space of [S, Section 7], with

$$H = d_0^{-1/2} U \operatorname{diag}(h, h^{-1}) U^*, \quad U = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

For a cut, write $x = u + v$, with u the tail. Let

$$a = Hu, \quad b = \overline{H}v, \quad g = \overline{H}H^{-1}, \quad F = b + (g - cI)a.$$

The convention here makes the inner product identity explicit: $\langle u, v \rangle_H = \langle ga, b \rangle$. The exact deficit is

$$M^2 \|x\|_H^2 - \|u\|_H^2 = M^2 \left(\|F\|_2^2 + \int \Delta \|a\|^2 dm \right). \quad (\text{O.34})$$

The functions a, b are orthogonal because their Fourier supports, including the specified constant-coordinate subspaces, are disjoint.

For the boundary component $s \in \{1, 2\}$, consider the three functionals

$$x_0 = \sqrt{d_0}(a_0)_s, \quad y_0 = \sqrt{d_0}(b_0)_s, \quad z_0 = \int \langle a, H e_s \rangle dm.$$

On the Hilbert space with norm squared $\|F\|_2^2 + \int \Delta \|a\|^2$, their Gram matrix is exactly

$$\mathcal{G} = \begin{pmatrix} A & -D & C \\ -D & \rho^2 A & -(1-c)C \\ C & -(1-c)C & B \end{pmatrix}. \quad (\text{O.35})$$

To verify the less immediate entries, use $(g - cI)^*(g - cI) = (\rho^2 - \Delta)I$, $(g - cI)H = \overline{H} - cH$, and $(g - cI)_{ss} = \Delta/(2c)$. The circle integrals are real by conjugation symmetry of the weight.

At a cut including the boundary coordinate, $y_0 = 0$. At the cut excluding it, $x_0 = 0$. Imposing these genuine constraints improves the functional bounds by a Schur complement. Define

$$J_1 = \begin{pmatrix} A & C \\ C & B \end{pmatrix} - \frac{1}{\rho^2 A} \begin{pmatrix} -D \\ -(1-c)C \end{pmatrix} \begin{pmatrix} -D & -(1-c)C \end{pmatrix}, \quad (\text{O.36})$$

$$J_0 = \begin{pmatrix} \rho^2 A & -(1-c)C \\ -(1-c)C & B \end{pmatrix} - \frac{1}{A} \begin{pmatrix} -D \\ C \end{pmatrix} \begin{pmatrix} -D & C \end{pmatrix}. \quad (\text{O.37})$$

Both are positive semidefinite. They are the exact Gram matrices for these functionals on the larger Hilbert spaces with the indicated single constraint. Further analytic-support constraints are not used; this is why the resulting certificate is sufficient rather than a claim of global optimality.

O.9.3 The two quadratic forms that determine replication

Write a vector as $v + ze_i$, with $\beta_i(v) = 0$, and minimize each endpoint deficit over the coefficient z , as in [S, Lemma 6.1]. Let the minimized values be m_0, m_1, m_- for the active values 0, 1, -1 . Direct expansion gives

$$m_0 - m_- = q_0(\beta_i(x_0), \langle Qx_0, e_i \rangle), \quad m_1 - m_- = q_1(\beta_i(x_1), \langle Qx_1, e_i \rangle),$$

where the Hermitian matrices of these forms are

$$Q_0 = \frac{1}{k-1} \begin{pmatrix} k & -k \\ -k & 1 \end{pmatrix}, \quad Q_1 = \frac{1}{k-1} \begin{pmatrix} 4k & -2(k+1) \\ -2(k+1) & 4 \end{pmatrix}. \quad (\text{O.38})$$

For example, writing $u = \langle v, e_i \rangle$, $w = \langle Qv, e_i \rangle$, the first difference is $(k|u|^2 + 2k \operatorname{Re}(u\overline{w}) + |w|^2)/(k-1)$. At the zero endpoint the minimizing coefficient is $-u$, which produces exactly Q_0 . At the one endpoint the substitution $w = z_0 - \beta_i(x_1)$, $ku = z_0 - k\beta_i(x_1)$ produces Q_1 .

Define

$$\Lambda_i = \frac{1}{M^2} \lambda_{\max}(J_i^{1/2} Q_i J_i^{1/2}), \quad \Lambda = \max(0, \Lambda_0, \Lambda_1). \quad (\text{O.39})$$

These are only two-dimensional eigenvalue calculations. No matrix square root need actually be computed:

$$\lambda_{\max}(J_i^{1/2} Q_i J_i^{1/2}) = \frac{\operatorname{tr}(Q_i J_i) + \sqrt{\operatorname{tr}(Q_i J_i)^2 - 4 \det(Q_i) \det(J_i)}}{2}, \quad (\text{O.40})$$

with $\det Q_0 = -k/(k-1)$, $\det Q_1 = -4$. Formula (34) implies

$$m_0 - m_- \leq \Lambda_0 m_0, \quad m_1 - m_- \leq \Lambda_1 m_1. \quad (\text{O.41})$$

Choose the smallest positive integer strictly exceeding the certificate:

$$\boxed{r_M(\nu, \eta) = 1 + \lfloor \Lambda \rfloor}. \quad (\text{O.42})$$

Then $m_- + (r-1)m_\sigma \geq (r-\Lambda)m_\sigma > 0$ for $\sigma = 0, 1$. A transition with a mixture of zero and one endpoints is a convex combination of these two cases. The replication construction therefore has a uniform positive phase margin of polynomial order in the degree. The original finite-degree coercivity argument supplies that polynomial order, since $\Delta \asymp |\theta|^\nu$ near zero and $\alpha + \nu < 1$.

The Gram matrices are real. For a real positive Gram matrix, the norm of the one-active-phase multiplier over $|z| \leq 1$ is the maximum of the norms at $z = 1, -1$: minimize over the preceding

coordinates and take a scalar Schur complement; on $|z| = 1$ it is affine in $\operatorname{Re} z$. Convexity handles the disk interior. Thus the two real endpoint checks above control all phases.

The integer in (42) is the smallest certified by these two quadratic bounds. It is not asserted to be the actual smallest replication count compatible with the original weighted basis. Among replications using this certificate and possibly unequal positive copy weights, equal weights minimize the largest copy weight; every active copy must have weight below $1/\Lambda$. Thus unequal copy weights do not improve the certified integer count.

O.10 The symbolic lower optimization

Let $\mathcal{P}_\alpha$ be the parameter region in (32). The optimized coefficient furnished by the beta-profile and phase-certificate families is

$$\mathcal{L}_M = S_\alpha \sup_{(\nu, \eta) \in \mathcal{P}_\alpha} \frac{e^{-2\eta}}{[2r_M(\nu, \eta) \log 2]^\alpha}, \quad \underline{K}_M \geq \mathcal{L}_M > 0. \quad (\text{O.43})$$

Every quantity in (43) is specified by scalar integrals, Gamma functions, and the two explicit matrices (36)–(40). The polynomial phase-margin coefficient and the small spectral-normalization parameter affect the threshold, but not (43).

There is a useful finite-integer organization of this optimization. For each positive integer r , define

$$\eta_r = \inf\{\eta : \exists \nu, (\nu, \eta) \in \mathcal{P}_\alpha, \Lambda(M, \nu, \eta) < r\},$$

with $\eta_r = +\infty$ for an empty set. Then

$$\mathcal{L}_M = \frac{S_\alpha}{(2 \log 2)^\alpha} \sup_{r \geq 1} \frac{e^{-2\eta_r}}{r^\alpha}. \quad (\text{O.44})$$

If a baseline feasible pair has parameters (η_0, r_0) , no integer $r > r_0 e^{2\eta_0/\alpha}$ can improve its coefficient, since $e^{-2\eta} \leq 1$. With the original choice $\eta_0 = \alpha/4$, the integer search need only extend to $r_0 \sqrt{e}$. Infima at the boundary are understood through admissible approximating parameters; their increasing threshold costs do not invalidate the liminf statement.

One can enlarge the test-profile or analytic-weight families further. Formula (43) is an optimized lower certificate within the stated families, not a representation of the true optimal lower constant $\underline{K}_M$.

O.11 A simpler explicit lower recipe

The two small eigenvalue problems can be replaced by a cruder but much better bound than the old replication estimate.

At either boundary cut, the constant component belongs to exactly one side. Thus the source's boundary estimate with coefficient $1 + M^{-2}(8d_0 I_0 + I_1/(2d_0))$ can be replaced by

$$B_{\text{bdry}} \leq \frac{A + B}{M^2}. \quad (\text{O.45})$$

For the included endpoint, both relevant functionals depend on a , and their sum of squares is at most $(A + B) \int \Delta \|a\|^2$. For the excluded endpoint, the squared norm of the coefficient functional is $\rho^2 A$, rather than A ; (45) is still a valid common bound.

The largest eigenvalue of Q_1 is

$$q_M = \frac{2(M^2 + 1) + 2\sqrt{2(M^4 + 1)}}{M^2 - 1}, \quad (\text{O.46})$$

and it dominates the largest eigenvalue of Q_0 . Therefore any integer

$$r > q_M B_{\text{bdry}} \quad (\text{O.47})$$

suffices. The main improvement here comes from retaining the cancellation in $m_\sigma - m_-$, rather than separately bounding the large negative term in m_- .

For the original fixed parameters

$$\nu = (1 - \alpha)/2, \quad \eta = \alpha/4, \quad d_* = \frac{2}{\pi M} (1 - M^{-1}) \sin(\pi\nu/4),$$

write $J(s) = \Gamma(1+s)/\Gamma(1+s/2)^2$. The elementary moment bounds on the rounded weight give

$$\boxed{\bar{B}_M = \frac{e^{2\eta}}{2M^2 d_*} [(J(-\alpha) + J(\alpha))J(-\nu) + J(-\alpha - \nu) + J(\alpha - \nu)] \geq B_{\text{bdry}}.} \quad (\text{O.48})$$

Here $d_0 \geq 1$, $d_0 \leq e^{2\eta}(J(-\alpha) + J(\alpha))/2$, and $\Delta \geq d_*|1 - z|^\nu$ prove the assertion. All Gamma arguments are positive.

Consequently,

$$r_M^{\text{elem}} = 1 + \lfloor q_M \bar{B}_M \rfloor, \quad \boxed{\underline{K}_M \geq \ell_M^{\text{elem}} := \frac{e^{-\alpha/2} S_\alpha}{[2r_M^{\text{elem}} \log 2]^\alpha}.} \quad (\text{O.49})$$

Replacing S_α by its explicit value $\mathcal{D}_\alpha(0)$ is valid when no scalar maximization is desired. This gives a closed Gamma-function recipe for every $M > 1$.

Some useful behavior of these **chosen bound formulas**, not of the exact unknown leading constants, is as follows.

At $M \downarrow 1$, the original parameter choice has a polynomial replication cost, whereas $\alpha \asymp \sqrt{M-1}$. Hence $(r_M^{\text{elem}})^\alpha \rightarrow 1$. The Dirichlet version of (49) therefore tends to one, in contrast to the small constant produced by the earlier all-time normalization.

At $M \rightarrow \infty$,

$$\bar{B}_M \sim \frac{3}{2} \pi \sqrt{e} M, \quad q_M \rightarrow 2(1 + \sqrt{2}), \quad r_M^{\text{elem}} \sim 3\pi \sqrt{e} (1 + \sqrt{2}) M.$$

The choice $\sigma = 1/2$ has $\mathcal{D}_\alpha(1/2) \sim [8(1 - \alpha)]^{-1/2}$. Thus this explicit subfamily alone gives

$$\liminf_{M \rightarrow \infty} \sqrt{M} \underline{K}_M \geq \frac{1}{24e\sqrt{\pi}(1 + \sqrt{2}) \log 2}. \quad (\text{O.50})$$

This is a lower bound, not a claim of equality or an assertion of joint uniformity in arbitrary coupled limits of M, t .

For comparison, the Gaussian upper recipe has

$$U_M^G \sim \pi(M-1)^{-1/2} \quad (M \downarrow 1), \quad U_M^G \sim \frac{16\sqrt{2}}{\log(4/3)} M^7 \quad (M \rightarrow \infty).$$

The power M^7 in this upper certificate is not claimed optimal.

O.12 Fully explicit representative at $M = 2$

Here $\alpha = 2/3$, $\nu = 1/6$, $\eta = 1/6$, and

$$d_* = \frac{\sin(\pi/24)}{2\pi}, \quad q_2 = \frac{2(5 + \sqrt{34})}{3} < \frac{29}{4}.$$

For $-1 < s < 0$, the chord bound $2 \sin(\theta/2) \geq 2\theta/\pi$ gives

$$J(s) \leq \frac{2^s}{1+s};$$

for $s \geq 0$, use $J(s) \leq 2^s$. It follows that the bracket in (48) is smaller than

$$\frac{18}{5} \frac{6}{5} + \frac{18}{5} + \frac{3}{2} = \frac{471}{50}.$$

Also $e^{1/3} < 7/5$, $\pi/24 < 1/7$, and $\sin x \geq x - x^3/6 > (293/294)x$ on the relevant interval. Therefore

$$\overline{B}_2 < \frac{7}{5} \frac{1764}{293} \frac{471}{50}, \quad q_2 \overline{B}_2 < \frac{29}{4} \frac{7}{5} \frac{1764}{293} \frac{471}{50} < 576. \quad (\text{O.51})$$

Thus **576 replicas suffice** for the original rounded weight; no numerical matrix or Gamma optimization is needed. The sharper certificate (39) can only improve this sufficient choice.

Already with the Dirichlet profile,

$$\boxed{K_2 \geq \ell_2^{(576)} := \frac{e^{-1/3} \sqrt{\Gamma(4/3)/\Gamma(8/3)}}{[1152 \log 2]^{2/3}} > \frac{1}{200}.} \quad (\text{O.52})$$

Here is an entirely elementary verification of the last strict inequality. Jensen's inequality for a Gamma-distributed variable gives $\Gamma(x+s)/\Gamma(x) \leq x^s$ for $x > 0$, $0 < s < 1$. At $x = s = 1/3$,

$$\frac{\Gamma(4/3)}{\Gamma(8/3)} = \frac{3}{10} \frac{\Gamma(1/3)}{\Gamma(2/3)} \geq \frac{3}{10} 3^{1/3} > \left(\frac{16}{25}\right)^2.$$

Since $e^{-1/3} > 5/7$, the numerator of (52) exceeds $16/35$. Since $\log 2 < 7/10$, its denominator is smaller than 90. Hence

$$\ell_2^{(576)} > \frac{8}{1575} > \frac{1}{200}.$$

For the upper side, use (24), $2^{1/6} < 9/8$, $\sqrt{3} < 7/4$, and $\log(4/3) > 2/7$. The numerator of (24) is smaller than $25 \cdot 18 \cdot 225/16 = 50625/8$. Because

$$\left(\frac{50625/8}{15000}\right)^3 = \left(\frac{27}{64}\right)^3 < \left(\frac{2}{7}\right)^2,$$

one obtains $U_2^G < 15000$.

Both comparisons have strict slack. The definitions of $\liminf$ and $\limsup$ therefore imply that one finite threshold t_2 works simultaneously with

$$\boxed{c_2 = 1/200, \quad C_2 = 15000.}$$

The attached standard-library checker verifies all rational comparisons in this section, together with finite exact tests of the Poisson inverse formula and the quadratic-form determinants. Those arithmetic checks do not verify the infinite-dimensional analytic lemmas.

O.13 Threshold conventions and what has actually been optimized

For every fixed $M > 1$ and $0 < \epsilon < 1$, the bounds proved above yield a finite threshold such that

$$(1 - \epsilon) \mathcal{L}_M L(t)^\alpha \leq G_M(t) \leq (1 + \epsilon) \mathcal{U}_M L(t)^\alpha \quad (t \geq t_M(\epsilon)). \quad (\text{O.53})$$

The threshold may depend on the admissible parameter choices approaching the suprema and infima. It must not depend on the generator or Hilbert space. All upper remainders used here

are uniform over that class: their scalar symbol bounds, transference estimates, and resolvent bounds involve only the fixed auxiliary choices and M .

No assertion is made that the endpoints $\mathcal{L}_M, \mathcal{U}_M$ themselves, without strict slack, are valid tail constants. No assertion is made that the new constants hold at threshold zero.

The optimizations with exact answers inside their respective relaxations are the Poisson-radius choice, the quadratic majorization problem, the vanishing augmentation cost, the residue-class counting factor, the limiting block density, the two quadratic spectral certificates, and the smallest integer allowed by those certificates. The remaining scalar infimum (2) and lower parameter optimization (43) are explicitly specified rather than replaced by large anonymous constants.

Several potential sources of genuine slack remain. The scalar concentration factor in (3) need not be optimal; the product argument relaxes the full operator geometry to coefficient constraints; $\|W_j\| \leq 2M$ may be wasteful; the phase certificate discards most Hardy-support constraints; and neither the weight family nor the beta-profile family has been proved extremal for the full lower problem. These are substantive issues rather than finite-threshold bookkeeping.

In particular, this note does not prove $G_M(t) \sim K_M L(t)^\alpha$. It gives substantially better upper and lower certificates for the proposed sharp order.

O.A The beta-profile moments

Let $f_\sigma(s) = [s(1-s)]^\sigma$, $\sigma \geq 0$, and $\hat{f}_\sigma(x) = \int_0^1 f_\sigma(s) e^{ixs} ds$. The change of variables $s = (1+v)/2$ and the Poisson integral for Bessel functions give, for $x > 0$,

$$\hat{f}_\sigma(x) = e^{ix/2} \sqrt{\pi} \Gamma(\sigma+1) x^{-\sigma-1/2} J_{\sigma+1/2}(x/2).$$

For $-1 < u < 2\sigma+1$, the resulting Mellin integral is

$$\begin{aligned} I_u(\sigma) &:= \int_{\mathbb{R}} |\hat{f}_\sigma(x)|^2 |x|^u dx \\ &= \sqrt{\pi} 2^{u-2\sigma} \Gamma(\sigma+1)^2 \frac{\Gamma(\sigma+(1-u)/2) \Gamma((1+u)/2)}{\Gamma(\sigma+1-u/2) \Gamma(2\sigma+3/2-u/2)}. \end{aligned} \tag{O.54}$$

Taking $I_{-\alpha}/I_\alpha$ yields (29).

For completeness, the equal-order Mellin identity used here is

$$\int_0^\infty x^{-\lambda} J_\mu(x)^2 dx = \frac{\Gamma(\lambda) \Gamma(\mu+(1-\lambda)/2)}{2^\lambda \Gamma((1+\lambda)/2)^2 \Gamma(\mu+(1+\lambda)/2)}, \quad 0 < \lambda < 2\mu+1.$$

It follows from the product integral $J_\mu(x)^2 = (2/\pi) \int_0^{\pi/2} J_{2\mu}(2x \cos v) dv$, the single-Bessel Mellin integral, and a beta integral. The interchange is absolutely justified first for $\lambda > 1/2$, and analytic continuation within the convergence strip gives the full range. These are the classical Bessel integral identities recorded in NIST DLMF, Sections 10.9 and 10.22. At $u=0$, (54) agrees with Parseval and the ordinary beta integral, which also checks its Fourier normalization.

At $\sigma = 1/2$, $\hat{f}_\sigma(0) = \pi/8$ and $I_1(1/2) = \pi^2/4$. Therefore

$$I_{-\alpha}(1/2) \sim \frac{\pi^2}{32(1-\alpha)}, \quad \mathcal{D}_\alpha(1/2) \sim [8(1-\alpha)]^{-1/2},$$

which is used in (50).

Part IX

Latest tail-coefficient construction

Source and scope. Full pre-SDP-asymptotics report MF-17-tail-constants-research.md.

Sections T.1–T.4 are an optional finite representation and can be skipped on an analytic continuation. Sections T.5–T.13 contain the latest upper and pure-power lower constructions. Sections T.14–T.16 state exactly what remains unresolved. The subsequent SDP-asymptotics report is not reproduced or used.

Status

This report continues the proposed fixed-bound inverse-semigroup argument in **MF-17-sharp-fixed-bound-rate.pdf** [S] and **optimized-constants-proof.md** [O]. The estimates proved here are new deductions, not claims that the preceding reports have undergone independent peer review. The arithmetic certificate checks finite arithmetic implications only.

The equality of the lower and upper tail constants is not proved in this report. We distinguish an exact finite-dimensional description of the true envelope, new bounds on the two tail constants, and optimization of a particular lower construction. In particular, an optimized certificate is not relabeled as the true asymptotic constant.

Fix $M > 1$ and put

$$G_M(t) = \sup_{(H,A) \in \mathcal{A}_M} \|e^{tA^{-1}}\|, \quad \|e^{sA}\| \leq Me^{-s}, \quad s \geq 0,$$

$$L(t) = \log \log(t + e^e), \quad \alpha = \frac{2}{\pi} \arccos(M^{-1}), \quad \rho = \sqrt{1 - M^{-2}}, \quad c = M^{-1}.$$

The two objects requested are

$$\underline{K}_M = \liminf_{t \rightarrow \infty} \frac{G_M(t)}{L(t)^\alpha}, \quad \overline{K}_M = \limsup_{t \rightarrow \infty} \frac{G_M(t)}{L(t)^\alpha}.$$

The finite-semidefinite-program representation in Part I is independent of the proposed sharp exponent. Parts II and III give stronger upper and lower certificates. Their combination, with strict numerical margins at $M = 2$, gives an eventual inequality with a much smaller coefficient ratio. It does not prove that the normalized envelope converges.

Part I. An exact finite-dimensional description of the true envelope

T.1 Yosida–Poisson normalization

For an integer $m \geq 2$ set $r = 1 - 1/m$ and let $\mathcal{P}_{M,r}$ consist of bounded operators S on arbitrary complex Hilbert spaces satisfying

$$\|S^k\| \leq Mr^k \quad (k \geq 0).$$

Define

$$F_{M,m}(t) = \sup_{S \in \mathcal{P}_{M,r}} \|f_{t/m}(S)\|, \quad f_a(z) = \exp\left(-\frac{a}{1-z}\right).$$

All the power series below converge in operator norm because the power bound has $r < 1$.

Proposition 1. For every $t \geq 0$,

$$F_{M,m}(t) \leq G_M(t) \leq e^{t(M+1)/m} F_{M,m}(t). \quad (\text{T.1.1})$$

Proof. Given $S \in \mathcal{P}_{M,r}$, the bounded generator $A_m = m(S - I)$ satisfies

$$\|e^{sA_m}\| \leq e^{-ms} \sum_{k \geq 0} \frac{(ms)^k}{k!} Mr^k = Me^{-s}.$$

Moreover $e^{tA_m^{-1}} = f_{t/m}(S)$, which proves the first inequality.

Conversely, let $A \in \mathcal{A}_M$ and write $B = A^{-1}$. Exponential stability gives $\|B\| \leq M$. Set

$$S = (m-1)(m-1-A)^{-1}.$$

The Laplace formula and its k -fold convolution give, for $k \geq 1$,

$$\|S^k\| \leq \frac{M(m-1)^k}{(k-1)!} \int_0^\infty s^{k-1} e^{-ms} ds = Mr^k.$$

The inverse of its Poissonized generator is exactly

$$B_m = [m(S - I)]^{-1} = \frac{m-1}{m}B - \frac{1}{m}I.$$

Thus $B = B_m + (B + I)/m$, and all these bounded operators commute. It follows that

$$\|e^{tB}\| \leq e^{t(M+1)/m} \|e^{tB_m}\|.$$

Taking the supremum proves (1.1). No uniform approximation of the original unbounded generators is being assumed. $\square$

The multiplier in (1.1) can be improved to $e^{t(M+1)/(2m)}$ using the numerical-abscissa estimate for A^{-1} from the preceding work, but the simpler version already suffices for the conclusions below.

T.2 A compact semidefinite program

Let $J_N e_j = e_{j+1}$ for $0 \leq j < N-1$, and $J_N e_{N-1} = 0$. Write

$$f_a(z) = \sum_{j \geq 0} c_j(a) z^j,$$

where

$$c_0(a) = e^{-a}, \quad c_j(a) = e^{-a} \sum_{k=1}^j \frac{(-a)^k}{k!} \binom{j-1}{k-1} \quad (j \geq 1). \quad (\text{T.2.1})$$

Set $\mathbf{c} = (c_0(a), \dots, c_{N-1}(a))^T$, with $a = t/m$. Define $V_{M,m,N}(t)$ by

$$\begin{aligned} & \text{maximize} && \mathbf{c}^* Q \mathbf{c}, \\ & \text{subject to} && Q \succeq 0, \quad Q_{00} = 1, \\ & && J_N^{*k} Q J_N^k \preceq M^2 r^{2k} Q, \quad 1 \leq k < N. \end{aligned} \quad (\text{T.2.2})$$

There are only finitely many matrix inequalities, all of order N . Since $\mathbf{c}$ is real, averaging a feasible Hermitian Q with its entrywise conjugate shows that real symmetric matrices suffice.

The feasible set is compact. Indeed, applying the $k = j$ inequality to e_0 gives

$$0 \leq Q_{jj} \leq M^2 r^{2j},$$

and positivity bounds every off-diagonal entry. It is nonempty and strictly feasible relative to $Q_{00} = 1$: $Q = \text{diag}(1, r^2, \dots, r^{2N-2})$ makes every required inequality strict when $M > 1$.

Every feasible Q defines a Hilbert space on the quotient of $\mathbb{C}^N$ by $\ker Q$. The inequalities imply that J_N preserves $\ker Q$, and its induced powers have norms at most Mr^k . The vector represented by e_0 has norm one, and

$$\|f_a(J_N)e_0\|_Q^2 = \mathbf{c}^* Q \mathbf{c}.$$

Therefore

$$\sqrt{V_{M,m,N}(t)} \leq F_{M,m}(t) \leq G_M(t). \quad (\text{T.2.3})$$

Allowing singular Q is useful: it exactly handles dependent cyclic vectors, rather than making a false linear-independence assumption.

The dual program is

$$\begin{aligned} & \text{minimize} && \lambda, \\ & \text{subject to} && Y_0, Y_1, \dots, Y_{N-1} \succeq 0, \\ & && \lambda E_{00} - \mathbf{c} \mathbf{c}^* = Y_0 + \sum_{k=1}^{N-1} \left(M^2 r^{2k} Y_k - J_N^k Y_k J_N^{*k} \right). \end{aligned} \quad (\text{T.2.4})$$

A feasible primal point gives a lower certificate; a feasible dual point gives an upper certificate. Strict primal feasibility and compactness give equality of the primal and dual optimum. For rational M, t , the common factor $e^{-2t/m}$ can be removed from the objective, leaving entirely rational program data.

T.2.1 A parameter-independent feasible set

For computation and asymptotic analysis, put $D_r = \text{diag}(1, r, \dots, r^{N-1})$ and $Q = D_r P D_r$. The program is equivalently

$$V_{M,m,N}(t) = \max_{P \in \mathcal{Q}_{M,N}} \mathbf{d}^* P \mathbf{d}, \quad d_j = r^j c_j(t/m), \quad (\text{T.2.5})$$

where

$$\mathcal{Q}_{M,N} = \{P \succeq 0 : P_{00} = 1, \quad J_N^{*k} P J_N^k \preceq M^2 P \ (1 \leq k < N)\}. \quad (\text{T.2.6})$$

This feasible set depends only on M, N , not on either time parameter. Its diagonal entries are at most M^2 . The entire oscillatory problem lies in the explicit objective vector $\mathbf{d}$. This normalization also avoids tiny diagonal entries r^{2j} in numerical implementations.

For fixed m, t , $V_{M,m,N}(t)$ is nondecreasing in N : append a zero row and column to a feasible Q . Its singular quotient is exactly the original shift model.

The power-bounded envelopes are also monotone in the discretization parameter. If $m' \geq m$ and $S \in \mathcal{P}_{M,1-1/m}$, then

$$S' = \left(1 - \frac{m}{m'}\right) I + \frac{m}{m'} S \in \mathcal{P}_{M,1-1/m'}, \quad m'(S' - I) = m(S - I).$$

The binomial expansion proves the power bound, including its $k = 0$ term because $1 \leq M$. Thus $F_{M,m}(t) \leq F_{M,m'}(t)$.

T.3 A uniform truncation error

There are three useful explicit error bounds. The third, obtained from the semigroup energy identity, is substantially better. Define

$$\delta_{M,m,N}^{\text{flat}}(t) = \frac{M e^{-t/(2m)} r}{(1 - r^2) \sqrt{N}}, \quad (\text{T.3.1})$$

and

$$\delta_{M,m,N}^{\text{sine}}(t) = 2 \sin \frac{\pi}{4N+2} \frac{M e^{-t/(2m)} r \sqrt{1+r^2}}{(1-r^2)^{3/2}}. \quad (\text{T.3.2})$$

Finally set

$$\delta_{M,m,N}^{\text{en}} = mM^3r \sin \frac{\pi}{4N+2}. \quad (\text{T.3.2a})$$

Let δ be the minimum of these three expressions. Unlike the first two, (3.2a) is independent of the observation time.

Theorem 2. The finite program satisfies the uniform sandwich

$$\boxed{\sqrt{V_{M,m,N}(t)} \leq G_M(t) \leq e^{t(M+1)/m} \left(\sqrt{V_{M,m,N}(t)} + \delta_{M,m,N}(t) \right)}. \quad (\text{T.3.3})$$

The error is uniform over the Hilbert space and the generator.

Proof. First observe the exact coefficient identity

$$\sum_{j \geq 0} |c_j(a)|^2 = e^{-a}. \quad (\text{T.3.4})$$

Indeed,

$$f_a(z) = e^{-a/2} \exp \left[-\frac{a}{2} \frac{1+z}{1-z} \right].$$

The second factor is bounded analytic with radial boundary modulus one almost everywhere. Parseval and dominated convergence give (3.4).

For $S \in \mathcal{P}_{M,r}$, amplify it to $\tilde{S} = S \otimes J_N$. Its powers have the same bound. Test it at $x \otimes v$, with $\|x\| = \|v\| = 1$. The cyclic span of this vector has dimension at most N , and its cyclic Gram matrix is feasible in (2.2), whether or not the orbit is independent. Hence

$$\|f_a(\tilde{S})(x \otimes v)\| \leq \sqrt{V}.$$

For the flat vector $v = N^{-1/2}(1, \dots, 1)$,

$$\|J_N^j v - v\| = \sqrt{\min(j/N, 1)} \leq \sqrt{j/N}.$$

Using (3.4) and $\sum_{j \geq 0} j r^{2j} = r^2/(1-r^2)^2$ shows

$$\|f_a(\tilde{S})(x \otimes v) - f_a(S)x \otimes v\| \leq \delta^{\text{flat}}.$$

For a better large- N estimate, choose the normalized positive vector with entries

$$v_j \propto \sin \frac{(j+1)\pi}{2N+1}, \quad 0 \leq j < N.$$

It is a least-singular-vector of $I - J_N$, and

$$\|(I - J_N)v\| = 2 \sin \frac{\pi}{4N+2}.$$

Since J_N is a contraction, telescoping gives

$$\|J_N^j v - v\| \leq j 2 \sin \frac{\pi}{4N+2} \quad (j \geq 0).$$

Now use (3.4) and

$$\sum_{j \geq 0} j^2 r^{2j} = \frac{r^2(1+r^2)}{(1-r^2)^3}$$

to obtain δ^{sine} . For the energy estimate, first let A be any bounded generator satisfying $\|e^{sA}\| \leq Me^{-s}$ and put $B = A^{-1}$. The positive operator

$$P = \int_0^\infty e^{sA^*} e^{sA} ds$$

satisfies $0 \leq P \leq M^2 I/2$ and $B^*P + PB = -B^*B$. Differentiating $\langle Pe^{uB}x, e^{uB}x \rangle$ yields

$$\int_0^\infty \|Be^{uB}x\|^2 du \leq \frac{M^2}{2} \|x\|^2.$$

Apply the same argument separately to A^* . The semigroup law and Cauchy–Schwarz give

$$\|tB^2e^{tB}\| \leq \frac{M^2}{2}. \quad (\text{T.3.4a})$$

Indeed pair its integral representation $\int_0^t Be^{uB}Be^{(t-u)B} du$ with two unit vectors and use the two energy estimates.

Now interpolate

$$S_\theta = S \otimes [(1 - \theta)I + \theta J_N], \quad 0 \leq \theta \leq 1.$$

Its powers satisfy the same bound Mr^k , and $A_\theta = m(S_\theta - I)$ is admissible. Since all the interpolating operators commute,

$$\frac{d}{d\theta} f_a(S_\theta) = f'_a(S_\theta)[S \otimes (J_N - I)], \quad \|f'_a(S_\theta)\| = m \|tA_\theta^{-2}e^{tA_\theta^{-1}}\| \leq \frac{mM^2}{2}.$$

Apply this derivative to $x \otimes v$, with the sine vector above, and use $\|S\| \leq Mr$. Integration in θ gives (3.2a). Thus $F_{M,m}(t) \leq \sqrt{V} + \delta$. Combining with Proposition 1 proves the theorem. $\square$

This is an approximation to the actual envelope, not to an auxiliary basis problem. It is not proposed as a cheap algorithm at the enormous times relevant to this conjecture.

T.3.1 Consequences of the finite program

Theorem 2 implies the exact pointwise identities

$$F_{M,m}(t) = \sup_N \sqrt{V_{M,m,N}(t)}, \quad G_M(t) = \sup_{m \geq 2, N \geq 1} \sqrt{V_{M,m,N}(t)}. \quad (\text{T.3.5})$$

In particular, generators with a **single negative eigenvalue** suffice for the pointwise envelope: they are the finite Jordan generators $m(J_N - I)$ in the feasible Hilbert metrics. A singular feasible metric may be replaced by a convex combination with D_r^2 and a limit, so positive-definite metrics suffice. This statement concerns the supremum, not attainment by one finite matrix.

For each fixed M , a single separable direct-sum generator also realizes $G_M(t)$ simultaneously for all $t \geq 0$. Take all integer m, N and a countable dense collection of strictly feasible rational metrics in the normalized form (2.6). The corresponding semigroups have the common bound Me^{-s} , so their direct sum is strongly continuous. Their inverses have norm at most M , and the direct-sum inverse exponential has norm equal to the supremum in (3.5). No assertion about the spectrum of the infinite direct sum is inferred merely from the spectra of the finite blocks.

There are simple independent checks on the representation. For $N = 2$,

$$\sqrt{V_{M,m,2}(t)} = e^{-a} \frac{\sqrt{4 + (Mra)^2} + Mra}{2}, \quad a = t/m. \quad (\text{T.3.6})$$

Indeed write $Q = \begin{pmatrix} 1 & z \\ z & b \end{pmatrix}$. Feasibility is $0 \leq b \leq M^2 r^2$ and $z^2 \leq b - b^2/(M^2 r^2)$. Maximizing $e^{-2a}(1 - 2az + a^2b)$ gives (3.6).

For $M = 1$, the program is the contraction case, and

$$\sqrt{V_{1,m,N}(t)} = \|f_a(rJ_N)\|. \quad (\text{T.3.7})$$

To see the upper inequality, any nilpotent contraction $T^N = 0$ has an isometric intertwining into $J_N \otimes I$: use the vector $(DT^{N-1}x, \dots, Dx)$ with $D = (I - T^*T)^{1/2}$. The reverse inequality follows by taking a maximizing unit vector for $f_a(rJ_N)$ and its cyclic Gram matrix.

Finally, (3.4) by itself gives

$$F_{M,m}(t) \leq e^{-a} + M\sqrt{e^{-a} - e^{-2a}} \frac{r}{\sqrt{1-r^2}}.$$

Letting $m \rightarrow \infty$ recovers the elementary uniform estimate

$$G_M(t) \leq 1 + M\sqrt{t/2}.$$

This supplies a cross-check on the normalization without using any sharp-rate theorem.

T.4 Exact expressions for the two requested constants

For integer $n \geq 2$, take

$$t = n, \quad k_n = \lceil \log_2(n+1) \rceil, \quad m = nk_n, \quad N = nk_n^2,$$

and let $W_{M,n}$ be the program (2.2) after removing the common factor e^{-2/k_n} from its objective. Its objective coefficients are rational; all its constraint data are rational when M is rational. The matrix size is $nk_n^2 = O(n \log^2 n)$. Put

$$b_{M,n} = \frac{e^{-1/k_n} \sqrt{W_{M,n}}}{L(n)^\alpha}. \quad (\text{T.4.1})$$

The energy truncation error satisfies $\delta^{\text{en}} \leq \pi M^3/(4k_n) = o(1)$, and the multiplicative error is $e^{(M+1)/k_n} = 1 + o(1)$. Consequently,

$$\boxed{\underline{K}_M = \liminf_{n \rightarrow \infty} b_{M,n}, \quad \overline{K}_M = \limsup_{n \rightarrow \infty} b_{M,n}.} \quad (\text{T.4.2})$$

The passage from integer to real times uses monotonicity of G_M and $L(n+1)/L(n) \rightarrow 1$. Monotonicity follows by replacing A with qA , $q \geq 1$.

Equations (2.2), (2.4), and (4.2) are exact finite-dimensional variational expressions for each requested quantity. They do **not** assert convergence of $b_{M,n}$. Establishing that convergence, or finding asymptotically matching primal and dual certificates, would close the actual gap. Optimizing a prefactor in a different inequality does not do so.

Part II. A stronger upper coefficient

T.5 A better scalar concentration bound

We improve the quantitative scalar Hardy-space lemma used in [S] and [O]. The analytic-disk characterization underlying this argument is the weighted-projection equivalence in Kalton–Le Merdy [KLM, Proposition 2.3]; the quantitative concentration estimate below is a deduction, not a quotation of their theorem. Let $w > 0$ be a trigonometric polynomial for which the Riesz projection has norm at most M on $L^2(w)$. At any prescribed point on the circle,

$$(P_r * w)(\theta) \leq M^2(1+\rho)^2 \left(\frac{1+r}{1-r} \right)^\alpha \int w \, dm. \quad (\text{T.5.1})$$

The previous coefficient was $M^3(1 + \rho)^2$.

Proof. Rotate the evaluation point to 1 and average $w(e^{i\theta})$ with $w(e^{-i\theta})$. Conjugate reflection preserves Fourier signs and the Riesz-projection bound. Convex averaging preserves the weighted norm inequality, while both the desired Poisson value and the mean of w are unchanged. Thus we may assume w is even.

Its zero-free Fejér–Riesz factor h can be chosen with real coefficients and $h(0) > 0$. Put $g = \bar{h}/h$. The projection-angle/Nehari argument in [S] supplies a bounded analytic approximant f with $\|g - f\|_\infty \leq \rho$. Symmetrize f , then smooth radially. For any $\rho' > \rho$ sufficiently close to ρ , this gives f_s with real coefficients and

$$|f_s h^2 - w| \leq \rho' w.$$

The analytic function $q = f_s h^2$ has positive real part, $q(0) > 0$ real, and $|\arg q| \leq \arcsin \rho'$. Also

$$w \leq \frac{\Re q}{1 - \rho'}, \quad q(0) \leq (1 + \rho') \int w \, dm.$$

Set $\alpha' = 2 \arcsin(\rho')/\pi$. The analytic function $(q/q(0))^{1/\alpha'}$ has positive real part and value one at zero. Applying the disk-to-half-plane Schwarz inequality yields

$$|q(r)| \leq q(0) \left(\frac{1+r}{1-r} \right)^{\alpha'}.$$

Taking Poisson means in the first preceding inequality and letting $\rho' \downarrow \rho$ proves (5.1), because $(1 + \rho)/(1 - \rho) = M^2(1 + \rho)^2$. $\square$

Combine (5.1) with the exact finite-degree Poisson Toeplitz comparison from [O]. The resulting coefficient is

$$D_M = \frac{M^2(1 + \rho)^2}{\alpha^\alpha(1 - \alpha)^{1-\alpha}}. \quad (\text{T.5.2})$$

In its useful quadratic form, all-cut synthesis says

$$\left\| \sum_{j=1}^n u_j \right\|^2 \leq D_M n^\alpha \sum_{j=1}^n \|u_j\|^2. \quad (\text{T.5.3})$$

It follows by stationarizing $n^{-1} \|\sum_j u_j z^j\|^2$; replacing the average energy by the largest vector norm would unnecessarily discard information.

T.6 Polar factorization and direct row estimates

Suppose the operators B_j have norms at most B , and a bounded resolvent R at each ordered cut approximately annihilates the earlier B_j and preserves the later ones. Write

$$B_j = U_j |B_j| = Y_j X_j, \quad Y_j = |B_j^*|^{1/2}, \quad X_j = U_j |B_j|^{1/2}.$$

Then $\|Y_j\|, \|X_j\| \leq \sqrt{B}$. If R commutes with B_j , the elementary square-root inequalities give

$$\|RY_j\| \leq \sqrt{\|R\| \|RB_j\|}, \quad \|R^* X_j^*\| \leq \sqrt{\|R\| \|B_j R\|}. \quad (\text{T.6.1})$$

The same statements hold with $I - R$. For example,

$$\| |B_j^*|^{1/2} R^* y \|^2 \leq \| |B_j^*| R^* y \| \| R^* y \| = \| B_j^* R^* y \| \| R^* y \|.$$

Thus square-root-summable cut errors give approximate cuts for precisely the two families needed below: Y_j and X_j^* .

Apply (5.3) to $Y_j x_j$ for arbitrary tuples (x_j) . In the approximate-cut case augment these vectors by $c_n \|x_j\| \delta_j$; the cut correction has norm bounded by the original operator-norm error divided by c_n , uniformly in the tuple. Taking

$$c_n = (\log(n+e))^{-1}, \quad D_Q \leq (\log(n+e))^{-3}$$

makes the augmented cut norm $M + O_M((\log n)^{-4})$ and makes the added energy negligible. The operator row of the Y_j therefore has norm at most

$$[\sqrt{D_M B} + o(1)]n^{\alpha/2}.$$

The same holds for the row of X_j^* , which is the adjoint of the column of X_j . Multiplication gives

$$\left\| \sum_{j=1}^n B_j \right\| \leq [D_M B + o(1)]n^\alpha. \quad (\text{T.6.2})$$

This removes both the factor β^2/α from the rearrangement argument and the factor $2M$ from introducing separate resolvent windows. The cut errors used for inverse-semigroup blocks decay super-exponentially in the first retained block index, so their square roots have the same required summability. A discarded prefix of $o(n^\alpha)$ suffices.

T.7 One real oscillatory block costs $\kappa_* M^3$, not $2\kappa_* M^3$

The scalar constant retained from [O] is

$$\kappa_* = \inf_{\substack{g \in C_c^\infty(0,1), \\ \|g\|_2=1}} \frac{\|g\|_1^2}{\|g * g\|_2^2}, \quad 1 \leq \kappa_* < \sqrt{2}.$$

For a fixed scale exponent $1 < p < 4/3$, the real Bessel leading block has kernel

$$\pi^{-1/2} a(u) \cos(2\sqrt{u} - 3\pi/4),$$

where $0 \leq a(u) \leq u^{-3/4}$, with the symbol derivative bounds and support between N and a constant times N^p . Its operator norm is

$$\leq [\kappa(g) + o(1)]M^3, \quad (\text{T.7.1})$$

uniformly over the bounded semigroup and the damping parameter in a .

The shifted-window factorization in [O] expresses the norm bound as M^3 times a supremum over Fourier frequencies of a sum of scalar localized transforms. At a fixed Fourier frequency, at most one of the two phases $\pm 2\sqrt{u}$ has a stationary point. The one-phase stationary coefficient is $2\sqrt{\pi}$, and its coefficient in the real Bessel kernel is $1/(2\sqrt{\pi})$, so their product is one.

The other phase is uniformly negligible in the sum of the localized transforms. On a window of length comparable to N centered at U , its phase derivative has magnitude at least a constant times $U^{-1/2}$. Two integrations by parts give a bound $CU^{1/4}/N$ for that transform. Summing windows up to $U \leq CN^p$ costs

$$O(N^{5p/4-2}) = o(1).$$

The translated partition in [O] has limiting sum constant $\kappa(g)$, proving (7.1). Both small-curvature errors and nonstationary errors are uniform because $N^2 U^{-3/2} \rightarrow \infty$ for fixed $p < 4/3$.

The Bessel remainder is integrable and does not enter the coefficient of $L(t)^\alpha$.

T.8 Averaging shifted logarithmic grids removes the residue-class loss

There is no need to pay $2^{1-\alpha}$ for two residue classes.

Fix $h < \log(4/3)$ and choose $\varphi \in C_c^\infty(0, h)$ with $0 \leq \varphi \leq 1$. Write $w_\varphi = \int_0^h \varphi(x) dx$. For every real x ,

$$1 = \frac{1}{w_\varphi} \int_0^h \sum_{j \in \mathbb{Z}} \varphi(x - s - jh) ds. \quad (\text{T.8.1})$$

Apply this identity with $x = \log \log u$ above a fixed initial interval. For each fixed shift s , the kernel decomposes into a single ordered family of real blocks with amplitudes

$$u^{-3/4} e^{-u/t} \varphi(\log \log u - s - jh).$$

Because the support of φ stays a positive distance from 0 and h , consecutive block supports have a genuine power-sized gap. All cuts of this one family have super-exponentially small resolvent errors; no parity split is required. The support ratio within a block is N^p with fixed $p < e^h < 4/3$. All bounds are uniform in $s \in [0, h]$.

There are $L(t)/h + O(1)$ relevant blocks. By (6.2) and (7.1), each shifted family contributes at most

$$[D_M \kappa(g) M^3 + o(1)] (L(t)/h)^\alpha.$$

Integrate (8.1). The cost of the averaging is h/w_φ . Taking first $t \rightarrow \infty$, then $w_\varphi \uparrow h$, then $h \uparrow \log(4/3)$, and finally the infimum in g , yields

$$\overline{K}_M \leq \mathcal{V}_M := \frac{\kappa_* M^5 (1 + \rho)^2}{\alpha^\alpha (1 - \alpha)^{1-\alpha} [\log(4/3)]^\alpha}. \quad (\text{T.8.2})$$

Each auxiliary cutoff is fixed before the large-time limit. The order of limits is what permits arbitrarily narrow gaps without pretending their constants are uniform as the gap closes.

Relative to [O], the factor is

$$\frac{\mathcal{V}_M}{\mathcal{U}_M^{\text{old}}} = \frac{\alpha}{2^{1-\alpha} (1 + \alpha)^2 M^2}. \quad (\text{T.8.3})$$

At $M = 2$, taking $\kappa_* \leq \sqrt{2}$ gives

$$\overline{K}_2 \leq 3 \cdot 2^{17/6} (2 + \sqrt{3})^2 [\log(4/3)]^{-2/3} < 700. \quad (\text{T.8.4})$$

The unrounded expression is about 683.375606. It is a bound on $\overline{K}_2$, not its evaluated value.

Part III. A pure-power lower construction with much less replication

T.9 Retaining the Hardy-space information

The earlier lower proof replaced the Hardy-space angle by a pointwise sector margin. With a pure power weight, some resulting reciprocal-margin integrals diverge. It introduced a bounded analytic rounding factor to repair those integrals, at the cost of attenuation and a large phase-replication count.

Here the pure power is retained:

$$v = \alpha/2, \quad 0 < v < 1/2, \quad h(z) = (1 - z)^{-v}.$$

The actual Hardy-space deficit has finite boundary-functional norms even though the cruder reciprocal pointwise margin does not. Those norms can be evaluated using the spectral representation of the generalized Hilbert matrix.

The paired polynomial space uses

$$H(z) = U \operatorname{diag}(h(z), h(z)^{-1}) U^T, \quad U = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and norm $\int \|Hp\|^2 dm$. Each coordinate vector has squared norm

$$d_0 = \frac{J_+ + J_-}{2}, \quad J_+ = \frac{\Gamma(1-2v)}{\Gamma(1-v)^2}, \quad J_- = \frac{\Gamma(1+2v)}{\Gamma(1+v)^2}.$$

We divide the coordinate basis by $\sqrt{d_0}$.

Let $g = \bar{H}H^{-1}$. For a cut write $x = u + v_-$, with u the tail, and put

$$a = Hu, \quad b = \bar{H}v_-.$$

The positive and negative Hardy supports, with complementary allowed constant components, are orthogonal. The exact normalized deficit is

$$E(u, b) := \|x\|^2 - M^{-2}\|u\|^2 = \rho^2\|a\|_2^2 + \|b\|_2^2 + 2\Re\langle ga, b \rangle. \quad (\text{T.9.1})$$

Its strict finite-degree positivity follows also from the original pointwise sector estimate, whose pure-power margin is bounded below by

$$\Delta(\theta) \geq \frac{2c(1-c)}{\pi} |1 - e^{i\theta}|.$$

The degree estimates in [S] therefore give a positive margin at least a constant times $n^{-(1+2\alpha)}$. That is enough for the all-time semigroup realization. The exact Hardy calculation below is used for the boundary functionals, not to replace this finite-degree margin by zero.

T.10 The six exact moments for one Hardy branch

For $\theta > 1/2$ write $\mathbf{H}_\theta = (1/(j+k+\theta))_{j,k \geq 0}$. Its spectral representation is [KS, Section 3, specialized to $a = b = \theta, c = 1$]

$$\begin{aligned} \mathbf{H}_\theta &\longleftrightarrow \frac{\pi}{\cosh(\pi x)}, \\ d\mu_\theta(x) &= \frac{2x \tanh(\pi x) |\Gamma(\theta - \frac{1}{2} + ix)|^2}{\Gamma(\theta)^2} dx, \quad x > 0. \end{aligned} \quad (\text{T.10.1})$$

This imported result is a known unitary diagonalization; the moment calculations that follow are deductions from it.

The two scalar branches have $\theta_\pm = 1 \pm v$ and Hankel operators $\pm(\rho/\pi)\mathbf{H}_{\theta_\pm}$. Define their critical energies

$$A_\pm = \rho^2 \left[I - (\mathbf{H}_{\theta_\pm}/\pi)^2 \right].$$

These operators are injective but are not bounded below. Their inverse forms on the particular feature vectors below are nevertheless finite.

Let $E = e_0$, let $H_\pm$ be the coefficient vector of $(1-z)^{\mp v}$, and let $C_\pm$ be the coefficient vector of the functional $a \mapsto (g_\pm a)_0$. Put

$$e_\pm = \frac{\Gamma(\frac{1}{2} \pm v) \Gamma(\frac{3}{2} \pm v)}{\rho^2 \Gamma(1 \pm v)^2}, \quad u = \frac{1/(2\rho v) - 1}{c^2}. \quad (\text{T.10.2})$$

The Gram matrix of $(E, H_\pm, C_\pm)$ for the inverse critical-energy form is

$$V_\pm = \begin{pmatrix} e_\pm & u & 1 + c^2 u \\ u & (e_\mp - J_\pm)/c^2 & e_\mp \\ 1 + c^2 u & e_\mp & 1 + c^2 e_\mp \end{pmatrix}. \quad (\text{T.10.3})$$

No quadrature is needed to evaluate these entries.

T.10.1 Derivation and endpoint integrability

The continuous dual Hahn polynomials for (10.1) are

$$P_n(x) = {}_3F_2(-n, \frac{1}{2} + ix, \frac{1}{2} - ix; \theta, 1; 1).$$

For $w = \pm v$ and $\theta = 1 + w$, the transform of the coefficient vector $(w)_n/n!$ is

$$F_w(x) = \frac{\Gamma(1+w)}{\Gamma(1-w)} \frac{|\Gamma(\frac{1}{2} - w + ix)|^2}{|\Gamma(\frac{1}{2} + ix)|^2}. \quad (\text{T.10.4})$$

To verify this, integrate against P_n . Multiplying the measure by the proposed transform changes one continuous-dual-Hahn parameter from $1/2$ to $1/2 - w$. Its moments of $(1/2 + ix)_k(1/2 - ix)_k$ become $(1 - w)_k(1 + w)_k$. Expanding the terminating ${}_3F_2$ and applying the finite Chu–Vandermonde identity gives $(w)_n/n!$. Completeness establishes (10.4).

The EE inverse moment follows from the continuous-dual-Hahn beta integral with parameters $(0, 1, \theta - 1/2)$, interpreted as a positive limit. The EH moment reduces, by the Gamma reflection identity, to an elementary integration by parts involving

$$\int_0^\infty \log \frac{\cosh y - a}{\cosh y + a} dy = -\pi \arcsin a \quad (|a| < 1).$$

It equals u in (10.2). The HH moment uses

$$F_{\pm v}(x)^2 d\mu_{1\pm v}(x) = \frac{d\mu_{1\mp v}(x)}{1 - \rho^2 \operatorname{sech}^2(\pi x)}.$$

Partial fractions in the critical denominator, followed by the same beta integral with parameters $(1/2 \mp v, 1/2 \mp v, 1/2 \pm v)$, give $(e_\mp - J_\pm)/c^2$.

Finally, the Toeplitz identity and $T_{g_\pm} H_\pm = 1$ give

$$C_\pm = (c^2 I + A_\pm) H_\pm, \quad \langle C_\pm, H_\pm \rangle = 1.$$

These identities give the remaining three entries in (10.3). All inverse-form integrals converge: at $x = 0$ the spectral density is $O(x^2)$, whereas the critical denominator is comparable to x^2 ; the transformed features are bounded there for fixed $0 < v < 1/2$. At infinity the Gamma factors give exponential decay. Thus the calculation does not silently invert a noncoercive operator on its whole Hilbert space.

T.11 A finite matrix recipe for the exact boundary certificates

Use feature order

$$(E_0, E_1, C_0, C_1, T_0, T_1),$$

where T_s represents $a \mapsto \langle a, He_s / \sqrt{d_0} \rangle$. Define

$$\tau = (0, 0, 2, 2, 1, 1), \quad d = (1, 1, 1, 1, d_0^{-1/2}, d_0^{-1/2}),$$

and the real symmetric 6×6 matrix

$$S_{ij} = \frac{d_i d_j}{2} \left[(V_+)_{\tau_i \tau_j} + (-1)^{i+j} (V_-)_{\tau_i \tau_j} \right]. \quad (\text{T.11.1})$$

Indices are zero-based throughout this recipe.

For active coordinate type $s \in \{0, 1\}$ and endpoint $i \in \{0, 1\}$, let

$$F = \{j \in \{0, 1\} : j < s + 1 - i\},$$

so F has $s + 1 - i$ entries. Thus the four cases are $\{0\}, \emptyset, \{0, 1\}, \{0\}$ in the order $(s, i) = (0, 0), (0, 1), (1, 0), (1, 1)$.

Start with $R = S$. When $F \neq \emptyset$, first impose the forbidden tail constants:

$$R \leftarrow S - S_{:,F} S_{F,F}^{-1} S_{F,:}.$$

Then put $C_F = \{2 + j : j \in F\}$ and perform the positive rank update

$$R \leftarrow R + R_{:,C_F} (I - R_{C_F,C_F})^{-1} R_{C_F,:}. \quad (\text{T.11.2})$$

The first operation is an orthogonal restriction in the critical-energy space. The second accounts exactly for allowing the corresponding constants on the negative side of the cut.

Form a 2×2 matrix Z as follows. For the included endpoint $i = 1$,

$$Z = \text{diag}(\sqrt{d_0}, 1) R_{\{s,4+s\},\{s,4+s\}} \text{diag}(\sqrt{d_0}, 1).$$

For the excluded endpoint $i = 0$,

$$Z = \begin{pmatrix} d_0(1 + R_{2+s,2+s}) & -\sqrt{d_0} R_{2+s,4+s} \\ -\sqrt{d_0} R_{4+s,2+s} & R_{4+s,4+s} \end{pmatrix}. \quad (\text{T.11.3})$$

There is one further genuine constraint: the phase-replication quotient is evaluated at the minimizing boundary coefficient. Orthogonality to that coefficient in the deficit inner product gives

$$Z \leftarrow \begin{cases} Z - \text{diag}(1, 0), & i = 0, \\ Z - \rho^{-2} \mathbf{1}\mathbf{1}^T, & i = 1. \end{cases} \quad (\text{T.11.4})$$

This was not retained in the preceding optimization. It is a restriction, not a subtraction of an arbitrary error: the boundary vector has feature pair $(1, 0)$ and energy one at the excluded endpoint, and feature pair $(1, 1)$ and energy ρ^2 at the included endpoint.

Let

$$Q_0 = \frac{1}{M^2 - 1} \begin{pmatrix} M^2 & -M^2 \\ -M^2 & 1 \end{pmatrix}, \quad Q_1 = \frac{1}{M^2 - 1} \begin{pmatrix} 4M^2 & -2(M^2 + 1) \\ -2(M^2 + 1) & 4 \end{pmatrix}.$$

Define

$$\Lambda_{s,i} = \frac{1}{M^2} \lambda_{\max}(Z^{1/2} Q_i Z^{1/2}). \quad (\text{T.11.5})$$

It can be evaluated by trace and determinant, with no matrix square root. The matrices Z are positive semidefinite, as their construction as restricted functional Gram matrices shows.

T.11.1 Why these finite Schur complements describe the genuine Hardy deficit

Let $Ka = P_-(ga)$, and let $C_F a$ consist of the allowed constant coordinates of ga on the initial side of the cut. In the unnormalized coefficient space, write the initial Hardy vector as $b = b_- + t$, where t is supported on F . Completing squares in (9.1) gives

$$E = \langle Aa, a \rangle - \|C_F a\|^2 + \|b_- + Ka\|^2 + \|t + C_F a\|^2, \quad a_0|_F = 0. \quad (\text{T.11.6})$$

The first Schur complement in (11.2) imposes the genuine constraint $a_0|_F = 0$ in the inverse- A form. The rank update is the exact Woodbury identity for replacing A by $A - C_F^* C_F$ on that constrained space. For an included endpoint the relevant features depend only on a . For an excluded endpoint, put $z = t + C_F a$; the normalized boundary coefficient is $\sqrt{d_0}(z_s - C_s a)$. Its independent z_s term is the 1 in the top-left entry of (11.3).

At the minimizer over the active coefficient, x is orthogonal to the boundary basis vector in the deficit inner product. Restricting functionals to this orthogonal complement subtracts their

outer product on that vector divided by its energy. The two energies are respectively 1 and ρ^2 , with feature pairs (1, 0) and (1, 1). This proves (11.4), including its normalization. All the larger spaces used here give upper bounds on the finite polynomial spaces; no forbidden analyticity constraint is silently imposed on the original finite problem.

T.11.2 Why the inverse in (11.2) exists

For $F = \{0, 1\}$, Hadamard diagonalization makes its eigenvalues $1/(\rho^2 e_+)$ and $1/(\rho^2 e_-)$, both positive. For a one-element F , write

$$t_v = \frac{\Gamma(\frac{1}{2} + v)\Gamma(1 - v)}{\Gamma(1 + v)\Gamma(\frac{1}{2} - v)}.$$

The scalar Schur complement is positive precisely when

$$(\frac{1}{2} + v)t_v + (\frac{1}{2} - v)t_v^{-1} < 1.$$

Monotonicity of $\Gamma(x + 1/2)/\Gamma(x)$, which follows by differentiating its logarithm and using the positive digamma difference, gives

$$\frac{1 - 2v}{1 + 2v} < t_v < 1.$$

Those are exactly the two roots of the corresponding quadratic inequality. This proves strict positivity for every $0 < v < 1/2$.

T.12 Different replication numbers for the two coordinate types

Choose integers

$$r_s > \max\{0, \Lambda_{s,0}, \Lambda_{s,1}\}, \quad s = 0, 1. \quad (\text{T.12.1})$$

The orthogonal replication argument in [S] allows r_s to differ between coordinate types: the proof at a cut uses only the replicas of its own boundary coordinate, and the other auxiliary summands carry endpoint multipliers 0 or 1.

For clarity, if m_0, m_1, m_- are the minimized deficits for active values 0, 1, -1 , the two quadratic matrices above say

$$m_i - m_- \leq \Lambda_{s,i} m_i.$$

Let $m_{\min} = \min(m_0, m_1)$. If m_- is negative, compare it with the endpoint attaining this minimum. At a cut inside r_s replicas,

$$m_- + \sum_{a=1}^{r_s-1} m_{\sigma_a} \geq [r_s - \max_i \Lambda_{s,i}] m_{\min} > 0.$$

The Gram matrix is real, so the worst phase on the unit disk occurs at one of the two real endpoints, as proved in [S, Appendix C]. The strict finite-degree margin noted after (9.1) gives a polynomially small positive phase margin, sufficient for the scalar eigenvalue separation.

The total dimension for n frequency pairs is now

$$N = (r_0 + r_1)n,$$

not twice the larger replication count times n .

The no-attenuation normalization of [S, Appendix B.1] gives admissible finite-dimensional generators with

$$L(t_n) = (r_0 + r_1)n \log 2 + o(n),$$

while their inverse multipliers approximate the repeated sign multiplier with relative error tending to zero. Therefore

$$\boxed{\underline{K}_M \geq \frac{\mathcal{D}_\alpha(\sigma)}{[(r_0 + r_1) \log 2]^\alpha} \quad (\sigma \geq 0).} \quad (\text{T.12.2})$$

Here the beta-profile coefficient from [O] is

$$\begin{aligned} \mathcal{D}_\alpha(\sigma)^2 &= 2^{-2\alpha} \frac{\Gamma((1-\alpha)/2) \Gamma(\sigma + (1+\alpha)/2)}{\Gamma((1+\alpha)/2) \Gamma(\sigma + (1-\alpha)/2)} \\ &\quad \times \frac{\Gamma(\sigma + 1 - \alpha/2) \Gamma(2\sigma + 3/2 - \alpha/2)}{\Gamma(\sigma + 1 + \alpha/2) \Gamma(2\sigma + 3/2 + \alpha/2)}. \end{aligned} \quad (\text{T.12.3})$$

One may maximize (12.3) over σ , or use any fixed value. The more general supported-profile variational problem can only improve this lower coefficient; it is not needed for the certified example below.

T.13 A strictly verified numerical consequence at $M = 2$

Set $\alpha = 2/3$ and $v = 1/3$. The exact Gamma formulas give the following rationally enclosed phase certificates:

Coordinate type	Excluded endpoint	Included endpoint	Admissible replicas
$s = 0$	2.758412831017...	1.994135786094...	3
$s = 1$	1.689473457489...	1.486785601107...	2

These are not estimates fitted to large matrices. They are evaluations of (10.2)–(11.5), enclosed with exact rational interval arithmetic.

For $\sigma = 1/4$, the lower coefficient in (12.2) lies in

$$[0.3883692806892978, 0.3883692806893222]. \quad (\text{T.13.1})$$

In particular it is strictly larger than $3/8$. The explicit Gaussian upper expression (8.4) is strictly smaller than 700. Hence there exists a finite t_2 such that

$$\boxed{\frac{3}{8}L(t)^{2/3} \leq G_2(t) \leq 700L(t)^{2/3} \quad (t \geq t_2).} \quad (\text{T.13.2})$$

The unrounded certificates imply $\bar{K}_2/\underline{K}_2 < 1760$. The displayed eventual coefficient ratio is $5600/3 \approx 1866.67$, compared with 3,000,000 in [O]. This is an improvement by a factor larger than 1607 in the certified eventual ratio. It is not a claim that the actual ratio $\bar{K}_2/\underline{K}_2$ is that large, or that it equals one.

The attached Python certificate uses only the standard library. It encloses Gamma values by the positive-real Stirling expansion with its signed first-omitted-term remainder; encloses logarithms, exponentials and square roots by elementary series and integer arithmetic; and performs the small Schur complements with outward-rounded rational intervals. Every displayed upper decimal is rounded outward in the machine-readable result. Its assertions verify the strict replica inequalities and the two eventual coefficients. The analytic lemmas preceding the arithmetic remain necessary.

Part IV. What the exact gap question still requires

T.14 What has been determined

The report gives two different kinds of explicit expression:

1. **For the actual envelope and actual tail constants:** the finite primal/dual programs (2.2), (2.4), the uniform error estimate (3.3), and the exact limiting formulas (4.2). These do not replace G_M by an easier surrogate.
2. **For improved certificates:** the upper coefficient (8.2) and the pure-power Gamma-matrix lower coefficient (12.2). These can be evaluated without any semigroup optimization and give (13.2).

Neither item is an evaluation of $\underline{K}_M$ or $\overline{K}_M$ in closed form. In particular, the proof does not show that the sequence in (4.2) converges.

T.15 A structural bound on the possible tail oscillation

The envelope has more regularity than mere monotonicity. There is a finite constant C_M , independent of t_1, t_2 , such that

$$0 \leq G_M(t_2) - G_M(t_1) \leq C_M \left[1 + \log \frac{\log t_2}{\log t_1} \right]^\alpha \quad (e^e \leq t_1 \leq t_2). \quad (\text{T.15.1})$$

This follows from the oscillatory-block estimates rather than from the numerical prefactors in Section 13.

Proof. For an admissible A , put $C = A + I$ and $b = t_2/t_1$. Then $\|e^{sC}\| \leq M$, and $A_b = C/b - I$ is admissible. Consequently

$$e^{t_2(C-bI)^{-1}} = e^{t_1 A_b^{-1}}.$$

Subtract its Bessel representation from that of $e^{t_2(C-I)^{-1}}$. With $T(u) = e^{(u/t_2)C}$, the difference is

$$- \int_0^\infty \frac{J_1(2\sqrt{u})}{\sqrt{u}} (e^{-u/t_2} - e^{-u/t_1}) T(u) du. \quad (\text{T.15.2})$$

The integrable Bessel remainder and a fixed initial interval cost $O(M)$. Decompose the leading cosine kernel into the fixed power-sized blocks used above. Blocks wholly below t_1 have symbol bounds multiplied by $C N_{j+1}/t_1$, because $e^{-u/t_2} - e^{-u/t_1} \leq u/t_1$; their operator norms sum to $O_M(1)$. Blocks wholly above t_2 are controlled by $C e^{-N_j/t_2}$ and also sum to $O_M(1)$. The remaining blocks, including the bounded number of boundary-crossing blocks, form an ordered interval with at most

$$C \left[1 + \log \frac{\log t_2}{\log t_1} \right]$$

indices. Their amplitude derivatives have uniform symbol bounds. The all-cut synthesis/augmentation argument applies to an interval starting at any index, with the same constant: discarding a fixed-size prefix in the relative index only improves the absolute sparse-tail errors. Its sum is bounded by C_M times the number of indices to the power α . This proves the upper bound in (15.1) after taking the supremum over A . The lower inequality is monotonicity. $\square$

For every fixed $q > 0$, (15.1) implies

$$|G_M(t^q) - G_M(t)| = O_{M,q}(1), \quad \frac{G_M(t^q)}{G_M(t)} \longrightarrow 1. \quad (\text{T.15.3})$$

The second statement uses the positive lower growth estimate. Thus multiplying $\log t$ by a fixed factor cannot generate a nontrivial relative oscillation. However, oscillations on a multiplicative scale in $L(t)$ are not ruled out.

More precisely, for $R \rightarrow \infty$ the profiles

$$F_R(x) = R^{-\alpha} G_M(\exp(\exp(Rx))), \quad x > 0,$$

are locally uniformly bounded and satisfy

$$|F_R(x) - F_R(y)| \leq C_M(|x - y| + R^{-1})^\alpha.$$

Every sequence $R \rightarrow \infty$ therefore has a locally uniformly convergent subsequence. Its limit is nondecreasing, locally α -Hölder, and bounded above and below by positive multiples of x^α . To prove equality of the requested tail constants one would need to show that these limit profiles all take the same value at $x = 1$ (and hence, by scaling the subsequence, that every profile is the same constant times x^α). This compactness conclusion alone does not establish uniqueness.

T.16 Why shrinking the certificate interval is not a convergence proof

Monotonicity of G_M , the semigroup property, and an order estimate $G_M \asymp_M L^\alpha$ are insufficient by themselves to imply convergence of G_M/L^α . For example, at the purely scalar level,

$$F(x) = x^\alpha[1 + \varepsilon \sin(\log x)]$$

is increasing for sufficiently small positive ε (it suffices that $\varepsilon\sqrt{1 + \alpha^2} < \alpha$), and has positive finite but unequal normalized liminf and limsup. Composing it with $x = L(t)$ shows the limitation of monotonicity and growth order. This example is not claimed to be an inverse-semigroup envelope.

Closing the requested gap needs a further theorem specific to the extremal semigroup problem: for example, convergence of the normalized primal–dual values in (4.2), or a limiting variational problem whose upper and lower realizations have the same coefficient. Merely calling one of the sufficient certificates “optimal” would not establish that theorem.

Part X

Continuation and reproducibility

Source and scope. Preparation record, file availability, and a continuation protocol based on the retained arguments.

Arithmetic reruns below were actually performed while compiling this handoff. They do not validate the universal analytic theorem.

Z.1 Continuation protocol

Z.1.1 The immediate objective

The retained mathematical target is no longer “find another smaller displayed upper constant.” It is to audit the proposed fixed- M rate and then determine the genuine asymptotic coefficient, or show that the normalized envelope has no limit. The best retained bounds are certificates, not an identification of $\underline{K}_M$ or $\overline{K}_M$.

The finite SDP representation in T is available if it helps, but the latest SDP-asymptotics branch is outside this handoff’s working path. No successor is required to continue that branch. The analytic alternatives already present are all-cut extremal geometry, translated-window transference, phase-robust conditional bases, and compactness of rescaled envelope profiles.

Z.1.2 A practical order of work

Begin with the baseline source S, not with the most improved decimal coefficients. Check its upper and lower dependency chains independently. An error in the baseline can be isolated without the generalized Hilbert-matrix machinery. Conversely, a problem confined to the newest pure-power moment calculation need not invalidate the rounded-weight exponent construction.

Next rederive the two finite-dimensional geometric objects that both sides exploit: the all-cut synthesis profile and the phase-replication quadratic forms. The main candidate exponent comes from an exact angle. This is structurally different from estimating each oscillatory block separately.

Then examine whether the latest upper and lower relaxations can arise from a common extremal object. T’s upper proof still bounds a scalar concentration problem and several transfer steps separately. Its lower proof uses a particular weighted basis, a profile family, and sufficient phase certificates. Equality of those bounds is not automatic, even if each local calculation is optimized.

For a leading-coefficient theorem, a promising *task formulation* in the record is uniqueness of the rescaled profile limit. T’s modulus provides compactness, but not uniqueness. Any proposed argument must exclude oscillation on a multiplicative scale of $L(t)$, not merely show $G_M(t^q)/G_M(t) \rightarrow 1$ for fixed q .

These are continuation tasks suggested by the existing structure, not results proved in this document.

Z.1.3 Guardrails against repeating the same failures

Keep M fixed throughout every construction and every limiting passage used for its rate. Record explicitly when a parameter is fixed before $t \rightarrow \infty$, when a threshold may depend on it, and when optimization is taken afterwards. Never silently replace M by $M + \varepsilon$ on the lower side and retain the endpoint exponent unchanged.

In the upper proof, retain strong-integral and domain justifications and do not assume polar factors commute merely because the original blocks do. In the lower proof, a coordinate-cut estimate is not a one-active-phase estimate; a one-active-phase estimate is not yet an all-time semigroup bound without a strict margin and a summable approximation error.

Do not infer growth from a finite witness, direct-sum addition, a large scalar calculus norm, or a fit at accessible times. Distinguish exact arithmetic, floating-point diagnostic agreement, a written analytic argument, and independent mathematical verification. Save the matrices, parameters, inequalities, and logs needed to reproduce any finite claim.

A useful future progress report should say exactly which of the following changed: a proof premise, a uniformity statement, an exponent, an eventual coefficient, a threshold, an extremal characterization, or the actual equality of the tail constants. Improvements to one do not automatically improve the others.

Z.2 Evidence, retained artifacts, and reproducibility

Z.2.1 What was rerun while preparing this handoff

Four existing exact-arithmetic checkers were executed from copied files. Their assertions completed successfully. This verifies the executions and their finite arithmetic conclusions; it does not constitute a new proof audit of the analytic prerequisites.

Check	Rerun result	Scope
R1 finite witness	48 positive-definite cones; smallest integer dominance margin 999856920846456106; squared test norm > 6.348	Exact integer/rational proof of the supplied 48-dimensional example and $G_2(100) > 5/2$.
Q all-time rounded constants	All exact-rational assertions passed	Arithmetic implications of Q's already-assumed analytic estimates; $c_2 = 10^{-6}$, $C_2 = 2 \cdot 10^{45}$, $t_2 = 0$.
O eventual constants	All rational comparisons passed	Replication count 576 and the eventual coefficients $1/200$, 15000, conditional on O's analytic argument.
T tail constants	All directed interval assertions passed	Four phase-certificate bounds, replica counts 3 and 2, lower profile coefficient $> 3/8$, upper Gaussian coefficient < 700 , ratio bound < 1760 , conditional on T's moment and transference formulas.

The T rerun encloses the lower formula between approximately 0.3883692806892978 and 0.3883692806893222, and the upper formula between approximately 683.3756054891403 and 683.3756054891405. These are enclosures of **formulas in the proof**, not numerical evaluations of $\underline{K}_2$ and $\overline{K}_2$ themselves.

The scripts do not use an optimizer or floating-point tolerances for their supporting assertions. Some JSON fields contain outward-rounded decimal displays; the rational assertions and rational interval endpoints are the evidence. The archived formula in T's JSON named `analytic_scope` describes the source's asserted analytic scope, not an independent verification by this handoff.

Z.2.2 Portable commands

From the root of the companion bundle:

```
pdflatex -interaction=nonstopmode -halt-on-error MF-17-research-handoff.tex
pdflatex -interaction=nonstopmode -halt-on-error MF-17-research-handoff.tex
pdflatex -interaction=nonstopmode -halt-on-error MF-17-research-handoff.tex

python evidence/finite48/verify_certificate.py evidence/finite48/certificate.json
python evidence/alltime/verify_M2_rational.py
python evidence/optimized/verify_exact_constants.py
python evidence/tail/certify_new_constants.py
```

The LaTeX document is standalone: it has no external figure, font-file, source-file, or bibliography-database dependency. A normal TeX distribution with the listed common packages is sufficient. The last two arithmetic scripts write their JSON output beside the script; copy them before rerunning when preserving original evidence bytes matters. Fresh outputs from this preparation are under each checker’s **rerun/** directory.

The exact 48-dimensional matrix is not printed as 2304 integers in the main text. Its complete integer Gram matrix, congruence witnesses, rational exponential bound, and verifier are included in **evidence/finite48/**. Thus the finite construction remains reproducible without enlarging the readable proof by pages of raw integers.

Z.2.3 Availability and nonavailability

The original initial and first-follow-up ZIP archives contained actual reduction diagnostics, radial-variation diagnostics, the finite witness, and its search code. These have been retained. They are not evidence for the universal sharp exponent. R0’s earlier missing dimension-24 candidate remains missing and uncertified.

Source S refers to additional quadrature and finite-Gram scripts in a companion archive. Only its PDF and TeX are mounted here; the omitted diagnostic archive is not recreated from its prose. Source U’s auxiliary exploratory scripts are likewise not mounted. Source Q’s full LaTeX source, general high-precision calculator, and some larger Gram-certificate scripts are not mounted, although its PDF, JSON dependency ledger, and independent $M = 2$ rational checker are available. Historical reports of their execution are preserved as reports, not reruns.

O’s arithmetic checker and output are present in its supplied ZIP. T’s checker and exact interval output are mounted and reproduced. The final SDP-asymptotics archive is intentionally not incorporated into the working proof or evidence package; its study was explicitly deferred by the user.

The companion **SOURCE_MANIFEST.json** lists each retained source’s path, byte size, and SHA-256 hash. **SHA256SUMS** covers the deliverable files other than itself. Source copies are kept separate from rerun outputs so the original record can be distinguished from this compilation’s actions.

Z.3 Final state to preserve

The records contain a complete **proposed** fixed- M exponent proof with

$$a(M) = \frac{2}{\pi} \arccos(M^{-1}),$$

a baseline rounded-weight construction, a newer pure-power construction using exact Hardy inverse-energy features, and a series of increasingly sharp certified prefactors. The most recent retained convenient eventual pair at $M = 2$ is $3/8$ and 700, under the analytic assumptions and proofs in T and its dependencies.

The records do not identify the optimal leading coefficient, do not prove equality of the normalized $\liminf$ and $\limsup$, and have not received an independent universal proof audit. The document preserves that distinction. A future agent should continue from the mathematical constructions and their explicit checks, not from the confidence of earlier wording or the number of preceding continuations.

Sources and recorded external dependencies

The entries below reproduce bibliographic information and theorem attributions recorded in the supplied sources. They are not a fresh literature or status check. Source P’s historical date and references remain historical. An external theorem used as a proof input must be checked in the cited primary source before an independent validation is claimed.

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